Pearson Edexcel A-Level Mathematics (9MA0)

Module 1

Core Foundation & Coordinate Geometry

Paper 1 & 2 (Pure) · 87 practice questions · answers and mark schemes at the back

Subtopic	Questions	Done
1.1 Algebraic Expressions	18	____ / 18
1.2 Quadratics	18	____ / 18
1.3 Equations and Inequalities	18	____ / 18
1.4 Straight Line Graphs	15	____ / 15
1.5 Circles	18	____ / 18

How to use this workbook

1. Read the key facts and watch one video for the subtopic. Tick the skills checklist as you go.
2. Attempt each question in the squared working space without notes. Mark it with the scheme at the back.
3. Tick **Todo** when you have attempted it and **Mastered** only when you can redo it cold a week later.
4. Copy any mistakes into the notes box at the end of the subtopic. Rebuild the workbook with a new seed for fresh questions.

Name _____ Started _____ Built 14 Oct 2026, seed 2026

1.1 Algebraic Expressions

Manipulating powers, brackets and surds is the grammar of every later topic. Most marks are lost to sign slips and to not writing answers in the exact form asked for.

Key facts

- $a^m \times a^n = a^{m+n}$, $a^m \div a^n = a^{m-n}$, $(a^m)^n = a^{mn}$
- $a^{-n} = \frac{1}{a^n}$, $a^{\frac{1}{n}} = \sqrt[n]{a}$, $a^{\frac{m}{n}} = (\sqrt[n]{a})^m$, $a^0 = 1$
- $\sqrt{ab} = \sqrt{a}\sqrt{b}$, $\sqrt{\frac{a}{b}} = \frac{\sqrt{a}}{\sqrt{b}}$
- Rationalise $\frac{1}{a+\sqrt{b}}$ by multiplying top and bottom by $a - \sqrt{b}$
- $a^2 - b^2 = (a-b)(a+b)$

Method

- Write every term as a power of the same base before combining.
- Turn roots into fractional powers before differentiating or integrating.
- Factorise fully: take out the HCF first, then look for a quadratic or difference of two squares.
- For surds, find the largest square factor: $\sqrt{72} = \sqrt{36}\sqrt{2} = 6\sqrt{2}$.

Exam traps

- $(3x)^2 = 9x^2$, not $3x^2$.
- $\sqrt{a+b} \neq \sqrt{a} + \sqrt{b}$.
- $\frac{1}{\sqrt{x}} = x^{-\frac{1}{2}}$ (the minus sign is the most dropped mark in Paper 1).
- 'Factorise fully' means no common factor left inside any bracket.

In the exam: Appears as quick 2-4 mark openers on Paper 1 and Paper 2, and inside almost every calculus question.

Specification checklist

Skill	Learnt	Practised	Mastered
Index laws	☐	☐	☐
Expanding brackets	☐	☐	☐
Factorising	☐	☐	☐
Negative and fractional indices	☐	☐	☐
Surds	☐	☐	☐
Rationalising denominators	☐	☐	☐

Videos (click to open)

☐ **Recap** — Algebraic Expressions in less than 20 minutes • A-Level Maths, Pure Year 1, Chapter 1 (Bicen Maths)
☐ Index laws — Indices: The Laws of Indices (TLMaths)
☐ Expanding brackets — Polynomials: Expanding Three or More Brackets (TLMaths)
☐ Factorising — Quadratics: Factorising Quadratics of the form ax^2 + bx + c (TLMaths)
☐ Negative and fractional indices — Indices: Examples of Negative Indices (TLMaths)
☐ Surds — Surds: Introducing Surds and Simplifying Surds (TLMaths)
☐ Rationalising denominators — Surds: Introducing Rationalising the Denominator Part 1 (TLMaths)

Q1.1.01 Expanding brackets ●○○ [3 marks] ☐ Todo ☐ Mastered

Expand and simplify $(x+2)(x-1)(2x-5)$.

(3)

WORKING

Q1.1.02 Surds ●○○ [2 marks] ☐ Todo ☐ Mastered

Simplify $\sqrt{96}-\sqrt{24}$, giving your answer in the form $k\sqrt{6}$.

(2)

WORKING

Q1.1.03 Negative and fractional indices ●○○ [2 marks] ☐ Todo ☐ Mastered

Without using a calculator, find the exact value of $27^{\frac{1}{3}}$.

(2)

WORKING

Q1.1.04 Factorising ●○○ [3 marks] ☐ Todo ☐ Mastered

Factorise fully $2x^3+6x^2-56x$.

(3)

WORKING

Q1.1.05 Expanding brackets ●○○ [3 marks] ☐ Todo ☐ Mastered

Expand and simplify $(x+2)(x-5)(x+4)$.

(3)

WORKING

Q1.1.06 Surds ●○○ [2 marks] ☐ Todo ☐ Mastered

Simplify $\sqrt{54} - \sqrt{216}$, giving your answer in the form $k\sqrt{6}$.

(2)

WORKING

Q1.1.07 Negative and fractional indices ●○○ [2 marks] ☐ Todo ☐ Mastered

Without using a calculator, find the exact value of $49^{-\frac{1}{2}}$.

(2)

WORKING

Q1.1.08 Factorising ●○○ [3 marks] ☐ Todo ☐ Mastered

Factorise fully $27x^3 - 75x$.

(3)

WORKING

Q1.1.09 Expanding brackets ●○○ [3 marks] ☐ Todo ☐ Mastered

Expand and simplify $(x + 6)(x - 1)(x + 2)$.

(3)

WORKING

Q1.1.10 **Surds** ●○○ **[2 marks]** ☐ Todo ☐ Mastered

Simplify $\sqrt{27} + \sqrt{75}$, giving your answer in the form $k\sqrt{3}$.

(2)

WORKING

Q1.1.11 **Negative and fractional indices** ●○○ **[2 marks]** ☐ Todo ☐ Mastered

Without using a calculator, find the exact value of $16^{\frac{1}{2}}$.

(2)

WORKING

Q1.1.12 **Index laws** ●○○ **[2 marks]** ☐ Todo ☐ Mastered

Simplify fully $\frac{(4x^4)^3}{3x^5}$.

(2)

WORKING

Q1.1.13 **Rationalising denominators** ●○○ **[3 marks]** ☐ Todo ☐ Mastered

Express $\frac{2}{5+\sqrt{6}}$ in the form $a + b\sqrt{6}$, where a and b are rational.

(3)

WORKING

Q1.1.14 Factorising ●○○ [3 marks] ☐ Todo ☐ Mastered

Factorise fully $x^3 + 7x^2 + 6x$.

(3)

WORKING

Q1.1.15 Index laws ●●○ [3 marks] ☐ Todo ☐ Mastered

Solve $9^x \times 3^{x+4} = 9^{-2}$.

(3)

WORKING

Q1.1.16 Rationalising denominators ●●○ [4 marks] ☐ Todo ☐ Mastered

Show that $\frac{5+4\sqrt{7}}{3+3\sqrt{7}}$ can be written as $p + q\sqrt{7}$, where p and q are rational, and find p and q.

(4)

WORKING

Q1.1.17 Index laws ●●○ [3 marks] ☐ Todo ☐ Mastered

Solve $8^x \times 2^{x+1} = 4^3$.

(3)

WORKING

Q1.1.18 Rationalising denominators ●●○ [4 marks] ☐ Todo ☐ Mastered

Show that $\frac{1+1\sqrt{5}}{4+3\sqrt{5}}$ can be written as $p+q\sqrt{5}$, where p and q are rational, and find p and q.

(4)

WORKING

MY NOTES — 1.1 ALGEBRAIC EXPRESSIONS: MISTAKES TO REMEMBER

1.2 Quadratics

Quadratics are solved by factorising, completing the square or the formula. The discriminant tells you how many roots there are and powers many 'find k' questions.

Key facts

- $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$
- $x^2 + bx + c = \left(x + \frac{b}{2}\right)^2 - \frac{b^2}{4} + c$
- Turning point of $y = a(x + p)^2 + q$ is $(-p, q)$
- $b^2 - 4ac > 0$: two distinct roots; $= 0$: one repeated root; < 0: no real roots
- Line $y = mx + c$ is a tangent to a curve when the resulting quadratic has $b^2 - 4ac = 0$

Method

- To complete the square with $a \neq 1$, take a out of the x^2 and x terms first.
- For 'find the range of k', write the discriminant condition, then solve the resulting inequality (often itself a quadratic: sketch it!).
- Hidden quadratics: substitute $u = x^2$, $u = \sqrt{x}$, $u = e^x$ or $u = 2^x$ and solve for u first.
- Sketching: roots, y-intercept, turning point, and the shape from the sign of a.

Exam traps

- Forgetting the 'show that' needs every line of algebra.
- Not rejecting impossible values of u (e.g. $e^x = -2$).
- Mixing up $>$ and $\geq$ for 'real roots' (which includes equal roots: ≥ 0).

In the exam: Discriminant 'find k' questions are 4-5 marks; completing the square often leads into a graph or a transformation question.

Specification checklist

Skill	Learnt	Practised	Mastered
Solving by factorising	☐	☐	☐
Quadratic formula	☐	☐	☐
Completing the square	☐	☐	☐
The discriminant	☐	☐	☐
Quadratic functions and graphs	☐	☐	☐
Hidden quadratics	☐	☐	☐

Videos (click to open)

☐ **Recap** — Quadratics in less than 28 minutes • A-Level Maths, Pure Year 1, Chapter 2 (Bicen Maths)
☐ Solving by factorising — Quadratics: Factorising Quadratics using the Difference of Two Squares (TLMaths)
☐ Quadratic formula — Quadratics: Where the Quadratic Formula Comes From (TLMaths)
☐ Completing the square — Quadratics: Examples of Completing the Square with the form ax^2 + bx + c (TLMaths)
☐ The discriminant — Quadratics: Using the Discriminant to Find How Many Roots a Quadratic Has (TLMaths)
☐ Quadratic functions and graphs — Graphs: Sketching Quadratics, Cubics, Quartics and Quintics (TLMaths)
☐ Hidden quadratics — Exponential Equations: Spotting Hidden Quadratics (TLMaths)

Q1.2.01 Quadratic functions and graphs ●○○ [6 marks] ☐ Todo ☐ Mastered

The curve C has equation $y = x^2 - 9$.

(a) Find the coordinates of the points where C meets the coordinate axes. **(3)**

(b) Find the coordinates of the turning point of C, stating whether it is a maximum or a minimum. **(3)**

WORKING

Q1.2.02 Quadratic formula ●○○ [3 marks] ☐ Todo ☐ Mastered

Solve $2x^2 - 2x - 9 = 0$, giving your answers in exact simplified surd form.

(3)

WORKING

Q1.2.03 Solving by factorising ●○○ [3 marks] ☐ Todo ☐ Mastered

Solve $2x^2 - 4x - 30 = 0$.

(3)

WORKING

Q1.2.04 Quadratic functions and graphs ●○○ [6 marks] ☐ Todo ☐ Mastered

The curve C has equation $y = 25 - x^2$.

(a) Find the coordinates of the points where C meets the coordinate axes. **(3)**

(b) Find the coordinates of the turning point of C, stating whether it is a maximum or a minimum. **(3)**

WORKING

Q1.2.05 Quadratic formula ●○○ [3 marks] ☐ Todo ☐ Mastered

Solve $x^2 - 2x - 7 = 0$, giving your answers in exact simplified surd form.

(3)

WORKING

Q1.2.06 Solving by factorising ●○○ [3 marks] ☐ Todo ☐ Mastered

Solve $2x^2 + 6x + 4 = 0$.

(3)

WORKING

Q1.2.07 Completing the square ●○○ [3 marks] ☐ Todo ☐ Mastered

(a) Write $x^2 - x$ in the form $(x + p)^2 + q$, where p and q are constants. **(2)**

(b) Hence state the coordinates of the turning point of $y = x^2 - x$. **(1)**

WORKING

Q1.2.08 Quadratic functions and graphs ●○○ [6 marks] ☐ Todo ☐ Mastered

The curve C has equation $y = -x^2 + 9x - 20$.

(a) Find the coordinates of the points where C meets the coordinate axes. **(3)**

(b) Find the coordinates of the turning point of C, stating whether it is a maximum or a minimum. **(3)**

WORKING

Q1.2.09 Quadratic formula ●○○ [3 marks] ☐ Todo ☐ Mastered

Solve $3x^2 + 6x - 7 = 0$, giving your answers in exact simplified surd form.

(3)

WORKING

Q1.2.10 Solving by factorising ●○○ [3 marks] ☐ Todo ☐ Mastered

Solve $x^2 + 3x - 10 = 0$.

(3)

WORKING

Q1.2.11 Completing the square ●●○ [4 marks] ☐ Todo ☐ Mastered

Express $-2x^2 - 4x + 1$ in the form $a(x + b)^2 + c$, where a, b and c are integers. Hence state the maximum value of $-2x^2 - 4x + 1$.

(4)

WORKING

Q1.2.12 **The discriminant** ●●○ **[3 marks]** ☐ Todo ☐ Mastered

The equation $x^2 + 3x + k = 0$, where k is a constant, has no real roots. Find the range of possible values of k.

(3)

WORKING

Q1.2.13 **Hidden quadratics** ●●○ **[4 marks]** ☐ Todo ☐ Mastered

Solve $x - 5\sqrt{x} + 4 = 0$.

(4)

WORKING

Q1.2.14 **Completing the square** ●●○ **[4 marks]** ☐ Todo ☐ Mastered

Express $-x^2 - 6x - 9$ in the form $a(x + b)^2 + c$, where a, b and c are integers. Hence state the maximum value of $-x^2 - 6x - 9$.

(4)

WORKING

Q1.2.15 The discriminant ●●○ [3 marks] ☐ Todo ☐ Mastered

The equation $2x^2 - 6x + k = 0$, where k is a constant, has two distinct real roots. Find the range of possible values of k.

(3)

WORKING

Q1.2.16 Hidden quadratics ●●○ [4 marks] ☐ Todo ☐ Mastered

Find all real solutions of $x^4 - 5x^2 + 4 = 0$.

(4)

WORKING

Q1.2.17 Hidden quadratics ●●○ [4 marks] ☐ Todo ☐ Mastered

Find all real solutions of $x^4 - 17x^2 + 16 = 0$.

(4)

WORKING

Q1.2.18 **The discriminant** ●●● **[6 marks]** ☐ **Todo** ☐ **Mastered**

The equation $x^2 + (k+1)x + (k+3) = 0$, where k is a constant, has no real roots.

(a) Show that $k^2 - 2k - 11 < 0$. **(3)**

(b) Hence find the set of possible values of k. **(3)**

WORKING

MY NOTES — 1.2 QUADRATICS: MISTAKES TO REMEMBER

1.3 Equations and Inequalities

Solve linear/quadratic simultaneous equations by substitution, and solve inequalities using a sketch so you pick the right region. Answers often need set notation.

Key facts

- Quadratic inequality: find critical values, sketch, choose inside ($a < x < b$) or outside ($x < a$ or $x > b$)
- Set notation: $\{x : -2 < x < 5\}$, $\{x : x < 1\} \cup \{x : x > 4\}$
- Dashed lines for $<$ or $>$; solid lines for $\leq$ or $\geq$
- Multiplying or dividing an inequality by a negative reverses its sign

Method

- Linear + quadratic: rearrange the linear equation for one variable and substitute into the quadratic.
- Always find both coordinates of every intersection point.
- For 'both inequalities hold', draw both solution sets on one number line and take the overlap.

Exam traps

- Writing $2 < x > 5$ (meaningless).
- Using 'and' when you mean 'or' for outside regions.
- Dividing by a variable that could be negative.

In the exam: Typical 5-7 mark question: simultaneous equations followed by an inequality or region shading.

Specification checklist

Skill	Learnt	Practised	Mastered
Linear simultaneous equations	☐	☐	☐
Linear and quadratic simultaneous equations	☐	☐	☐
Linear inequalities	☐	☐	☐
Quadratic inequalities	☐	☐	☐
Inequalities on graphs and regions	☐	☐	☐
Set notation for solutions	☐	☐	☐

Videos (click to open)

☐ **Recap** — Equations & Inequalities in less than 20 minutes • A-Level Maths, Pure Year 1, Chapter 3 (Bicen Maths)
☐ Linear simultaneous equations — Simultaneous Equations: Examples of Using the Elimination Method (TLMaths)
☐ Linear and quadratic simultaneous equations — Simultaneous Equations: Examples of One Linear and One Quadratic Equation (TLMaths)
☐ Linear inequalities — Inequalities: Solving Linear Inequalities Graphically (TLMaths)
☐ Quadratic inequalities — Inequalities: Examples of Solving Quadratic Inequalities (TLMaths)
☐ Inequalities on graphs and regions — Inequalities: Draw and Shade a Region (TLMaths)
☐ Set notation for solutions — Inequalities: Introducing Inequalities, Set Notation and Interval Notation (TLMaths)

Q1.3.01 Quadratic inequalities ●○○ [3 marks] ☐ Todo ☐ Mastered

Find the set of values of x for which $2x^2 - 6x - 20 < 0$. Give your answer in set notation.

(3)

WORKING

Q1.3.02 Linear simultaneous equations ●○○ [3 marks] ☐ Todo ☐ Mastered

Solve the simultaneous equations

$$-2x - 4y = 0$$

$$5x + y = -27$$

(3)

WORKING

Q1.3.03 Linear inequalities ●○○ [2 marks] ☐ Todo ☐ Mastered

Find the set of values of x for which $7x + 1 > 5x + 7$.

(2)

WORKING

Q1.3.04 Quadratic inequalities ●○○ [3 marks] ☐ Todo ☐ Mastered

Find the set of values of x for which $x^2 - 4x - 12 \leq 0$. Give your answer in set notation.

(3)

WORKING

Q1.3.05 **Linear simultaneous equations** ●○○ **[3 marks]** ☐ **Todo** ☐ **Mastered**

Solve the simultaneous equations

$$5x + 4y = -23$$

$$2x + y = -8$$

(3)

WORKING

Q1.3.06 **Linear inequalities** ●○○ **[2 marks]** ☐ **Todo** ☐ **Mastered**

Find the set of values of x for which $2x - 9 > 1x + 9$.

(2)

WORKING

Q1.3.07 **Quadratic inequalities** ●○○ **[3 marks]** ☐ **Todo** ☐ **Mastered**

Find the set of values of x for which $x^2 - 8x + 15 \leq 0$. Give your answer in set notation.

(3)

WORKING

Q1.3.08 Linear simultaneous equations ●○○ [3 marks] ☐ Todo ☐ Mastered

Solve the simultaneous equations

$$-4x - 5y = -42$$

$$-2x + 4y = 18$$

(3)

WORKING

Q1.3.09 Linear inequalities ●○○ [2 marks] ☐ Todo ☐ Mastered

Find the set of values of x for which $4x + 2 > 6x - 6$.

(2)

WORKING

Q1.3.10 Set notation for solutions ●●○ [4 marks] ☐ Todo ☐ Mastered

Find the set of values of x for which both $x - 8 < 0$ and $x^2 - 11x + 30 < 0$. Give your answer in set notation.

(4)

WORKING

Q1.3.11 Inequalities on graphs and regions ●●○ [4 marks] ☐ Todo ☐ Mastered

On a sketch, shade the region R satisfying $y > 2x^2 - 5$ and $y < 2x + 5$.
Use dashed or solid lines appropriately and find the x-coordinates of the points where the boundaries meet.

(4)

WORKING

Q1.3.12 **Linear and quadratic simultaneous equations** ●●○ **[6 marks]** ☐ **Todo** ☐ **Mastered**

Solve the simultaneous equations

$$y + x - 2 = 0$$

$$y = x^2 + 6x + 12$$

(6)

WORKING

Q1.3.13 **Set notation for solutions** ●●○ **[4 marks]** ☐ **Todo** ☐ **Mastered**

Find the set of values of x for which both $x + 2 < 0$ and $x^2 + 3x < 0$. Give your answer in set notation.

(4)

WORKING

Q1.3.14 Inequalities on graphs and regions ●●○ [4 marks] ☐ Todo ☐ Mastered

On a sketch, shade the region R satisfying $y > x^2 - 7$ and $y < x + 7$.
Use dashed or solid lines appropriately and find the x-coordinates of the points where the boundaries meet.

(4)

WORKING

Q1.3.15 Linear and quadratic simultaneous equations ●●○ [6 marks] ☐ Todo ☐ Mastered

Solve the simultaneous equations

$$y - 4x - 3 = 0$$

$$y = x^2 + x + 5$$

(6)

WORKING

Q1.3.16 Set notation for solutions ●●○ [4 marks] ☐ Todo ☐ Mastered

Find the set of values of x for which both $x - 4 < 0$ and $x^2 - 3x + 2 < 0$. Give your answer in set notation.

(4)

WORKING

Q1.3.17 **Inequalities on graphs and regions** ●●○ **[4 marks]** ☐ **Todo** ☐ **Mastered**

On a sketch, shade the region R satisfying $y > x^2 - 8$ and $y < -x + 8$.
Use dashed or solid lines appropriately and find the x-coordinates of the points where the boundaries meet.

(4)

WORKING

Q1.3.18 **Linear and quadratic simultaneous equations** ●●○ **[6 marks]** ☐ **Todo** ☐ **Mastered**

Solve the simultaneous equations

$$y - 3x - 3 = 0$$

$$y = x^2 + 5x$$

(6)

WORKING

MY NOTES — 1.3 EQUATIONS AND INEQUALITIES: MISTAKES TO REMEMBER

1.4 Straight Line Graphs

Straight lines: gradient, equations in different forms, parallel and perpendicular lines, distances and midpoints, and using lines as simple models.

Key facts

- $m = \frac{y_2 - y_1}{x_2 - x_1}$, $y - y_1 = m(x - x_1)$
- Parallel lines: $m_1 = m_2$; perpendicular lines: $m_1 m_2 = -1$
- Midpoint $\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right)$, length $\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$
- $ax + by + c = 0$ with a, b, c integers is a common required form

Method

- Perpendicular bisector: midpoint + negative reciprocal gradient.
- Area of a triangle formed with the axes: find both intercepts first.
- In modelling questions, interpret the gradient as a rate and the intercept as a starting value.

Exam traps

- Leaving fractions when 'integers' are demanded.
- Swapping x and y differences in the gradient.

In the exam: Usually combined with circles or as part (a) of a longer coordinate geometry question.

Specification checklist

Skill	Learnt	Practised	Mastered
Gradient and y = mx + c	☐	☐	☐
y - y1 = m(x - x1) and ax + by + c = 0	☐	☐	☐
Parallel and perpendicular lines	☐	☐	☐
Length and midpoint	☐	☐	☐
Modelling with straight lines	☐	☐	☐

Videos (click to open)

☐ **Recap** — Straight Line Graphs in less than 22 minutes • A-Level Maths, Pure Year 1, Chapter 5 (Bicen Maths)
☐ Gradient and y = mx + c — Coordinate Geometry: Finding the Equation of a Line in the form ax+by+c=0 (TLMaths)
☐ y - y1 = m(x - x1) and ax + by + c = 0 — Coordinate Geometry: The Equation of a Line in the form y-y1 = m(x - x1) (TLMaths)
☐ Parallel and perpendicular lines — Coordinate Geometry: Parallel and Perpendicular Lines (TLMaths)
☐ Length and midpoint — Coordinate Geometry: Examples of Finding the Midpoint between Two Points (TLMaths)
☐ Modelling with straight lines — Coordinate geometry (straight lines) : C1 Edexcel January 2011 Q9 (ExamSolutions)

Q1.4.01 Parallel and perpendicular lines ●○○ [3 marks] ☐ Todo ☐ Mastered

The line l_1 has equation $y = -2x + 3$.
The line l_2 is perpendicular to l_1 and passes through $P(0, 5)$. Find an equation of l_2 in the form $y = mx + c$.

(3)

WORKING

Q1.4.02 Length and midpoint ●○○ [3 marks] ☐ Todo ☐ Mastered

A is the point $(-2, 7)$ and B is the point $(-10, -8)$.

(a) Find the midpoint of AB. **(1)**

(b) Find the exact length of AB, simplifying your answer. **(2)**

WORKING

Q1.4.03 Gradient and y = mx + c ●○○ [4 marks] ☐ Todo ☐ Mastered

The points $A(5, 2)$ and $B(-5, -8)$ lie on the line l. Find an equation for l in the form $ax + by + c = 0$, where a, b and c are integers.

(4)

WORKING

Q1.4.04 Parallel and perpendicular lines ●○○ [3 marks] ☐ Todo ☐ Mastered

The line l_1 has equation $y = 1x + 7$.
The line l_2 is perpendicular to l_1 and passes through $P(-2, -3)$. Find an equation of l_2 in the form $y = mx + c$.

(3)

WORKING

Q1.4.05 Length and midpoint ●○○ [3 marks] ☐ Todo ☐ Mastered

A is the point $(-3, -2)$ and B is the point $(-4, 5)$.

(a) Find the midpoint of AB. **(1)**

(b) Find the exact length of AB, simplifying your answer. **(2)**

WORKING

Q1.4.06 Gradient and y = mx + c ●○○ [4 marks] ☐ Todo ☐ Mastered

The points $A(0, -1)$ and $B(5, 2)$ lie on the line l. Find an equation for l in the form $ax + by + c = 0$, where a, b and c are integers.

(4)

WORKING

Q1.4.07 Parallel and perpendicular lines ●○○ [3 marks] ☐ Todo ☐ Mastered

The line l_1 has equation $y = 5x + 3$.

The line l_2 is parallel to l_1 and passes through $P(4, -3)$. Find an equation of l_2 in the form $y = mx + c$.

(3)

WORKING

Q1.4.08 Length and midpoint ●○○ [3 marks] ☐ Todo ☐ Mastered

A is the point $(4, 8)$ and B is the point $(8, 12)$.

(a) Find the midpoint of AB. **(1)**

(b) Find the exact length of AB, simplifying your answer. **(2)**

WORKING

Q1.4.09 Gradient and y = mx + c ●○○ [4 marks] ☐ Todo ☐ Mastered

The points $A(-6, -1)$ and $B(-4, -4)$ lie on the line l. Find an equation for l in the form $ax + by + c = 0$, where a, b and c are integers.

(4)

WORKING

Q1.4.10 Modelling with straight lines ●●○ [6 marks] ☐ Todo ☐ Mastered

A plumber charges a fixed call-out fee plus an hourly rate. A job lasting 4 hours costs £154 and a job lasting 12 hours costs £346. The cost C is modelled as a linear function of the time t.

(a) Find an equation linking C and t. **(3)**

(b) Interpret the gradient and the intercept in context. **(2)**

(c) State a limitation of the model. **(1)**

WORKING

Q1.4.11 y - y1 = m(x - x1) and ax + by + c = 0 ●●○ [5 marks] ☐ Todo ☐ Mastered

Find the equation of the perpendicular bisector of the line segment joining $P(6, 12)$ and $Q(-12, -2)$. Give your answer in the form $ax + by + c = 0$, where a, b and c are integers.

(5)

WORKING

Q1.4.12 Modelling with straight lines ●●○ [6 marks] ☐ Todo ☐ Mastered

A plumber charges a fixed call-out fee plus an hourly rate. A job lasting 5 hours costs £90 and a job lasting 12 hours costs £153. The cost C is modelled as a linear function of the time t.

(a) Find an equation linking C and t. **(3)**

(b) Interpret the gradient and the intercept in context. **(2)**

(c) State a limitation of the model. **(1)**

WORKING

Q1.4.13 y - y1 = m(x - x1) and ax + by + c = 0 ●●○ [5 marks] ☐ Todo ☐ Mastered

Find the equation of the perpendicular bisector of the line segment joining $P(4, 4)$ and $Q(8, -10)$. Give your answer in the form $ax + by + c = 0$, where a, b and c are integers.

(5)

WORKING

Q1.4.14 Modelling with straight lines ●●○ [6 marks] ☐ Todo ☐ Mastered

The cost of hiring a van is a fixed fee plus a charge per day. A job lasting 4 days costs £68 and a job lasting 9 days costs £113. The cost C is modelled as a linear function of the time t.

(a) Find an equation linking C and t. **(3)**

(b) Interpret the gradient and the intercept in context. **(2)**

(c) State a limitation of the model. **(1)**

WORKING

Q1.4.15 y - y1 = m(x - x1) and ax + by + c = 0 ●●○ [5 marks] ☐ Todo ☐ Mastered

Find the equation of the perpendicular bisector of the line segment joining $P(8, -10)$ and $Q(-2, 4)$. Give your answer in the form $ax + by + c = 0$, where a, b and c are integers.

(5)

WORKING

MY NOTES — 1.4 STRAIGHT LINE GRAPHS: MISTAKES TO REMEMBER

1.5 Circles

Circles: write the equation from centre and radius, complete the square to find centre and radius, and use circle properties (tangent perpendicular to radius, perpendicular bisector of a chord passes through the centre).

Key facts

- $(x-a)^2+(y-b)^2=r^2$, centre (a, b), radius r
- $x^2+y^2+2fx+2gy+c=0$ has centre $(-f, -g)$, $r=\sqrt{f^2+g^2-c}$
- Tangent at P is perpendicular to the radius at P
- Perpendicular bisector of a chord passes through the centre; angle in a semicircle is 90°
- Line meets circle: substitute, then b^2-4ac tells you secant (>0), tangent ($=0$), miss (<0)

Method

- To find a circle through 3 points, intersect two perpendicular bisectors.
- For tangent lengths from an external point, use Pythagoras with the radius.
- To check if a point is inside, compare its distance from the centre with r.

Exam traps

- Forgetting to square root r^2.
- Sign errors when completing the square: $(x-3)^2$ means centre $x=+3$.

In the exam: Paper 1 or 2, 6-10 marks: often centre/radius then a tangent or an intersection with a line.

Specification checklist

Skill	Learnt	Practised	Mastered
Equation of a circle	☐	☐	☐
Completing the square to find centre and radius	☐	☐	☐
Tangents and normals to circles	☐	☐	☐
Chords and perpendicular bisectors	☐	☐	☐
Intersections of lines and circles	☐	☐	☐
Circle theorems in coordinate geometry	☐	☐	☐

Videos (click to open)

☐ **Recap** — Circles in less than 22 minutes • A-Level Maths, Pure Year 1, Chapter 6 (Bicen Maths)
☐ Equation of a circle — Circles: Introducing the Equation of the Circle (TLMaths)
☐ Completing the square to find centre and radius — Circles: Using Completing the Square to find the Centre & Radius (TLMaths)
☐ Tangents and normals to circles — Circles: Finding Tangents & Normals (TLMaths)
☐ Chords and perpendicular bisectors — Circles: Perpendicular Bisector of Two Points on a Circle (TLMaths)
☐ Intersections of lines and circles — Circles: Finding where a Line Intersects a Circle (TLMaths)
☐ Circle theorems in coordinate geometry — Circles: The Angle in a Semicircle is a Right Angle (TLMaths)

Q1.5.01 Completing the square to find centre and radius ●○○ [4 marks] ☐ Todo ☐ Mastered

The circle C has equation $x^2 + y^2 + 4x - 8y - 30 = 0$. Find the coordinates of the centre of C and the exact radius of C.

(4)

WORKING

Q1.5.02 Equation of a circle ●○○ [4 marks] ☐ Todo ☐ Mastered

The points $A(-5, 3)$ and $B(3, 5)$ are the ends of a diameter of the circle C. Find an equation for C.

(4)

WORKING

Q1.5.03 Completing the square to find centre and radius ●○○ [4 marks] ☐ Todo ☐ Mastered

The circle C has equation $x^2 + y^2 + 12x + 14y + 77 = 0$. Find the coordinates of the centre of C and the exact radius of C.

(4)

WORKING

Q1.5.04 Equation of a circle ●○○ [4 marks] ☐ Todo ☐ Mastered

The points $A(5, 3)$ and $B(3, -1)$ are the ends of a diameter of the circle C. Find an equation for C.

(4)

WORKING

Q1.5.05 Completing the square to find centre and radius ●○○ [4 marks] ☐ Todo ☐ Mastered

The circle C has equation $x^2 + y^2 - 2x - 12y + 19 = 0$. Find the coordinates of the centre of C and the exact radius of C.

(4)

WORKING

Q1.5.06 Equation of a circle ●○○ [4 marks] ☐ Todo ☐ Mastered

The points $A(-2, 0)$ and $B(6, 8)$ are the ends of a diameter of the circle C. Find an equation for C.

(4)

WORKING

Q1.5.07 Tangents and normals to circles ●●○ [6 marks] ☐ Todo ☐ Mastered

The circle C has centre $(-2, 3)$ and passes through the point $P(0, 4)$.

(a) Find an equation for C. **(2)**

(b) Find an equation of the tangent to C at P, giving your answer in the form $ax + by + c = 0$. **(4)**

WORKING

Q1.5.08 Intersections of lines and circles ●●○ [5 marks] ☐ Todo ☐ Mastered

The circle C has equation $(x+2)^2+(y-2)^2=5$ and the line l has equation $y=-x+k$, where k is a constant. Given that l is a tangent to C, find the exact possible values of k.

(5)

WORKING

Q1.5.09 Circle theorems in coordinate geometry ●●○ [6 marks] ☐ Todo ☐ Mastered

The points $A(-5,\ -1)$, $B(5,\ -1)$ and $C(3, 3)$ lie on a circle.

(a) Show that AC is perpendicular to BC. **(3)**

(b) Hence explain why AB is a diameter and find an equation of the circle. **(3)**

WORKING

Q1.5.10 Tangents and normals to circles ●●○ [6 marks] ☐ Todo ☐ Mastered

The circle C has centre $(-5, -4)$ and passes through the point $P(-2, -3)$.

(a) Find an equation for C. **(2)**

(b) Find an equation of the tangent to C at P, giving your answer in the form $ax + by + c = 0$. **(4)**

WORKING

Q1.5.11 Intersections of lines and circles ●●○ [5 marks] ☐ Todo ☐ Mastered

The circle C has equation $(x-4)^2 + (y+4)^2 = 5$ and the line l has equation $y = -x + k$, where k is a constant. Given that l is a tangent to C, find the exact possible values of k.

(5)

WORKING

Q1.5.12 Circle theorems in coordinate geometry ●●○ [6 marks] ☐ Todo ☐ Mastered

The points $A(-8, 2)$, $B(2, 2)$ and $C(0, 6)$ lie on a circle.

(a) Show that AC is perpendicular to BC. **(3)**

(b) Hence explain why AB is a diameter and find an equation of the circle. **(3)**

WORKING

Q1.5.13 Tangents and normals to circles ●●○ [6 marks] ☐ Todo ☐ Mastered

The circle C has centre $(-4, 2)$ and passes through the point $P(-8, 5)$.

(a) Find an equation for C. **(2)**

(b) Find an equation of the tangent to C at P, giving your answer in the form $ax + by + c = 0$. **(4)**

WORKING

Q1.5.14 Intersections of lines and circles ●●○ [5 marks] ☐ Todo ☐ Mastered

The circle C has equation $(x-4)^2 + (y-2)^2 = 10$ and the line l has equation $y = -x + k$, where k is a constant. Given that l is a tangent to C, find the exact possible values of k.

(5)

WORKING

Q1.5.15 Circle theorems in coordinate geometry ●●○ [6 marks] ☐ Todo ☐ Mastered

The points $A(-2, -3)$, $B(8, -3)$ and $C(6, 1)$ lie on a circle.

(a) Show that AC is perpendicular to BC. **(3)**

(b) Hence explain why AB is a diameter and find an equation of the circle. **(3)**

WORKING

Q1.5.16 Chords and perpendicular bisectors ●●● [7 marks] ☐ Todo ☐ Mastered

The points $P(4, 1)$, $Q(-1, 6)$ and $R(-4, -3)$ lie on a circle C.

(a) Find the equations of the perpendicular bisectors of PQ and QR. **(4)**

(b) Hence find the centre and radius of C. **(3)**

WORKING

Q1.5.17 Chords and perpendicular bisectors ●●● [7 marks] ☐ Todo ☐ Mastered

The points $P(5, 2)$, $Q(-2, 1)$ and $R(6, -5)$ lie on a circle C.

(a) Find the equations of the perpendicular bisectors of PQ and QR. **(4)**

(b) Hence find the centre and radius of C. **(3)**

WORKING

Q1.5.18 **Chords and perpendicular bisectors** ●●● **[7 marks]** ☐ **Todo** ☐ **Mastered**

The points $P(3, 1)$, $Q(-7, 1)$ and $R(1, 5)$ lie on a circle C.

(a) Find the equations of the perpendicular bisectors of PQ and QR. **(4)**

(b) Hence find the centre and radius of C. **(3)**

WORKING

MY NOTES — 1.5 CIRCLES: MISTAKES TO REMEMBER

Answers and mark schemes

M = method mark, A = accuracy mark (depends on the preceding M), B = independent mark, ft = follow-through, * = given answer. Numerical answers are to 3 s.f. unless stated.

1.1 Algebraic Expressions

Q1.1.01 **Expanding brackets · 3 marks**

$2x^3 - 3x^2 - 9x + 10$

M1 expands two brackets correctly · M1 multiplies by third bracket · A1 $2x^3 - 3x^2 - 9x + 10$

Q1.1.02 **Surds · 2 marks**

$2\sqrt{6}$

M1 $\sqrt{96} = 4\sqrt{6}$ or $\sqrt{24} = 2\sqrt{6}$ · A1 $2\sqrt{6}$

Q1.1.03 **Negative and fractional indices · 2 marks**

3

M1 $27^{1/3} = 3$ · A1 3

Q1.1.04 **Factorising · 3 marks**

$2x(x-4)(x+7)$

M1 takes out $2x$ · M1 attempts to factorise the quadratic · A1 $2x(x-4)(x+7)$

Q1.1.05 **Expanding brackets · 3 marks**

$x^3 + x^2 - 22x - 40$

M1 expands two brackets correctly · M1 multiplies by third bracket · A1 $x^3 + x^2 - 22x - 40$

Q1.1.06 **Surds · 2 marks**

$-3\sqrt{6}$

M1 $\sqrt{54} = 3\sqrt{6}$ or $\sqrt{216} = 6\sqrt{6}$ · A1 $-3\sqrt{6}$

Q1.1.07 **Negative and fractional indices · 2 marks**

$\frac{1}{7}$

M1 $49^{1/2} = 7$ · A1 $\frac{1}{7}$

Q1.1.08 **Factorising · 3 marks**

$3x(3x-5)(3x+5)$

M1 takes out a common factor $3x$ · M1 difference of two squares · A1 $3x(3x-5)(3x+5)$

Q1.1.09 **Expanding brackets · 3 marks**

$x^3 + 7x^2 + 4x - 12$

M1 expands two brackets correctly · M1 multiplies by third bracket · A1 $x^3 + 7x^2 + 4x - 12$

Q1.1.10 **Surds · 2 marks**

$8\sqrt{3}$

M1 $\sqrt{27} = 3\sqrt{3}$ or $\sqrt{75} = 5\sqrt{3}$ · A1 $8\sqrt{3}$

Q1.1.11 **Negative and fractional indices · 2 marks**

4

M1 $16^{1/2} = 4$ · A1 4

Q1.1.12 **Index laws · 2 marks**

$\frac{64x^7}{3}$

M1 uses $(x^m)^n = x^{mn}$ to get $64x^{12}$ · A1 $\frac{64x^7}{3}$

Q1.1.13 **Rationalising denominators · 3 marks**

$\frac{10}{19} - \frac{2\sqrt{6}}{19}$

M1 multiplies numerator and denominator by $5 - \sqrt{6}$ · M1 denominator $= 19$ · A1 $\frac{10}{19} - \frac{2\sqrt{6}}{19}$

Q1.1.14 **Factorising · 3 marks**

$x(x+1)(x+6)$

M1 takes out x · M1 attempts to factorise the quadratic · A1 $x(x+1)(x+6)$

Q1.1.15 **Index laws · 3 marks**

$x = -\frac{8}{3}$

M1 writes every term as a power of 3 · M1 $2x + (x+4) = -4$ · A1 $x = -\frac{8}{3}$

Q1.1.16 **Rationalising denominators · 4 marks**

$\frac{\sqrt{7}}{18} + \frac{23}{18}$

M1 multiplies by $3 - 3\sqrt{7}$ · M1 expands numerator (4 terms) · M1 denominator -54 · A1 $\frac{\sqrt{7}}{18} + \frac{23}{18}$

Q1.1.17 **Index laws · 3 marks**

$x = \frac{5}{4}$

M1 writes every term as a power of 2 · M1 $3x + (x+1) = 6$ · A1 $x = \frac{5}{4}$

Q1.1.18 **Rationalising denominators · 4 marks**

$\frac{11}{29} - \frac{\sqrt{5}}{29}$

M1 multiplies by $4 - 3\sqrt{5}$ · M1 expands numerator (4 terms) · M1 denominator -29 · A1 $\frac{11}{29} - \frac{\sqrt{5}}{29}$

1.2 Quadratics

Q1.2.01 **Quadratic functions and graphs · 6 marks**

(a) $(-3, 0), (3, 0), (0, -9)$

M1 factorises or solves $y = 0$ · A1 both x-intercepts · B1 y-intercept

(b) $(0, -9)$, minimum

M1 symmetry or completing the square for x · A1 coordinates · B1 nature

Q1.2.02 **Quadratic formula · 3 marks**

$x = \frac{1 \pm \sqrt{19}}{2}$

M1 correct substitution into the formula · M1 discriminant $= 76$ · A1 both roots, simplified

Q1.2.03 **Solving by factorising · 3 marks**

$x = -3,\ x = 5$

M1 factorises: $(2x + 6)(x - 5)$ · A1 one correct root · A1 both roots

Q1.2.04 **Quadratic functions and graphs · 6 marks**

(a) $(-5, 0), (5, 0), (0, 25)$

M1 factorises or solves $y = 0$ · A1 both x-intercepts · B1 y-intercept

(b) $(0, 25)$, maximum

M1 symmetry or completing the square for x · A1 coordinates · B1 nature

Q1.2.05 **Quadratic formula · 3 marks**

$x = 1 \pm 2\sqrt{2}$

M1 correct substitution into the formula · M1 discriminant $= 32$ · A1 both roots, simplified

Q1.2.06 **Solving by factorising · 3 marks**

$x = -2,\ x = -1$

M1 factorises: $(2x + 2)(x + 2)$ · A1 one correct root · A1 both roots

Q1.2.07 **Completing the square · 3 marks**

(a) $(x - \frac{1}{2})^2 - \frac{1}{4}$

B1 $p = -\frac{1}{2}$ · M1A1 $q = -\frac{1}{4}$

(b) $(\frac{1}{2}, -\frac{1}{4})$

B1ft

Q1.2.08 **Quadratic functions and graphs · 6 marks**

(a) $(4, 0), (5, 0), (0, -20)$

M1 factorises or solves $y = 0$ · A1 both x-intercepts · B1 y-intercept

(b) $(\frac{9}{2}, \frac{1}{4})$, maximum

M1 symmetry or completing the square for x · A1 coordinates · B1 nature

Q1.2.09 **Quadratic formula · 3 marks**

$x = \frac{-3 \pm \sqrt{30}}{3}$

M1 correct substitution into the formula · M1 discriminant $= 120$ · A1 both roots, simplified

Q1.2.10 **Solving by factorising · 3 marks**

$x = -5,\ x = 2$

M1 factorises: $(x - 2)(x + 5)$ · A1 one correct root · A1 both roots

Q1.2.11 **Completing the square · 4 marks**

$-2(x + 1)^2 + 3$; maximum value 3

B1 $a = -2$ · M1 $b = 1$ · A1 $c = 3$ · B1ft max value $= 3$

Q1.2.12 **The discriminant · 3 marks**

$k > \frac{9}{4}$

M1 uses $b^2 - 4ac$ with $a = 1,\ b = 3,\ c = k$ · M1 correct inequality < 0 · A1 $k > \frac{9}{4}$

Q1.2.13 **Hidden quadratics · 4 marks**

$x = 1,\ x = 16$

M1 lets $u = \sqrt{x}$ · M1 $(u - 1)(u - 4) = 0$ · M1 squares to find x · A1 both values

Q1.2.14 **Completing the square · 4 marks**

$-1(x+3)^2+0$; maximum value 0

B1 $a=-1$ · M1 $b=3$ · A1 $c=0$ · B1ft max value $=0$

Q1.2.15 **The discriminant · 3 marks**

$k<\frac{9}{2}$

M1 uses b^2-4ac with $a=2$, $b=-6$, $c=k$ · M1 correct inequality >0 · A1 $k<\frac{9}{2}$

Q1.2.16 **Hidden quadratics · 4 marks**

$x=\pm1$, $x=\pm2$

M1 lets $u=x^2$ · M1 $(u-1)(u-4)=0$ · M1 square roots · A1 all four

Q1.2.17 **Hidden quadratics · 4 marks**

$x=\pm1$, $x=\pm4$

M1 lets $u=x^2$ · M1 $(u-1)(u-16)=0$ · M1 square roots · A1 all four

Q1.2.18 **The discriminant · 6 marks**

(a) $k^2-2k-11<0$

M1 uses $b^2-4ac<0$ · M1 expands · A1* given answer, no errors

(b) $1-2\sqrt{3}<k<1+2\sqrt{3}$

M1 solves the quadratic to find critical values · M1 chooses the inside region · A1 $1-2\sqrt{3}<k<1+2\sqrt{3}$

1.3 Equations and Inequalities

Q1.3.01 **Quadratic inequalities · 3 marks**

$-2<x<5$, i.e. $\{x:-2<x<5\}$

M1 critical values $x=-2,5$ · M1 chooses inside region · A1 correct set notation

Q1.3.02 **Linear simultaneous equations · 3 marks**

$x=-6$, $y=3$

M1 eliminates one variable · A1 one value · A1 both

Q1.3.03 **Linear inequalities · 2 marks**

$x>3$

M1 collects terms · A1 $x>3$

Q1.3.04 **Quadratic inequalities · 3 marks**

$-2\le x\le 6$, i.e. $\{x:-2\le x\le 6\}$

M1 critical values $x=-2,6$ · M1 chooses inside region · A1 correct set notation

Q1.3.05 **Linear simultaneous equations · 3 marks**

$x=-3$, $y=-2$

M1 eliminates one variable · A1 one value · A1 both

Q1.3.06 **Linear inequalities · 2 marks**

$x>18$

M1 collects terms · A1 $x>18$

Q1.3.07 **Quadratic inequalities · 3 marks**

$3\le x\le 5$, i.e. $\{x:3\le x\le 5\}$

M1 critical values $x=3,5$ · M1 chooses inside region · A1 correct set notation

Q1.3.08 **Linear simultaneous equations · 3 marks**

$x=3$, $y=6$

M1 eliminates one variable · A1 one value · A1 both

Q1.3.09 **Linear inequalities · 2 marks**

$x>4$

M1 collects terms · A1 $x>4$

Q1.3.10 **Set notation for solutions · 4 marks**

$\{x:5<x<6\}$

B1 $x<8$ · M1A1 $5<x<6$ · A1 combined set

Q1.3.11 **Inequalities on graphs and regions · 4 marks**

Boundaries are dashed (strict). Intersections where $2x^2-2x-10=0$: $x=\frac{1}{2}-\frac{\sqrt{21}}{2}$, $x=\frac{1}{2}+\frac{\sqrt{21}}{2}$

B1 parabola and line sketched · B1 dashed lines · M1 solves for intersections · A1 region R shaded between the curves

Q1.3.12 **Linear and quadratic simultaneous equations · 6 marks**

$x=-5, y=7$; $x=-2, y=4$

M1 substitutes linear into quadratic · A1 $x^2+7x+10=0$ · M1 solves 3TQ · A1 both x values · M1 substitutes back · A1 both pairs

Q1.3.13 **Set notation for solutions · 4 marks**

$\{x : -3 < x < -2\}$

B1 $x < -2$ · M1A1 $-3 < x < 0$ · A1 combined set

Q1.3.14 **Inequalities on graphs and regions · 4 marks**

Boundaries are dashed (strict). Intersections where $x^2 - x - 14 = 0$: $x = \frac{1}{2} - \frac{\sqrt{57}}{2}$, $x = \frac{1}{2} + \frac{\sqrt{57}}{2}$

B1 parabola and line sketched · B1 dashed lines · M1 solves for intersections · A1 region R shaded between the curves

Q1.3.15 **Linear and quadratic simultaneous equations · 6 marks**

$x = 1, y = 7; x = 2, y = 11$

M1 substitutes linear into quadratic · A1 $x^2 - 3x + 2 = 0$ · M1 solves 3TQ · A1 both x values · M1 substitutes back · A1 both pairs

Q1.3.16 **Set notation for solutions · 4 marks**

$\{x : 1 < x < 2\}$

B1 $x < 4$ · M1A1 $1 < x < 2$ · A1 combined set

Q1.3.17 **Inequalities on graphs and regions · 4 marks**

Boundaries are dashed (strict). Intersections where $x^2 + x - 16 = 0$: $x = -\frac{1}{2} + \frac{\sqrt{65}}{2}$, $x = -\frac{\sqrt{65}}{2} - \frac{1}{2}$

B1 parabola and line sketched · B1 dashed lines · M1 solves for intersections · A1 region R shaded between the curves

Q1.3.18 **Linear and quadratic simultaneous equations · 6 marks**

$x = -3, y = -6; x = 1, y = 6$

M1 substitutes linear into quadratic · A1 $x^2 + 2x - 3 = 0$ · M1 solves 3TQ · A1 both x values · M1 substitutes back · A1 both pairs

1.4 Straight Line Graphs

Q1.4.01 **Parallel and perpendicular lines · 3 marks**

$y = \frac{1}{2}x + 5$

B1 gradient $\frac{1}{2}$ · M1 uses the point · A1 $y = \frac{1}{2}x + 5$

Q1.4.02 **Length and midpoint · 3 marks**

(a) $(-6, -\frac{1}{2})$

B1

(b) 17

M1 $\sqrt{8^2 + 15^2}$ · A1 17

Q1.4.03 **Gradient and y = mx + c · 4 marks**

$x - y - 3 = 0$

M1 gradient $= 1$ · M1 uses $y - y_1 = m(x - x_1)$ · A1 any correct form · A1 integer form

Q1.4.04 **Parallel and perpendicular lines · 3 marks**

$y = -1x - 5$

B1 gradient -1 · M1 uses the point · A1 $y = -1x - 5$

Q1.4.05 **Length and midpoint · 3 marks**

(a) $(-\frac{7}{2}, \frac{3}{2})$

B1

(b) $5\sqrt{2}$

M1 $\sqrt{1^2 + 7^2}$ · A1 $5\sqrt{2}$

Q1.4.06 **Gradient and y = mx + c · 4 marks**

$3x - 5y - 5 = 0$

M1 gradient $= \frac{3}{5}$ · M1 uses $y - y_1 = m(x - x_1)$ · A1 any correct form · A1 integer form

Q1.4.07 **Parallel and perpendicular lines · 3 marks**

$y = 5x - 23$

B1 gradient 5 · M1 uses the point · A1 $y = 5x - 23$

Q1.4.08 **Length and midpoint · 3 marks**

(a) $(6, 10)$

B1

(b) $4\sqrt{2}$

M1 $\sqrt{4^2 + 4^2}$ · A1 $4\sqrt{2}$

Q1.4.09 **Gradient and y = mx + c · 4 marks**

$3x + 2y + 20 = 0$

M1 gradient $= -\frac{3}{2}$ · M1 uses $y - y_1 = m(x - x_1)$ · A1 any correct form · A1 integer form

Q1.4.10 **Modelling with straight lines · 6 marks**

(a) $C = 24t + 58$

M1 gradient $= 24$ · M1 substitutes a point · A1 $C = 24t + 58$

(b) £24 per hour; fixed charge £58

B1 gradient · B1 intercept

(c) e.g. charges may be per part-hour / discounts for long jobs

B1 sensible limitation

Q1.4.11 **y - y1 = m(x - x1) and ax + by + c = 0 · 5 marks**

$9x + 7y - 8 = 0$

B1 midpoint $(-3, 5)$ · M1 gradient of PQ · M1 perpendicular gradient $-\frac{9}{7}$ · M1 line through midpoint · A1 integer form

Q1.4.12 **Modelling with straight lines · 6 marks**

(a) $C = 9t + 45$

M1 gradient $= 9$ · M1 substitutes a point · A1 $C = 9t + 45$

(b) £9 per hour; fixed charge £45

B1 gradient · B1 intercept

(c) e.g. charges may be per part-hour / discounts for long jobs

B1 sensible limitation

Q1.4.13 **y - y1 = m(x - x1) and ax + by + c = 0 · 5 marks**

$2x - 7y - 33 = 0$

B1 midpoint $(6, -3)$ · M1 gradient of PQ · M1 perpendicular gradient $\frac{2}{7}$ · M1 line through midpoint · A1 integer form

Q1.4.14 **Modelling with straight lines · 6 marks**

(a) $C = 9t + 32$

M1 gradient $= 9$ · M1 substitutes a point · A1 $C = 9t + 32$

(b) £9 per day; fixed charge £32

B1 gradient · B1 intercept

(c) e.g. charges may be per part-hour / discounts for long jobs

B1 sensible limitation

Q1.4.15 **y - y1 = m(x - x1) and ax + by + c = 0 · 5 marks**

$5x - 7y - 36 = 0$

B1 midpoint $(3, -3)$ · M1 gradient of PQ · M1 perpendicular gradient $\frac{5}{7}$ · M1 line through midpoint · A1 integer form

1.5 Circles

Q1.5.01 **Completing the square to find centre and radius · 4 marks**

centre $(-2, 4)$, radius $5\sqrt{2}$

M1 completes the square in x and y · A1 centre · M1 $r^2 = 4 + 16 - (-30)$ · A1 $r = 5\sqrt{2}$

Q1.5.02 **Equation of a circle · 4 marks**

$(x + 1)^2 + (y - 4)^2 = 17$

M1A1 centre $(-1, 4)$ · M1 $r^2 = 17$ · A1 equation

Q1.5.03 **Completing the square to find centre and radius · 4 marks**

centre $(-6, -7)$, radius $2\sqrt{2}$

M1 completes the square in x and y · A1 centre · M1 $r^2 = 36 + 49 - (77)$ · A1 $r = 2\sqrt{2}$

Q1.5.04 **Equation of a circle · 4 marks**

$(x - 4)^2 + (y - 1)^2 = 5$

M1A1 centre $(4, 1)$ · M1 $r^2 = 5$ · A1 equation

Q1.5.05 **Completing the square to find centre and radius · 4 marks**

centre $(1, 6)$, radius $3\sqrt{2}$

M1 completes the square in x and y · A1 centre · M1 $r^2 = 1 + 36 - (19)$ · A1 $r = 3\sqrt{2}$

Q1.5.06 **Equation of a circle · 4 marks**

$(x - 2)^2 + (y - 4)^2 = 32$

M1A1 centre $(2, 4)$ · M1 $r^2 = 32$ · A1 equation

Q1.5.07 **Tangents and normals to circles · 6 marks**

(a) $(x + 2)^2 + (y - 3)^2 = 5$

M1 radius^2 by Pythagoras · A1

(b) $2x + y - 4 = 0$

M1 gradient of radius $\frac{1}{2}$ · M1 tangent gradient -2 · M1 line through P · A1

Q1.5.08 **Intersections of lines and circles · 5 marks**

$k = -\sqrt{10}$ or $k = \sqrt{10}$

M1 substitutes the line into the circle · A1 correct 3TQ in x · M1 uses $b^2 - 4ac = 0$ · M1 solves for k · A1 both values

Q1.5.09 **Circle theorems in coordinate geometry · 6 marks**

(a) gradients $\frac{1}{2}$ and -2; product -1

M1 gradient of AC · M1 gradient of BC · A1 product $=-1$ so perpendicular

(b) angle in a semicircle is $90°$; $x^2+(y+1)^2=25$

B1 angle in semicircle · M1 centre = midpoint of AB · A1 equation

Q1.5.10 **Tangents and normals to circles · 6 marks**

(a) $(x+5)^2+(y+4)^2=10$

M1 radius² by Pythagoras · A1

(b) $3x+y+9=0$

M1 gradient of radius $\frac{1}{3}$ · M1 tangent gradient -3 · M1 line through P · A1

Q1.5.11 **Intersections of lines and circles · 5 marks**

$k=-\sqrt{10}$ or $k=\sqrt{10}$

M1 substitutes the line into the circle · A1 correct 3TQ in x · M1 uses $b^2-4ac=0$ · M1 solves for k · A1 both values

Q1.5.12 **Circle theorems in coordinate geometry · 6 marks**

(a) gradients $\frac{1}{2}$ and -2; product -1

M1 gradient of AC · M1 gradient of BC · A1 product $=-1$ so perpendicular

(b) angle in a semicircle is $90°$; $(x+3)^2+(y-2)^2=25$

B1 angle in semicircle · M1 centre = midpoint of AB · A1 equation

Q1.5.13 **Tangents and normals to circles · 6 marks**

(a) $(x+4)^2+(y-2)^2=25$

M1 radius² by Pythagoras · A1

(b) $4x-3y+47=0$

M1 gradient of radius $-\frac{3}{4}$ · M1 tangent gradient $\frac{4}{3}$ · M1 line through P · A1

Q1.5.14 **Intersections of lines and circles · 5 marks**

$k=6-2\sqrt{5}$ or $k=2\sqrt{5}+6$

M1 substitutes the line into the circle · A1 correct 3TQ in x · M1 uses $b^2-4ac=0$ · M1 solves for k · A1 both values

Q1.5.15 **Circle theorems in coordinate geometry · 6 marks**

(a) gradients $\frac{1}{2}$ and -2; product -1

M1 gradient of AC · M1 gradient of BC · A1 product $=-1$ so perpendicular

(b) angle in a semicircle is $90°$; $(x-3)^2+(y+3)^2=25$

B1 angle in semicircle · M1 centre = midpoint of AB · A1 equation

Q1.5.16 **Chords and perpendicular bisectors · 7 marks**

(a) two lines through the midpoints, perpendicular to PQ and QR

M1A1 first bisector · M1A1 second bisector

(b) centre $(-1, 1)$, radius 5

M1 solves simultaneously · A1 centre · B1 radius

Q1.5.17 **Chords and perpendicular bisectors · 7 marks**

(a) two lines through the midpoints, perpendicular to PQ and QR

M1A1 first bisector · M1A1 second bisector

(b) centre $(2,\ -2)$, radius 5

M1 solves simultaneously · A1 centre · B1 radius

Q1.5.18 **Chords and perpendicular bisectors · 7 marks**

(a) two lines through the midpoints, perpendicular to PQ and QR

M1A1 first bisector · M1A1 second bisector

(b) centre $(-2, 1)$, radius 5

M1 solves simultaneously · A1 centre · B1 radius

Pearson Edexcel A-Level Mathematics (9MA0)

Module 2

Functions, Expansions & Advanced Algebra

Paper 1 & 2 (Pure) · 90 practice questions · answers and mark schemes at the back

Subtopic	Questions	Done
2.1 Algebraic Methods	27	____ / 27
2.2 Functions and Graphs	24	____ / 24
2.3 Binomial Expansion	18	____ / 18
2.4 Exponentials and Logarithms	21	____ / 21

How to use this workbook

1. Read the key facts and watch one video for the subtopic. Tick the skills checklist as you go.
2. Attempt each question in the squared working space without notes. Mark it with the scheme at the back.
3. Tick **Todo** when you have attempted it and **Mastered** only when you can redo it cold a week later.
4. Copy any mistakes into the notes box at the end of the subtopic. Rebuild the workbook with a new seed for fresh questions.

Name _____ Started _____ Built 14 Oct 2026, seed 2026

2.1 Algebraic Methods

Proof (deduction, exhaustion, contradiction, counter-example), algebraic fractions, polynomial division, the factor theorem and partial fractions.

Key facts

- Even: $2n$; odd: $2n+1$; consecutive: $n, n+1$
- Factor theorem: $f(a)=0 \Leftrightarrow (x-a)$ is a factor; $f\left(\frac{b}{a}\right)=0 \Leftrightarrow (ax-b)$ is a factor
- $\frac{px+q}{(x-a)(x-b)} \equiv \frac{A}{x-a}+\frac{B}{x-b}$
- Repeated factor: $\frac{\ldots}{(x-a)^2(x-b)} \equiv \frac{A}{x-a}+\frac{B}{(x-a)^2}+\frac{C}{x-b}$
- Improper fraction (degree top $\geq$ degree bottom): divide first

Method

- Proof by contradiction: 'Assume the opposite...', reach an impossibility, then conclude.
- Partial fractions: multiply by the denominator, substitute the roots, compare coefficients for the rest.
- Proofs must end with a conclusion sentence that answers the question.

Exam traps

- Proof by example is not a proof.
- Not stating the assumption explicitly in a contradiction proof.
- Forgetting the $\frac{B}{(x-a)^2}$ term for a repeated factor.

In the exam: A proof question appears on nearly every Paper 1/2 (4-5 marks). Partial fractions feed into binomial expansion and integration.

Specification checklist

Skill	Learnt	Practised	Mastered
Proof by deduction	☐	☐	☐
Proof by exhaustion	☐	☐	☐
Proof by contradiction	☐	☐	☐
Disproof by counter-example	☐	☐	☐
Algebraic fractions	☐	☐	☐
Polynomial division	☐	☐	☐
Factor theorem	☐	☐	☐
Partial fractions	☐	☐	☐
Partial fractions with repeated factors	☐	☐	☐

Videos (click to open)

☐ **Recap** — Proof (Year 1) in less than 18 minutes • A-Level Maths, Pure Year 1, Chapter 7b (Bicen Maths)
☐ **Recap** — Proof by Contradiction in less than 18 minutes • A-Level Maths, Pure Year 2, Chapter 1a (Bicen Maths)
☐ **Recap** — Factor Theorem in less than 18 minutes • A-Level Maths, Pure Year 1, Chapter 7a (Bicen Maths)
☐ Proof by deduction — Introducing Proof by Deduction (TLMaths)
☐ Proof by exhaustion — Introducing Proof by Exhaustion (TLMaths)
☐ Proof by contradiction — Introducing Proof by Contradiction (TLMaths)
☐ Disproof by counter-example — Disproof by Counter Example Examples (TLMaths)
☐ Algebraic fractions — Rational Expressions: Examples of Simplifying Algebraic Fractions (TLMaths)
☐ Polynomial division — Polynomials: Polynomial Division using the Long Division Method (TLMaths)
☐ Factor theorem — Polynomials: Introducing the Factor Theorem (TLMaths)
☐ Partial fractions — Algebraic Fractions: Introducing Partial Fractions via Substitution (TLMaths)
☐ Partial fractions with repeated factors — Algebraic Fractions: How we deal with Repeated Factors in the Denominator (TLMaths)

Q2.1.01 Partial fractions ●○○ [3 marks] ☐ Todo ☐ Mastered

Express $\frac{35-2x}{x^2+x-20}$ in partial fractions.

(3)

WORKING

Q2.1.02 Algebraic fractions ●○○ [3 marks] ☐ Todo ☐ Mastered

Simplify fully $\frac{3x^2-6x-9}{x^2+6x+5}$.

(3)

WORKING

Q2.1.03 Proof by exhaustion ●○○ [2 marks] ☐ Todo ☐ Mastered

Prove that for all integers n with $1 \leq n \leq 6$, n^2+2 is not divisible by 4.

(2)

WORKING

Q2.1.04 Disproof by counter-example ●○○ [2 marks] ☐ Todo ☐ Mastered

Disprove: 'if $a > b$ then $a^2 > b^2$ for all real a, b'.

(2)

WORKING

Q2.1.05 Polynomial division ●○○ [4 marks] ☐ Todo ☐ Mastered

Divide $x^3 + 7x^2 + 4x - 12$ by $(x - 1)$ and hence factorise $x^3 + 7x^2 + 4x - 12$ completely.

(4)

WORKING

Q2.1.06 Proof by deduction ●○○ [3 marks] ☐ Todo ☐ Mastered

Prove that the sum of any three consecutive integers is divisible by 3.

(3)

WORKING

Q2.1.07 Partial fractions ●○○ [3 marks] ☐ Todo ☐ Mastered

Express $\frac{28}{x^2 - x - 12}$ in partial fractions.

(3)

WORKING

Q2.1.08 Algebraic fractions ●○○ [3 marks] ☐ Todo ☐ Mastered

Simplify fully $\frac{x^2 - 3x + 2}{x^2 - 5x + 6}$.

(3)

WORKING

Q2.1.09 **Factor theorem** ●○○ **[2 marks]** ☐ **Todo** ☐ **Mastered**

$f(x) = 3x^3 + x^2 + 2x + 4$.

Use the factor theorem to determine whether $(x - 1)$ is a factor of $f(x)$.

(2)

WORKING

Q2.1.10 **Proof by exhaustion** ●○○ **[4 marks]** ☐ **Todo** ☐ **Mastered**

Prove that the square of any integer is either a multiple of 3 or one more than a multiple of 3.

(4)

WORKING

Q2.1.11 **Polynomial division** ●○○ **[4 marks]** ☐ **Todo** ☐ **Mastered**

Divide $2x^3 - 16x^2 + 38x - 24$ by $(x - 4)$ and hence factorise $2x^3 - 16x^2 + 38x - 24$ completely.

(4)

WORKING

Q2.1.12 Proof by deduction ●○○ [2 marks] ☐ Todo ☐ Mastered

Prove that $(n+3)^2 - (n-3)^2$ is a multiple of 12 for all integers n.

(2)

WORKING

Q2.1.13 Partial fractions ●○○ [3 marks] ☐ Todo ☐ Mastered

Express $\frac{7x-11}{x^2-x-6}$ in partial fractions.

(3)

WORKING

Q2.1.14 Algebraic fractions ●○○ [3 marks] ☐ Todo ☐ Mastered

Simplify fully $\frac{x^2+5x+6}{x^2-3x-18}$.

(3)

WORKING

Q2.1.15 Factor theorem ●○○ [2 marks] ☐ Todo ☐ Mastered

$f(x) = 2x^3 - 4x^2 + 6x + 3$.

Use the factor theorem to determine whether $(x-2)$ is a factor of $f(x)$.

(2)

WORKING

Q2.1.16 Polynomial division ●○○ [4 marks] ☐ Todo ☐ Mastered

Divide $x^3 - 6x^2 - 7x + 60$ by $(x + 3)$ and hence factorise $x^3 - 6x^2 - 7x + 60$ completely.

(4)

WORKING

Q2.1.17 Proof by deduction ●○○ [3 marks] ☐ Todo ☐ Mastered

Prove that $x^2 - 6x + 11 > 0$ for all real values of x.

(3)

WORKING

Q2.1.18 Partial fractions ●○○ [3 marks] ☐ Todo ☐ Mastered

Express $\frac{17 - 4x}{x^2 - 9x + 20}$ in partial fractions.

(3)

WORKING

Q2.1.19 Algebraic fractions ●○○ [3 marks] ☐ Todo ☐ Mastered

Simplify fully $\frac{2x^2 - 14x + 24}{x^2 + x - 12}$.

(3)

WORKING

Q2.1.20 Factor theorem ●○○ [2 marks] ☐ Todo ☐ Mastered

$f(x) = x^3 - 4x^2 + 2x - 9$.

Use the factor theorem to determine whether $(x - 1)$ is a factor of $f(x)$.

(2)

WORKING

Q2.1.21 Proof by contradiction ●●○ [5 marks] ☐ Todo ☐ Mastered

Prove by contradiction that there are infinitely many prime numbers.

(5)

WORKING

Q2.1.22 Factor theorem ●●○ [6 marks] ☐ Todo ☐ Mastered

$f(x) = 2x^3 + 18x^2 + px + 48$, where p is a constant.

Given that $(x + 2)$ is a factor of $f(x)$,

(a) find the value of p, **(2)**

(b) hence factorise $f(x)$ completely. **(4)**

WORKING

Q2.1.23 Partial fractions with repeated factors ●●○ [4 marks] ☐ Todo ☐ Mastered

Given that $\frac{-x^2+7x+13}{(x-3)(x+2)^2} \equiv \frac{A}{x+2} + \frac{B}{(x+2)^2} + \frac{C}{x-3}$, find the values of A, B and C.

(4)

WORKING

Q2.1.24 Proof by contradiction ●●○ [5 marks] ☐ Todo ☐ Mastered

Prove by contradiction that $\sqrt{2}$ is irrational.

(5)

WORKING

Q2.1.25 Partial fractions with repeated factors ●●○ [4 marks] ☐ Todo ☐ Mastered

Given that $\frac{-x^2+7x-14}{(x-3)^2(x-1)} \equiv \frac{A}{x-3} + \frac{B}{(x-3)^2} + \frac{C}{x-1}$, find the values of A, B and C.

(4)

WORKING

Q2.1.26 **Partial fractions with repeated factors** ●●○ **[4 marks]** ☐ **Todo** ☐ **Mastered**

Given that $\frac{7x^2+23x-34}{(x-3)(x+4)^2} \equiv \frac{A}{x+4} + \frac{B}{(x+4)^2} + \frac{C}{x-3}$, find the values of A, B and C.

(4)

WORKING

Q2.1.27 **Proof by contradiction** ●●○ **[4 marks]** ☐ **Todo** ☐ **Mastered**

Prove by contradiction that if n^2 is even then n is even.

(4)

WORKING

MY NOTES — 2.1 ALGEBRAIC METHODS: MISTAKES TO REMEMBER

2.2 Functions and Graphs

Functions: domain/range, composite and inverse functions, the modulus function, and transformations of graphs.

Key facts

- $fg(x) = f(g(x))$: apply g first
- f^{-1} exists only if f is one-to-one; domain of f^{-1} = range of f; $y = f^{-1}(x)$ is the reflection of $y = f(x)$ in $y = x$
- $y = f(x) + a$: up a; $y = f(x + a)$: left a; $y = af(x)$: vertical stretch s.f. a; $y = f(ax)$: horizontal stretch s.f. $\frac{1}{a}$
- $y = |f(x)|$: reflect negative parts up; $y = f(|x|)$: reflect the right-hand half in the y-axis

Method

- Solve $|f(x)| = g(x)$ by sketching both graphs, then solve $f(x) = g(x)$ and $-f(x) = g(x)$ and check each answer.
- To find an inverse: write $y = f(x)$, rearrange for x, swap letters.
- Combined transformations: do changes inside the bracket (horizontal) and outside (vertical) separately.

Exam traps

- $f^{-1}(x) \neq \frac{1}{f(x)}$.
- $f(x + 2)$ moves the graph left, not right.
- Not rejecting modulus solutions that make the right-hand side negative.

In the exam: Functions questions are long (8-12 marks), mixing range, composite, inverse and a modulus sketch.

Specification checklist

Skill	Learnt	Practised	Mastered
Modulus function y = |x|	☐	☐	☐
Solving modulus equations and inequalities	☐	☐	☐
Mappings and functions, domain and range	☐	☐	☐
Composite functions	☐	☐	☐
Inverse functions	☐	☐	☐
Transformations y = af(x), f(x)+a, f(x+a), f(ax)	☐	☐	☐
Modulus graphs y = |f(x)| and y = f(|x|)	☐	☐	☐
Combining transformations	☐	☐	☐

Videos (click to open)

☐ **Recap** — Graphs & Transformations in less than 22 minutes • A-Level Maths, Pure Year 1, Chapter 4 (Bicen Maths)
☐ **Recap** — Functions & Graphs in less than 26 minutes • A-Level Maths, Pure Year 2, Chapter 2 (Bicen Maths)
☐ Modulus function y = |x| — Graphs: Introducing the Modulus Function (TLMaths)
☐ Solving modulus equations and inequalities — Graphs: Introducing Solving Modulus Equations (TLMaths)
☐ Mappings and functions, domain and range — Functions: Examples of Finding the Domain and Range (TLMaths)
☐ Composite functions — Functions: The Domain of a Composite Function (TLMaths)
☐ Inverse functions — Functions: Finding an Inverse Function (TLMaths)
☐ Transformations y = af(x), f(x)+a, f(x+a), f(ax) — Graph Transformations: Beginning an Investigation into Transformations (TLMaths)
☐ Modulus graphs y = |f(x)| and y = f(|x|) — Graphs: EDEXCEL Sketching y=f(|x|) (TLMaths)
☐ Combining transformations — Trigonometry | Combining Transformations of Trig Graphs y=f(ax+b) types (ExamSolutions)

Q2.2.01 Modulus function y = |x| ●○○ [4 marks] ☐ Todo ☐ Mastered

Sketch $y = |x + 4| - 1$, giving the coordinates of the vertex and of any points where the graph meets the axes.

(4)

WORKING

Q2.2.02 Solving modulus equations and inequalities ●○○ [3 marks] ☐ Todo ☐ Mastered

Solve $|x + 5| < 5$.

(3)

WORKING

Q2.2.03 Mappings and functions, domain and range ●○○ [3 marks] ☐ Todo ☐ Mastered

The function f is defined by $f(x) = x^2 + 8x + 25$, $x \in \mathbb{R}$. Find the range of f, and explain why f does not have an inverse.

(3)

WORKING

Q2.2.04 Composite functions ●○○ [6 marks] ☐ Todo ☐ Mastered

The functions f and g are defined by $f(x) = -2x - 2$, $x \in \mathbb{R}$ and $g(x) = 2x^2 + 5$, $x \in \mathbb{R}$.

(a) Find $fg(3)$. **(2)**

(b) Find $gf(x)$ in its simplest form. **(2)**

(c) Find an expression for $f^{-1}(x)$. **(2)**

WORKING

Q2.2.05 Transformations y = af(x), f(x)+a, f(x+a), f(ax) ●○○ [2 marks] ☐ Todo ☐ Mastered

The curve $y = f(x)$ has a maximum point at $(-1, 1)$. Write down the coordinates of the corresponding point on the curve with equation $y = f(-x)$, and describe the transformation.

(2)

WORKING

Q2.2.06 Modulus function y = |x| ●○○ [4 marks] ☐ Todo ☐ Mastered

Sketch $y = |x + 4| - 3$, giving the coordinates of the vertex and of any points where the graph meets the axes.

(4)

WORKING

Q2.2.07 Mappings and functions, domain and range ●○○ [3 marks] ☐ Todo ☐ Mastered

The function f is defined by $f(x) = x^2 + 2x + 9$, $x \in \mathbb{R}$. Find the range of f, and explain why f does not have an inverse.

(3)

WORKING

Q2.2.08 **Composite functions** ●○○ **[6 marks]** ☐ **Todo** ☐ **Mastered**

The functions f and g are defined by $f(x) = -x - 5$, $x \in \mathbb{R}$ and $g(x) = x^2 - 4$, $x \in \mathbb{R}$.

(a) Find $fg(2)$. **(2)**

(b) Find $gf(x)$ in its simplest form. **(2)**

(c) Find an expression for $f^{-1}(x)$. **(2)**

WORKING

Q2.2.09 **Transformations y = af(x), f(x)+a, f(x+a), f(ax)** ●○○ **[2 marks]** ☐ **Todo** ☐ **Mastered**

The curve $y = f(x)$ has a maximum point at $(0, 4)$. Write down the coordinates of the corresponding point on the curve with equation $y = f(x - 2)$, and describe the transformation.

(2)

WORKING

Q2.2.10 **Modulus function y = |x|** ●○○ **[4 marks]** ☐ **Todo** ☐ **Mastered**

Sketch $y = 2|x + 4| + 5$, giving the coordinates of the vertex and of any points where the graph meets the axes.

(4)

WORKING

Q2.2.11 Mappings and functions, domain and range ●○○ [3 marks] ☐ Todo ☐ Mastered

The function f is defined by $f(x) = x^2 - 10x + 29$, $x \in \mathbb{R}$. Find the range of f, and explain why f does not have an inverse.

(3)

WORKING

Q2.2.12 Composite functions ●○○ [6 marks] ☐ Todo ☐ Mastered

The functions f and g are defined by $f(x) = 1 - 5x$, $x \in \mathbb{R}$ and $g(x) = x^2 - 3$, $x \in \mathbb{R}$.

(a) Find $fg(-3)$. **(2)**

(b) Find $gf(x)$ in its simplest form. **(2)**

(c) Find an expression for $f^{-1}(x)$. **(2)**

WORKING

Q2.2.13 Transformations y = af(x), f(x)+a, f(x+a), f(ax) ●○○ [2 marks] ☐ Todo ☐ Mastered

The curve $y = f(x)$ has a maximum point at $(3, -2)$. Write down the coordinates of the corresponding point on the curve with equation $y = f(-x)$, and describe the transformation.

(2)

WORKING

Q2.2.14 Modulus graphs y = |f(x)| and y = f(|x|) ●●○ [7 marks] ☐ Todo ☐ Mastered

Given $f(x) = x^2 - 9x + 20$,

(a) sketch $y = |f(x)|$, stating the coordinates of any points where the graph meets the axes and the coordinates of its maximum turning point, **(4)**

(b) sketch $y = f(|x|)$, stating where it meets the axes. **(3)**

WORKING

Q2.2.15 Inverse functions ●●○ [4 marks] ☐ Todo ☐ Mastered

$f(x) = \frac{4x+3}{3x-1}$, $x \in \mathbb{R}$, $x \neq \frac{1}{3}$.

(a) Find $f^{-1}(x)$. **(3)**

(b) State the domain of f^{-1}. **(1)**

WORKING

Q2.2.16 Combining transformations ●●○ [3 marks] ☐ Todo ☐ Mastered

The curve $y = x^2$ is transformed onto $y = 3(x-4)^2 + 2$. Describe fully a sequence of transformations that maps the first curve onto the second.

(3)

WORKING

Q2.2.17 Modulus graphs y = |f(x)| and y = f(|x|) ●●○ [7 marks] ☐ Todo ☐ Mastered

Given $f(x) = x^2 - 2x - 3$,

(a) sketch $y = |f(x)|$, stating the coordinates of any points where the graph meets the axes and the coordinates of its maximum turning point, **(4)**

(b) sketch $y = f(|x|)$, stating where it meets the axes. **(3)**

WORKING

Q2.2.18 Solving modulus equations and inequalities ●●○ [4 marks] ☐ Todo ☐ Mastered

Solve $|2x - 6| = 2x + 1$.

(4)

WORKING

Q2.2.19 Inverse functions ●●○ [4 marks] ☐ Todo ☐ Mastered

$f(x) = \frac{4 - 3x}{x - 5}$, $x \in \mathbb{R}$, $x \neq 5$.

(a) Find $f^{-1}(x)$. **(3)**

(b) State the domain of f^{-1}. **(1)**

WORKING

Q2.2.20 Combining transformations ●●○ [3 marks] ☐ Todo ☐ Mastered

The curve $y = x^2$ is transformed onto $y = -1(x + 2)^2 + 1$. Describe fully a sequence of transformations that maps the first curve onto the second.

(3)

WORKING

Q2.2.21 Modulus graphs y = |f(x)| and y = f(|x|) ●●○ [7 marks] ☐ Todo ☐ Mastered

Given $f(x) = x^2 + 2x - 3$,

(a) sketch $y = |f(x)|$, stating the coordinates of any points where the graph meets the axes and the coordinates of its maximum turning point, **(4)**

(b) sketch $y = f(|x|)$, stating where it meets the axes. **(3)**

WORKING

Q2.2.22 Solving modulus equations and inequalities ●●○ [4 marks] ☐ Todo ☐ Mastered

Solve $|x + 1| = -2x - 3$.

(4)

WORKING

Q2.2.23 Inverse functions ●●○ [4 marks] ☐ Todo ☐ Mastered

$f(x) = \frac{3x+1}{3x-3}$, $x \in \mathbb{R}$, $x \neq 1$.

(a) Find $f^{-1}(x)$. **(3)**

(b) State the domain of f^{-1}. **(1)**

WORKING

Q2.2.24 Combining transformations ●●○ [3 marks] ☐ Todo ☐ Mastered

The curve $y = x^2$ is transformed onto $y = -1(x+4)^2 + 2$. Describe fully a sequence of transformations that maps the first curve onto the second.

(3)

WORKING

MY NOTES — 2.2 FUNCTIONS AND GRAPHS: MISTAKES TO REMEMBER

2.3 Binomial Expansion

Binomial expansion for positive integer powers (finite) and for any rational power (infinite, valid only for small x).

Key facts

- $(a+b)^n = \sum_{r=0}^{n} \binom{n}{r} a^{n-r} b^r$
- $(1+x)^n = 1 + nx + \frac{n(n-1)}{2!}x^2 + \frac{n(n-1)(n-2)}{3!}x^3 + \ldots$
- For rational n, $(1+bx)^n$ is valid for $|bx| < 1$, i.e. $|x| < \frac{1}{|b|}$
- $(a+bx)^n = a^n\left(1+\frac{b}{a}x\right)^n$, valid for $|x| < \frac{a}{|b|}$

Method

- Put brackets around the whole term, e.g. $(-3x)^2 = 9x^2$.
- For $(a+bx)^n$ take out a^n first, then expand.
- With partial fractions: expand each fraction separately; the overall validity is the narrower interval.

Exam traps

- Forgetting to raise a to the power n when factoring out.
- Not stating the range of validity.
- Sign errors with negative n: $(1-x)^{-1} = 1 + x + x^2 + \ldots$

In the exam: 4-6 marks; often followed by 'use your expansion to estimate...' with a substitution.

Specification checklist

Skill	Learnt	Practised	Mastered
(a + b)^n for positive integer n	☐	☐	☐
(1 + x)^n for rational n	☐	☐	☐
(a + bx)^n for rational n	☐	☐	☐
Range of validity	☐	☐	☐
Using partial fractions with expansions	☐	☐	☐
Approximations using expansions	☐	☐	☐

Videos (click to open)

☐ **Recap** — Binomial Expansion (Year 1) in less than 20 minutes • A-Level Maths, Pure Year 1, Chapter 8 (Bicen Maths)
☐ **Recap** — Binomial Expansion (Year 2) in less than 21 minutes • A-Level Maths, Pure Year 2, Chapter 4 (Bicen Maths)
☐ (a + b)^n for positive integer n — Binomial Expansion: Explaining where nCr comes from (TLMaths)
☐ (1 + x)^n for rational n — Binomial Expansion (partial fractions type) (ExamSolutions)
☐ (a + bx)^n for rational n — Binomial Expansion: Writing (a + bx)^n in the form p(1 + qx)^n (TLMaths)
☐ Range of validity — Binomial Expansion: Introducing the Range of Validity (TLMaths)
☐ Using partial fractions with expansions — Algebraic Fractions: Partial Fractions with Binomial Expansion (TLMaths)
☐ Approximations using expansions — Binomial Expansion: Approximating 1.03^8 WITHOUT a calculator (TLMaths)

Q2.3.01 Range of validity ●○○ [1 mark] ☐ Todo ☐ Mastered

State the set of values of x for which the binomial expansion of $(7+4x)^{-1}$ is valid.

(1)

WORKING

Q2.3.02 (a + b)^n for positive integer n ●○○ [4 marks] ☐ Todo ☐ Mastered

Find the first 4 terms, in ascending powers of x, of the binomial expansion of $(1+4x)^7$, giving each term in its simplest form.

(4)

WORKING

Q2.3.03 Range of validity ●○○ [1 mark] ☐ Todo ☐ Mastered

State the set of values of x for which the binomial expansion of $(9-3x)^{\frac{1}{2}}$ is valid.

(1)

WORKING

Q2.3.04 (a + b)^n for positive integer n ●○○ [4 marks] ☐ Todo ☐ Mastered

Find the first 4 terms, in ascending powers of x, of the binomial expansion of $(1-2x)^4$, giving each term in its simplest form.

(4)

WORKING

Q2.3.05 Range of validity ●○○ [1 mark] ☐ Todo ☐ Mastered

State the set of values of x for which the binomial expansion of $(5+6x)^{-\frac{1}{3}}$ is valid.

(1)

WORKING

Q2.3.06 (a + b)^n for positive integer n ●○○ [4 marks] ☐ Todo ☐ Mastered

Find the first 4 terms, in ascending powers of x, of the binomial expansion of $(2+4x)^5$, giving each term in its simplest form.

(4)

WORKING

Q2.3.07 Range of validity ●○○ [1 mark] ☐ Todo ☐ Mastered

State the set of values of x for which the binomial expansion of $(3-5x)^{\frac{1}{2}}$ is valid.

(1)

WORKING

Q2.3.08 Approximations using expansions ●●○ [5 marks] ☐ Todo ☐ Mastered

(a) Expand $\sqrt{1+4x}$ in ascending powers of x up to and including the term in x^2. **(3)**

(b) By substituting $x=\frac{1}{20}$, find an approximation for $\sqrt{\frac{6}{5}}$, giving your answer to 5 decimal places. **(2)**

WORKING

Q2.3.09 (1 + x)^n for rational n ●●○ [5 marks] ☐ Todo ☐ Mastered

(a) Find the binomial expansion of $(1+2x)^{-\frac{3}{2}}$, in ascending powers of x, up to and including the term in x^3. **(4)**

(b) State the range of values of x for which the expansion is valid. **(1)**

WORKING

Q2.3.10 Approximations using expansions ●●○ [5 marks] ☐ Todo ☐ Mastered

(a) Expand $\sqrt{1+2x}$ in ascending powers of x up to and including the term in x^2. **(3)**

(b) By substituting $x = \frac{1}{100}$, find an approximation for $\sqrt{\frac{51}{50}}$, giving your answer to 5 decimal places. **(2)**

WORKING

Q2.3.11 (1 + x)^n for rational n ●●○ [5 marks] ☐ Todo ☐ Mastered

(a) Find the binomial expansion of $(1+\frac{1}{2}x)^{-\frac{1}{3}}$, in ascending powers of x, up to and including the term in x^3. **(4)**

(b) State the range of values of x for which the expansion is valid. **(1)**

WORKING

Q2.3.12 (1 + x)^n for rational n ●●○ [5 marks] ☐ Todo ☐ Mastered

(a) Find the binomial expansion of $(1-2x)^{-\frac{1}{3}}$, in ascending powers of x, up to and including the term in x^3. **(4)**

(b) State the range of values of x for which the expansion is valid. **(1)**

WORKING

Q2.3.13 Using partial fractions with expansions ●●● [7 marks] ☐ Todo ☐ Mastered

$f(x) = \frac{-12x-2}{1-9x^2}$.

(a) Express $f(x)$ in partial fractions. **(3)**

(b) Hence find the series expansion of $f(x)$ in ascending powers of x up to and including the term in x^2, and state the range of validity. **(4)**

WORKING

Q2.3.14 (a + bx)^n for rational n ●●● [5 marks] ☐ Todo ☐ Mastered

Find the first three terms, in ascending powers of x, of the binomial expansion of $(4+x)^{-\frac{1}{2}}$, giving each coefficient as a simplified fraction, and state the range of validity.

(5)

WORKING

Q2.3.15 Using partial fractions with expansions ●●● [7 marks] ☐ Todo ☐ Mastered

$f(x) = \frac{-11x-2}{-6x^2+x+1}$.

(a) Express $f(x)$ in partial fractions. **(3)**

(b) Hence find the series expansion of $f(x)$ in ascending powers of x up to and including the term in x^2, and state the range of validity. **(4)**

WORKING

Q2.3.16 (a + bx)^n for rational n ●●● [5 marks] ☐ Todo ☐ Mastered

Find the first three terms, in ascending powers of x, of the binomial expansion of $(4-3x)^{-\frac{1}{2}}$, giving each coefficient as a simplified fraction, and state the range of validity.

(5)

WORKING

Q2.3.17 Using partial fractions with expansions ●●● [7 marks] ☐ Todo ☐ Mastered

$f(x) = \frac{-5x-7}{-3x^2+2x+1}$.

(a) Express $f(x)$ in partial fractions. **(3)**

(b) Hence find the series expansion of $f(x)$ in ascending powers of x up to and including the term in x^2, and state the range of validity. **(4)**

WORKING

Q2.3.18 (a + bx)^n for rational n ●●● [5 marks] ☐ Todo ☐ Mastered

Find the first three terms, in ascending powers of x, of the binomial expansion of $(9-3x)^{-\frac{1}{2}}$, giving each coefficient as a simplified fraction, and state the range of validity.

(5)

WORKING

MY NOTES — 2.3 BINOMIAL EXPANSION: MISTAKES TO REMEMBER

2.4 Exponentials and Logarithms

Exponentials and logarithms: laws of logs, solving equations, e^x and $\ln x$, growth/decay models and reducing curves to straight lines.

Key facts

- $\log_a x = n \Leftrightarrow a^n = x$
- $\log_a(xy) = \log_a x + \log_a y$, $\log_a \frac{x}{y} = \log_a x - \log_a y$, $\log_a x^k = k\log_a x$
- $\frac{d}{dx}e^{kx} = ke^{kx}$, $\ln(e^x) = x$, $e^{\ln x} = x$
- $y = ab^x \Rightarrow \log y = \log a + x\log b$ (plot $\log y$ against x)
- $y = ax^n \Rightarrow \log y = \log a + n\log x$ (plot $\log y$ against $\log x$)

Method

- Solve $a^x = b$ by taking logs: $x = \frac{\ln b}{\ln a}$.
- For $e^{2x} - 5e^x + 6 = 0$ use $u = e^x$.
- In models $N = Ae^{kt}$: A is the initial value; use one data point to find k.

Exam traps

- $\log(x + y) \neq \log x + \log y$.
- Forgetting to reject $\ln$ of a negative number.
- Leaving the answer in the wrong form (exact vs 3 s.f.).

In the exam: Modelling questions (8-10 marks) are common in Paper 2; log-linear graphs appear regularly.

Specification checklist

Skill	Learnt	Practised	Mastered
Exponential functions and graphs	☐	☐	☐
e^x and ln x	☐	☐	☐
Laws of logarithms	☐	☐	☐
Solving exponential equations	☐	☐	☐
Exponential growth and decay models	☐	☐	☐
Non-linear data: log y against x and log y against log x	☐	☐	☐
Gradient of e^kx	☐	☐	☐

Videos (click to open)

☐ **Recap** — Exponentials & Logarithms in less than 32 minutes • A-Level Maths, Pure Year 1, Chapter 14 (Bicen Maths)
☐ Exponential functions and graphs — Exponentials: Sketching y = e^x (TLMaths)
☐ e^x and ln x — Logarithms: Introducing The Natural Logarithm ln(x) (TLMaths)
☐ Laws of logarithms — Laws of Logarithms: Introducing the Laws of Logarithms (TLMaths)
☐ Solving exponential equations — Exponential Equations: Solve 2^(x+3) = 3^(3-x) (TLMaths)
☐ Exponential growth and decay models — Exponential Growth & Decay: Long-Term Behaviour (TLMaths)
☐ Non-linear data: log y against x and log y against log x — Reduction to Linear Form: Reducing y = ax^n to Linear Form (TLMaths)
☐ Gradient of e^kx — Exponential Model: The Gradient of e^(kx) (TLMaths)

Q2.4.01 Exponential functions and graphs ●○○ [3 marks] ☐ Todo ☐ Mastered

Sketch the curve $y = 5e^x - 4$, stating the coordinates of the y-intercept and the equation of the asymptote.

(3)

WORKING

Q2.4.02 Gradient of e^kx ●○○ [2 marks] ☐ Todo ☐ Mastered

The mass, m grams, of a substance is $m = 35e^{0.3t}$. Find the rate of change of m when $t = 8$, giving your answer to 3 significant figures.

(2)

WORKING

Q2.4.03 Laws of logarithms ●○○ [3 marks] ☐ Todo ☐ Mastered

Given that $p = \log_3 a$ and $q = \log_3 b$, express $\log_3 \left(\frac{a^3\sqrt{b}}{3b^1}\right)$ in terms of p and q.

(3)

WORKING

Q2.4.04 e^x and ln x ●○○ [2 marks] ☐ Todo ☐ Mastered

Solve $\ln(5x - 4) = 1$, giving your answer in exact form.

(2)

WORKING

Q2.4.05 Exponential functions and graphs ●○○ [3 marks] ☐ Todo ☐ Mastered

Sketch the curve $y = 4e^{-x} + 6$, stating the coordinates of the y-intercept and the equation of the asymptote.

(3)

WORKING

Q2.4.06 Gradient of e^kx ●○○ [2 marks] ☐ Todo ☐ Mastered

The mass, m grams, of a substance is $m = 32e^{0.3t}$. Find the rate of change of m when $t = 2$, giving your answer to 3 significant figures.

(2)

WORKING

Q2.4.07 e^x and ln x ●○○ [2 marks] ☐ Todo ☐ Mastered

Solve $\ln(2x - 4) = 1$, giving your answer in exact form.

(2)

WORKING

Q2.4.08 Exponential functions and graphs ●○○ [3 marks] ☐ Todo ☐ Mastered

Sketch the curve $y = 1e^{-2x} - 4$, stating the coordinates of the y-intercept and the equation of the asymptote.

(3)

WORKING

Q2.4.09 Gradient of e^kx ●○○ [2 marks] ☐ Todo ☐ Mastered

The mass, m grams, of a substance is $m = 11e^{0.05t}$. Find the rate of change of m when $t = 9$, giving your answer to 3 significant figures.

(2)

WORKING

Q2.4.10 Laws of logarithms ●○○ [3 marks] ☐ Todo ☐ Mastered

Given that $p = \log_3 a$ and $q = \log_3 b$, express $\log_3\left(\frac{a^4\sqrt{b}}{3b^2}\right)$ in terms of p and q.

(3)

WORKING

Q2.4.11 Solving exponential equations ●●○ [4 marks] ☐ Todo ☐ Mastered

Solve $2^{x+3} = 3^{x-3}$, giving your answer to 3 significant figures.

(4)

WORKING

Q2.4.12 e^x and ln x ●●○ [4 marks] ☐ Todo ☐ Mastered

Solve $e^{2x} - 9e^x + 18 = 0$, giving your answers as exact values.

(4)

WORKING

Q2.4.13 **Non-linear data: log y against x and log y against log x** ●●○ **[3 marks]** ☐ Todo ☐ Mastered

Two variables are believed to be related by $y = ax^n$. A graph of $\log_{10} y$ against $\log_{10} x$ is a straight line passing through $(0, 1.114)$ with gradient -1. Find a to 2 significant figures and state n.

(3)

WORKING

Q2.4.14 **Exponential growth and decay models** ●●○ **[6 marks]** ☐ Todo ☐ Mastered

The number of rabbits on an island, N, is modelled by $N = Ae^{kt}$, where t is the time in years. Initially $N = 200$, and when $t = 5$, $N = 257$.

(a) Find the value of k to 3 significant figures. **(3)**

(b) Find the time when N first exceeds 1000. **(2)**

(c) Give one reason why this model may not be realistic in the long term. **(1)**

WORKING

Q2.4.15 **Solving exponential equations** ●●○ **[4 marks]** ☐ Todo ☐ Mastered

Solve $3^{x+1} = 7^{x-3}$, giving your answer to 3 significant figures.

(4)

WORKING

Q2.4.16 **Non-linear data: log y against x and log y against log x** ●●○ **[4 marks]** ☐ Todo ☐ Mastered

Two variables are believed to be related by $y = ab^x$. A graph of $\log_{10} y$ against x is a straight line with gradient 0.079 and vertical intercept 1.176. Find the values of a and b to 2 significant figures.

(4)

WORKING

Q2.4.17 **Exponential growth and decay models** ●●○ **[6 marks]** ☐ Todo ☐ Mastered

The number of rabbits on an island, N, is modelled by $N = Ae^{kt}$, where t is the time in years. Initially $N = 800$, and when $t = 6$, $N = 1644$.

(a) Find the value of k to 3 significant figures. **(3)**

(b) Find the time when N first exceeds 1600. **(2)**

(c) Give one reason why this model may not be realistic in the long term. **(1)**

WORKING

Q2.4.18 **Solving exponential equations** ●●○ **[4 marks]** ☐ Todo ☐ Mastered

Solve $3^{x+1} = 5^{x-2}$, giving your answer to 3 significant figures.

(4)

WORKING

Q2.4.19 **Non-linear data: log y against x and log y against log x** ●●○ **[4 marks]** ☐ **Todo** ☐ **Mastered**

Two variables are believed to be related by $y = ab^x$. A graph of $\log_{10} y$ against x is a straight line with gradient 0.079 and vertical intercept 0.301. Find the values of a and b to 2 significant figures.

(4)

WORKING

Q2.4.20 **Exponential growth and decay models** ●●○ **[6 marks]** ☐ **Todo** ☐ **Mastered**

The number of bacteria in a culture, N, is modelled by $N = Ae^{kt}$, where t is the time in hours. Initially $N = 400$, and when $t = 9$, $N = 822$.

(a) Find the value of k to 3 significant figures. **(3)**

(b) Find the time when N first exceeds 1200. **(2)**

(c) Give one reason why this model may not be realistic in the long term. **(1)**

WORKING

Q2.4.21 Solving exponential equations ●●○ [4 marks] ☐ Todo ☐ Mastered

Solve $2^{x+3} = 5^{x-3}$, giving your answer to 3 significant figures.

(4)

WORKING

MY NOTES — 2.4 EXPONENTIALS AND LOGARITHMS: MISTAKES TO REMEMBER

Answers and mark schemes

M = method mark, A = accuracy mark (depends on the preceding M), B = independent mark, ft = follow-through, * = given answer. Numerical answers are to 3 s.f. unless stated.

2.1 Algebraic Methods

Q2.1.01 **Partial fractions · 3 marks**

$\frac{-5}{x+5}+\frac{3}{x-4}$

M1 correct form and factorised denominator · M1 substitutes or compares coefficients · A1 both constants

Q2.1.02 **Algebraic fractions · 3 marks**

$\frac{3x-9}{x+5}$

M1 factorises numerator · M1 factorises denominator · A1 cancelled form

Q2.1.03 **Proof by exhaustion · 2 marks**

Check $n = 1..6$: $3, 6, 11, 18, 27, 38$; none divisible by 4.

B1 sets up correctly (algebraic form / cases / assumption) · M1 valid algebra or reasoning · A1 conclusion stated in words

Q2.1.04 **Disproof by counter-example · 2 marks**

$a = 1$, $b = -2$: $a > b$ but $1 < 4$.

B1 sets up correctly (algebraic form / cases / assumption) · M1 valid algebra or reasoning · A1 conclusion stated in words

Q2.1.05 **Polynomial division · 4 marks**

quotient $x^2 + 8x + 12$; $(x-1)(x+2)(x+6)$

M1 long division to obtain $xx^2 + \ldots$ · A1 quotient · M1 factorises quadratic · A1 full factorisation

Q2.1.06 **Proof by deduction · 3 marks**

Let the integers be n, $n+1$, $n+2$. Sum $= 3n + 3 = 3(n+1)$, a multiple of 3.

B1 sets up correctly (algebraic form / cases / assumption) · M1 valid algebra or reasoning · A1 conclusion stated in words

Q2.1.07 **Partial fractions · 3 marks**

$\frac{4}{x-4}-\frac{4}{x+3}$

M1 correct form and factorised denominator · M1 substitutes or compares coefficients · A1 both constants

Q2.1.08 **Algebraic fractions · 3 marks**

$\frac{x-1}{x-3}$

M1 factorises numerator · M1 factorises denominator · A1 cancelled form

Q2.1.09 **Factor theorem · 2 marks**

$f(1) = 10$, so it is not a factor

M1 evaluates $f(1)$ · A1 value and conclusion

Q2.1.10 **Proof by exhaustion · 4 marks**

Cases $n = 3k$: $9k^2$; $n = 3k \pm 1$: $9k^2 \pm 6k + 1 = 3(3k^2 \pm 2k) + 1$.

B1 sets up correctly (algebraic form / cases / assumption) · M1 valid algebra or reasoning · A1 conclusion stated in words

Q2.1.11 **Polynomial division · 4 marks**

quotient $2x^2 - 8x + 6$; $2(x-4)(x-3)(x-1)$

M1 long division to obtain $2xx^2 + \ldots$ · A1 quotient · M1 factorises quadratic · A1 full factorisation

Q2.1.12 **Proof by deduction · 2 marks**

Expands to $12n$, so it is a multiple of 12.

B1 sets up correctly (algebraic form / cases / assumption) · M1 valid algebra or reasoning · A1 conclusion stated in words

Q2.1.13 **Partial fractions · 3 marks**

$\frac{2}{x-3}+\frac{5}{x+2}$

M1 correct form and factorised denominator · M1 substitutes or compares coefficients · A1 both constants

Q2.1.14 **Algebraic fractions · 3 marks**

$\frac{x+2}{x-6}$

M1 factorises numerator · M1 factorises denominator · A1 cancelled form

Q2.1.15 **Factor theorem · 2 marks**

$f(2) = 15$, so it is not a factor

M1 evaluates $f(2)$ · A1 value and conclusion

Q2.1.16 **Polynomial division · 4 marks**

quotient $x^2 - 9x + 20$; $(x-5)(x-4)(x+3)$

M1 long division to obtain $xx^2 + \ldots$ · A1 quotient · M1 factorises quadratic · A1 full factorisation

Q2.1.17 **Proof by deduction · 3 marks**

$x^2 - 6x + 11 = (x-3)^2 + 2$; $(x-3)^2 \geq 0$ so the expression $\geq 2 > 0$.

B1 sets up correctly (algebraic form / cases / assumption) · M1 valid algebra or reasoning · A1 conclusion stated in words

Q2.1.18 **Partial fractions · 3 marks**

$\frac{-3}{x-5} - \frac{1}{x-4}$

M1 correct form and factorised denominator · M1 substitutes or compares coefficients · A1 both constants

Q2.1.19 **Algebraic fractions · 3 marks**

$\frac{2x-8}{x+4}$

M1 factorises numerator · M1 factorises denominator · A1 cancelled form

Q2.1.20 **Factor theorem · 2 marks**

$f(1) = -10$, so it is not a factor

M1 evaluates $f(1)$ · A1 value and conclusion

Q2.1.21 **Proof by contradiction · 5 marks**

Assume finitely many $p_1, \ldots, p_n$; $N = p_1 p_2 \cdots p_n + 1$ leaves remainder 1 on division by each; so N is prime or has a prime factor not in the list; contradiction.

B1 sets up correctly (algebraic form / cases / assumption) · M1 valid algebra or reasoning · A1 conclusion stated in words

Q2.1.22 **Factor theorem · 6 marks**

(a) $p = 52$

M1 sets $f(-2) = 0$ · A1 $p = 52$

(b) $2(x+2)(x+3)(x+4)$

M1 divides by the factor · A1 quadratic factor · M1 factorises · A1

Q2.1.23 **Partial fractions with repeated factors · 4 marks**

$A = -2$, $B = 1$, $C = 1$

M1 multiplies through by the denominator · B1 $B = 1$ (substitute $x = -2$) · B1 $C = 1$ (substitute $x = 3$) · A1 A by comparing x^2 coefficients

Q2.1.24 **Proof by contradiction · 5 marks**

Assume $\sqrt{2} = \frac{a}{b}$ in lowest terms; $a^2 = 2b^2$ so a even, $a = 2c$; then $b^2 = 2c^2$ so b even; contradiction with lowest terms.

B1 sets up correctly (algebraic form / cases / assumption) · M1 valid algebra or reasoning · A1 conclusion stated in words

Q2.1.25 **Partial fractions with repeated factors · 4 marks**

$A = 1$, $B = -1$, $C = -2$

M1 multiplies through by the denominator · B1 $B = -1$ (substitute $x = 3$) · B1 $C = -2$ (substitute $x = 1$) · A1 A by comparing x^2 coefficients

Q2.1.26 **Partial fractions with repeated factors · 4 marks**

$A = 5$, $B = 2$, $C = 2$

M1 multiplies through by the denominator · B1 $B = 2$ (substitute $x = -4$) · B1 $C = 2$ (substitute $x = 3$) · A1 A by comparing x^2 coefficients

Q2.1.27 **Proof by contradiction · 4 marks**

Assume n odd, $n = 2k + 1$; $n^2 = 4k^2 + 4k + 1 = 2(2k^2 + 2k) + 1$ is odd; contradiction.

B1 sets up correctly (algebraic form / cases / assumption) · M1 valid algebra or reasoning · A1 conclusion stated in words

2.2 Functions and Graphs

Q2.2.01 **Modulus function y = |x| · 4 marks**

V-shape with vertex $(-4, -1)$; y-intercept $(0, 3)$; x-intercepts $(-5, 0)$, $(-3, 0)$

B1 V shape, correct orientation · B1 vertex · B1 y-intercept · B1 x-intercepts (or none)

Q2.2.02 **Solving modulus equations and inequalities · 3 marks**

$-10 < x < 0$

M1 $-5 < x + 5 < 5$ or equivalent critical values · A1 critical values · A1 correct region

Q2.2.03 **Mappings and functions, domain and range · 3 marks**

$f(x) \geq 9$; f is many-to-one

M1 completes the square · A1 $f(x) \geq 9$ · B1 many-to-one

Q2.2.04 **Composite functions · 6 marks**

(a) -48

M1 $g(3) = 23$ · A1

(b) $8x^2 + 16x + 13$

M1 substitutes f into g · A1

(c) $f^{-1}(x) = \frac{-x-2}{2}$

M1 rearranges $y = -2x - 2$ · A1

Q2.2.05 **Transformations y = af(x), f(x)+a, f(x+a), f(ax) · 2 marks**

$(1, 1)$

B1 coordinates · B1 transformation described fully

Q2.2.06 **Modulus function y = |x| · 4 marks**

V-shape with vertex $(-4, -3)$; y-intercept $(0, 1)$; x-intercepts $(-7, 0)$, $(-1, 0)$

B1 V shape, correct orientation · B1 vertex · B1 y-intercept · B1 x-intercepts (or none)

Q2.2.07 **Mappings and functions, domain and range · 3 marks**

$f(x) \geq 8$; f is many-to-one

M1 completes the square · A1 $f(x) \geq 8$ · B1 many-to-one

Q2.2.08 **Composite functions · 6 marks**

(a) -5

M1 $g(2) = 0$ · A1

(b) $x^2 + 10x + 21$

M1 substitutes f into g · A1

(c) $f^{-1}(x) = -x - 5$

M1 rearranges $y = -x - 5$ · A1

Q2.2.09 **Transformations y = af(x), f(x)+a, f(x+a), f(ax) · 2 marks**

$(2, 4)$

B1 coordinates · B1 transformation described fully

Q2.2.10 **Modulus function y = |x| · 4 marks**

V-shape with vertex $(-4, 5)$; y-intercept $(0, 13)$; no x-intercepts

B1 V shape, correct orientation · B1 vertex · B1 y-intercept · B1 x-intercepts (or none)

Q2.2.11 **Mappings and functions, domain and range · 3 marks**

$f(x) \geq 4$; f is many-to-one

M1 completes the square · A1 $f(x) \geq 4$ · B1 many-to-one

Q2.2.12 **Composite functions · 6 marks**

(a) -29

M1 $g(-3) = 6$ · A1

(b) $25x^2 - 10x - 2$

M1 substitutes f into g · A1

(c) $f^{-1}(x) = \frac{1-x}{5}$

M1 rearranges $y = 1 - 5x$ · A1

Q2.2.13 **Transformations y = af(x), f(x)+a, f(x+a), f(ax) · 2 marks**

$(-3, -2)$

B1 coordinates · B1 transformation described fully

Q2.2.14 **Modulus graphs y = |f(x)| and y = f(|x|) · 7 marks**

(a) touches x-axis at 4 and 5; y-intercept 20; maximum at $(\frac{9}{2}, \frac{1}{4})$

B1 shape (reflected part above axis) · B1 x-intercepts · B1 y-intercept · B1 turning point

(b) symmetric in the y-axis: reflect the part for $x \geq 0$; x-intercepts -5, -4, 4, 5; y-intercept 20

B1 symmetry in y-axis · B1 intercepts · B1 shape

Q2.2.15 **Inverse functions · 4 marks**

(a) $f^{-1}(x) = \frac{x+3}{3x-4}$

M1 swaps x and y or rearranges · M1 collects y terms · A1

(b) $x \in \mathbb{R}, x \neq \frac{4}{3}$

B1 range of f is all reals except $\frac{4}{3}$

Q2.2.16 **Combining transformations · 3 marks**

translation by $(4, 0)$; vertical stretch scale factor 3; translation by $(0, 2)$

B1 horizontal translation · B1 stretch (and reflection) · B1 vertical translation in a valid order

Q2.2.17 **Modulus graphs y = |f(x)| and y = f(|x|) · 7 marks**

(a) touches x-axis at -1 and 3; y-intercept 3; maximum at $(1, 4)$

B1 shape (reflected part above axis) · B1 x-intercepts · B1 y-intercept · B1 turning point

(b) symmetric in the y-axis: reflect the part for $x \geq 0$; x-intercepts -3, 3; y-intercept -3

B1 symmetry in y-axis · B1 intercepts · B1 shape

Q2.2.18 **Solving modulus equations and inequalities · 4 marks**

$x = \frac{5}{4}$

M1 solves $2x - 6 = 2x + 1$ · M1 solves $-(2x - 6) = 2x + 1$ · M1 checks the RHS is non-negative / rejects invalid roots · A1 $x = \frac{5}{4}$

Q2.2.19 **Inverse functions · 4 marks**
(a) $f^{-1}(x) = \frac{5x+4}{x+3}$
M1 swaps x and y or rearranges · M1 collects y terms · A1
(b) $x \in \mathbb{R}, x \neq -3$
B1 range of f is all reals except -3

Q2.2.20 **Combining transformations · 3 marks**
translation by $(-2, 0)$; vertical stretch scale factor -1 (including reflection in the x-axis); translation by $(0, 1)$
B1 horizontal translation · B1 stretch (and reflection) · B1 vertical translation in a valid order

Q2.2.21 **Modulus graphs y = |f(x)| and y = f(|x|) · 7 marks**
(a) touches x-axis at -3 and 1; y-intercept 3; maximum at $(-1, 4)$
B1 shape (reflected part above axis) · B1 x-intercepts · B1 y-intercept · B1 turning point
(b) symmetric in the y-axis: reflect the part for $x \geq 0$; x-intercepts -1, 1; y-intercept -3
B1 symmetry in y-axis · B1 intercepts · B1 shape

Q2.2.22 **Solving modulus equations and inequalities · 4 marks**
$x = -2$
M1 solves $x + 1 = -2x - 3$ · M1 solves $-(x + 1) = -2x - 3$ · M1 checks the RHS is non-negative / rejects invalid roots · A1 $x = -2$

Q2.2.23 **Inverse functions · 4 marks**
(a) $f^{-1}(x) = \frac{3x+1}{3x-3}$
M1 swaps x and y or rearranges · M1 collects y terms · A1
(b) $x \in \mathbb{R}, x \neq 1$
B1 range of f is all reals except 1

Q2.2.24 **Combining transformations · 3 marks**
translation by $(-4, 0)$; vertical stretch scale factor -1 (including reflection in the x-axis); translation by $(0, 2)$
B1 horizontal translation · B1 stretch (and reflection) · B1 vertical translation in a valid order

2.3 Binomial Expansion

Q2.3.01 **Range of validity · 1 marks**
$|x| < \frac{7}{4}$
B1

Q2.3.02 **(a + b)^n for positive integer n · 4 marks**
$1 + 28x + 336x^2 + 2240x^3$
B1 first term · M1 correct binomial coefficients with powers of $4x$ · A1 two terms correct · A1 all correct

Q2.3.03 **Range of validity · 1 marks**
$|x| < 3$
B1

Q2.3.04 **(a + b)^n for positive integer n · 4 marks**
$1 - 8x + 24x^2 - 32x^3$
B1 first term · M1 correct binomial coefficients with powers of $-2x$ · A1 two terms correct · A1 all correct

Q2.3.05 **Range of validity · 1 marks**
$|x| < \frac{5}{6}$
B1

Q2.3.06 **(a + b)^n for positive integer n · 4 marks**
$32 + 320x + 1280x^2 + 2560x^3$
B1 first term · M1 correct binomial coefficients with powers of $4x$ · A1 two terms correct · A1 all correct

Q2.3.07 **Range of validity · 1 marks**
$|x| < \frac{3}{5}$
B1

Q2.3.08 **Approximations using expansions · 5 marks**
(a) $1 + 2x - 2x^2$
M1 binomial with $n = \frac{1}{2}$ · A1 · A1
(b) 1.09500
M1 substitutes · A1

Q2.3.09 **(1 + x)^n for rational n · 5 marks**
(a) $1 - 3x + \frac{15x^2}{2} - \frac{35x^3}{2}$
M1 correct structure $1 + nx + \frac{n(n-1)}{2!}x^2 + \ldots$ · B1 $1 + \ldots x$ · A1 x^2 term · A1 x^3 term
(b) $|x| < \frac{1}{2}$
B1

Q2.3.10 Approximations using expansions · 5 marks
(a) $1 + x - \frac{x^2}{2}$
M1 binomial with $n = \frac{1}{2}$ · A1 · A1
(b) 1.00995
M1 substitutes · A1

Q2.3.11 (1 + x)^n for rational n · 5 marks
(a) $1 - \frac{x}{6} + \frac{x^2}{18} - \frac{7x^3}{324}$
M1 correct structure $1 + nx + \frac{n(n-1)}{2!}x^2 + \ldots$ · B1 $1 + \ldots x$ · A1 x^2 term · A1 x^3 term
(b) $|x| < 2$
B1

Q2.3.12 (1 + x)^n for rational n · 5 marks
(a) $1 + \frac{2x}{3} + \frac{8x^2}{9} + \frac{112x^3}{81}$
M1 correct structure $1 + nx + \frac{n(n-1)}{2!}x^2 + \ldots$ · B1 $1 + \ldots x$ · A1 x^2 term · A1 x^3 term
(b) $|x| < \frac{1}{2}$
B1

Q2.3.13 Using partial fractions with expansions · 7 marks
(a) $\frac{1}{1+3x} - \frac{3}{1-3x}$
M1 form · A1 one constant · A1 both
(b) $-2 - 12x - 18x^2$, $|x| < \frac{1}{3}$
M1 expands $(1+ax)^{-1}$ · M1 expands $(1-bx)^{-1}$ · A1 combined · B1 validity

Q2.3.14 (a + bx)^n for rational n · 5 marks
$\frac{1}{2} - \frac{x}{16} + \frac{3x^2}{256}$, valid for $|x| < 4$
M1 takes out $4^{-\frac{1}{2}}$ · M1 expands $(1+kx)^n$ · A1 two terms · A1 three terms · B1 validity

Q2.3.15 Using partial fractions with expansions · 7 marks
(a) $\frac{1}{1+3x} - \frac{3}{1-2x}$
M1 form · A1 one constant · A1 both
(b) $-2 - 9x - 3x^2$, $|x| < \frac{1}{3}$
M1 expands $(1+ax)^{-1}$ · M1 expands $(1-bx)^{-1}$ · A1 combined · B1 validity

Q2.3.16 (a + bx)^n for rational n · 5 marks
$\frac{1}{2} + \frac{3x}{16} + \frac{27x^2}{256}$, valid for $|x| < \frac{4}{3}$
M1 takes out $4^{-\frac{1}{2}}$ · M1 expands $(1+kx)^n$ · A1 two terms · A1 three terms · B1 validity

Q2.3.17 Using partial fractions with expansions · 7 marks
(a) $\frac{-4}{1+3x} - \frac{3}{1-x}$
M1 form · A1 one constant · A1 both
(b) $-7 + 9x - 39x^2$, $|x| < \frac{1}{3}$
M1 expands $(1+ax)^{-1}$ · M1 expands $(1-bx)^{-1}$ · A1 combined · B1 validity

Q2.3.18 (a + bx)^n for rational n · 5 marks
$\frac{1}{3} + \frac{x}{18} + \frac{x^2}{72}$, valid for $|x| < 3$
M1 takes out $9^{-\frac{1}{2}}$ · M1 expands $(1+kx)^n$ · A1 two terms · A1 three terms · B1 validity

2.4 Exponentials and Logarithms

Q2.4.01 Exponential functions and graphs · 3 marks
y-intercept $(0, 1)$; asymptote $y = -4$
B1 shape (increasing) · B1 intercept · B1 asymptote

Q2.4.02 Gradient of e^kx · 2 marks
116 g per unit time
M1 $\frac{dm}{dt} = 10.5e^{0.3t}$ · A1

Q2.4.03 Laws of logarithms · 3 marks
$3p - \frac{1}{2}q - 1$
M1 uses addition/subtraction law · M1 uses power law · A1

Q2.4.04 e^x and ln x · 2 marks
$x = \frac{e^1 + 4}{5}$
M1 $5x - 4 = e^1$ · A1

Q2.4.05 Exponential functions and graphs · 3 marks
y-intercept $(0, 10)$; asymptote $y = 6$
B1 shape (decreasing) · B1 intercept · B1 asymptote

Q2.4.06 **Gradient of e^kx · 2 marks**

17.5 g per unit time

M1 $\frac{dm}{dt} = 9.60e^{0.3t}$ · A1

Q2.4.07 **e^x and ln x · 2 marks**

$x = \frac{e^1 + 4}{2}$

M1 $2x - 4 = e^1$ · A1

Q2.4.08 **Exponential functions and graphs · 3 marks**

y-intercept $(0,\ -3)$; asymptote $y = -4$

B1 shape (decreasing) · B1 intercept · B1 asymptote

Q2.4.09 **Gradient of e^kx · 2 marks**

0.863 g per unit time

M1 $\frac{dm}{dt} = 0.550e^{0.05t}$ · A1

Q2.4.10 **Laws of logarithms · 3 marks**

$4p - \frac{3}{2}q - 1$

M1 uses addition/subtraction law · M1 uses power law · A1

Q2.4.11 **Solving exponential equations · 4 marks**

$x = 13.3$

M1 takes logs · M1 expands and collects x terms · M1 makes x the subject · A1

Q2.4.12 **e^x and ln x · 4 marks**

$x = \ln 6,\ x = \ln 3$

M1 lets $u = e^x$ · M1 $(u-6)(u-3) = 0$ · M1 takes ln · A1 both exact

Q2.4.13 **Non-linear data: log y against x and log y against log x · 3 marks**

$a = 13,\ n = -1$

M1 $\log y = \log a + n\log x$ · A1 n · A1 a

Q2.4.14 **Exponential growth and decay models · 6 marks**

(a) $k = 0.0502$

B1 $A = 200$ · M1 $257 = 200e^{5k}$ · A1

(b) $t = 32.1$ years

M1 sets up and takes logs · A1

(c) unlimited growth; limited food/space/resources

B1

Q2.4.15 **Solving exponential equations · 4 marks**

$x = 8.19$

M1 takes logs · M1 expands and collects x terms · M1 makes x the subject · A1

Q2.4.16 **Non-linear data: log y against x and log y against log x · 4 marks**

$a = 15,\ b = 1.2$

M1 $\log y = \log a + x\log b$ · M1 $\log a = 1.176$ · A1 a · A1 b

Q2.4.17 **Exponential growth and decay models · 6 marks**

(a) $k = 0.120$

B1 $A = 800$ · M1 $1644 = 800e^{6k}$ · A1

(b) $t = 5.77$ years

M1 sets up and takes logs · A1

(c) unlimited growth; limited food/space/resources

B1

Q2.4.18 **Solving exponential equations · 4 marks**

$x = 8.45$

M1 takes logs · M1 expands and collects x terms · M1 makes x the subject · A1

Q2.4.19 **Non-linear data: log y against x and log y against log x · 4 marks**

$a = 2.0,\ b = 1.2$

M1 $\log y = \log a + x\log b$ · M1 $\log a = 0.301$ · A1 a · A1 b

Q2.4.20 **Exponential growth and decay models · 6 marks**

(a) $k = 0.0800$

B1 $A = 400$ · M1 $822 = 400e^{9k}$ · A1

(b) $t = 13.7$ hours

M1 sets up and takes logs · A1

(c) unlimited growth; limited food/space/resources

B1

Q2.4.21 **Solving exponential equations · 4 marks**

$x = 7.54$

M1 takes logs · M1 expands and collects x terms · M1 makes x the subject · A1

Pearson Edexcel A-Level Mathematics (9MA0)

Module 3

Sequences & Trigonometry

Paper 1 & 2 (Pure) · 78 practice questions · answers and mark schemes at the back

Subtopic	Questions	Done
3.1 Sequences and Series	24	____ / 24
3.2 Radians & Trigonometry	21	____ / 21
3.3 Advanced Trigonometry & Modelling	18	____ / 18
3.4 Parametric Equations	15	____ / 15

How to use this workbook

1. Read the key facts and watch one video for the subtopic. Tick the skills checklist as you go.
2. Attempt each question in the squared working space without notes. Mark it with the scheme at the back.
3. Tick **Todo** when you have attempted it and **Mastered** only when you can redo it cold a week later.
4. Copy any mistakes into the notes box at the end of the subtopic. Rebuild the workbook with a new seed for fresh questions.

Name _____ Started _____ Built 14 Oct 2026, seed 2026

3.1 Sequences and Series

Arithmetic and geometric sequences and series, sum to infinity, sigma notation, recurrence relations and modelling with series.

Key facts

- AP: $u_n = a + (n-1)d$, $S_n = \frac{n}{2}(2a + (n-1)d) = \frac{n}{2}(a + l)$
- GP: $u_n = ar^{n-1}$, $S_n = \frac{a(1-r^n)}{1-r}$, $S_\infty = \frac{a}{1-r}$ for $|r| < 1$
- $\sum_{r=1}^{n} 1 = n$; a sum of a linear expression is an arithmetic series
- Recurrence $u_{n+1} = f(u_n)$: increasing if $u_{n+1} > u_n$ for all n; periodic if terms repeat

Method

- For 'find the first n such that...', form an inequality and use logs (GP) or a quadratic (AP).
- Count terms carefully: $\sum_{r=5}^{20}$ has 16 terms.
- In money problems, check whether the first year is $n = 1$ or $n = 0$.

Exam traps

- Using S_∞ when $|r| \geq 1$.
- Rounding n down when the question needs the first value exceeding a target.

In the exam: Paper 1 or 2, 6-10 marks; modelling contexts (salaries, savings, running distance) are popular.

Specification checklist

Skill	Learnt	Practised	Mastered
Arithmetic sequences	☐	☐	☐
Arithmetic series	☐	☐	☐
Geometric sequences	☐	☐	☐
Geometric series	☐	☐	☐
Sum to infinity	☐	☐	☐
Sigma notation	☐	☐	☐
Recurrence relations, increasing / decreasing / periodic	☐	☐	☐
Modelling with series	☐	☐	☐

Videos (click to open)

☐ **Recap** — Sequences & Series in 23 minutes • A-Level Maths, Pure Year 2, Chapter 3 (Bicen Maths)
☐ Arithmetic sequences — Arithmetic Sequences: Introducing Arithmetic Sequences (TLMaths)
☐ Arithmetic series — Arithmetic Sequences: Introducing Arithmetic Series (TLMaths)
☐ Geometric sequences — Geometric Sequences: Introducing Geometric Sequences (TLMaths)
☐ Geometric series — Geometric Sequences: Introducing Geometric Series (TLMaths)
☐ Sum to infinity — Geometric Sequences: Introducing Summing to Infinity (TLMaths)
☐ Sigma notation — Sigma Notation: Introducing Sigma Notation (TLMaths)
☐ Recurrence relations, increasing / decreasing / periodic — Sequences: Inductive Definitions and Recurrence Relations (TLMaths)
☐ Modelling with series — Modelling with Sequences: The Gardener Problem (TLMaths)

Q3.1.01 Geometric series ●○○ [2 marks] ☐ Todo ☐ Mastered

A geometric series has first term 29 and common ratio 1.2. Find the sum of the first 14 terms, giving your answer to 3 significant figures.

(2)

WORKING

Q3.1.02 Arithmetic sequences ●○○ [5 marks] ☐ Todo ☐ Mastered

The 8th term of an arithmetic sequence is 53 and the 21st term is 157.

(a) Find the first term and the common difference. **(3)**

(b) Find the sum of the first 30 terms. **(2)**

WORKING

Q3.1.03 Geometric sequences ●○○ [4 marks] ☐ Todo ☐ Mastered

The second term of a geometric sequence is 3 and the 5th term is $\frac{81}{64}$. Given that the common ratio is positive, find the common ratio and the first term.

(4)

WORKING

Q3.1.04 Geometric series ●○○ [2 marks] ☐ Todo ☐ Mastered

A geometric series has first term 7 and common ratio 0.8. Find the sum of the first 13 terms, giving your answer to 3 significant figures.

(2)

WORKING

Q3.1.05 Arithmetic sequences ●○○ [5 marks] ☐ Todo ☐ Mastered

The 5th term of an arithmetic sequence is 34 and the 13th term is 106.

(a) Find the first term and the common difference. **(3)**

(b) Find the sum of the first 47 terms. **(2)**

WORKING

Q3.1.06 Sigma notation ●○○ [3 marks] ☐ Todo ☐ Mastered

Find $\sum_{r=2}^{36}(6r+9)$.

(3)

WORKING

Q3.1.07 Geometric sequences ●○○ [4 marks] ☐ Todo ☐ Mastered

The second term of a geometric sequence is $\frac{87}{4}$ and the 5th term is $\frac{2349}{256}$. Given that the common ratio is positive, find the common ratio and the first term.

(4)

WORKING

Q3.1.08 **Recurrence relations, increasing / decreasing / periodic** ●○○ **[3 marks]** ☐ Todo ☐ Mastered

A sequence is defined by $u_1 = 2$, $u_{n+1} = -3u_n - 1$. Find u_2, u_3 and u_4, and state whether the sequence is increasing, decreasing or neither.

(3)

WORKING

Q3.1.09 **Geometric series** ●○○ **[2 marks]** ☐ Todo ☐ Mastered

A geometric series has first term 29 and common ratio 1.2. Find the sum of the first 8 terms, giving your answer to 3 significant figures.

(2)

WORKING

Q3.1.10 **Arithmetic sequences** ●○○ **[5 marks]** ☐ Todo ☐ Mastered

The 8th term of an arithmetic sequence is 57 and the 25th term is 176.

(a) Find the first term and the common difference. **(3)**

(b) Find the sum of the first 42 terms. **(2)**

WORKING

Q3.1.11 Geometric sequences ●○○ [4 marks] ☐ Todo ☐ Mastered

The second term of a geometric sequence is 78 and the 4th term is 702. Given that the common ratio is positive, find the common ratio and the first term.

(4)

WORKING

Q3.1.12 Modelling with series ●●○ [5 marks] ☐ Todo ☐ Mastered

Sam's salary in year 1 is £20,000. It increases by 4% each year.

(a) Find the first year in which Sam's salary exceeds £60,000. **(3)**

(b) Find the total Sam earns over the first 10 years, to the nearest £. **(2)**

WORKING

Q3.1.13 Recurrence relations, increasing / decreasing / periodic ●●○ [4 marks] ☐ Todo ☐ Mastered

A sequence is defined by $u_1 = 5$, $u_{n+1} = \frac{2}{u_n}$.

(a) Show that the sequence is periodic and state its order. **(2)**

(b) Find $\sum_{r=1}^{25} u_r$. **(2)**

WORKING

Q3.1.14 Sigma notation ●●○ [3 marks] ☐ Todo ☐ Mastered

Find the exact value of $\sum_{k=1}^{8} 8(3)^k$.

(3)

WORKING

Q3.1.15 Arithmetic series ●●○ [4 marks] ☐ Todo ☐ Mastered

An arithmetic series has first term 12 and common difference 7. The sum of the first n terms is 4335.
Show that $7n^2 + 17n - 8670 = 0$ and hence find n.

(4)

WORKING

Q3.1.16 Sum to infinity ●●○ [6 marks] ☐ Todo ☐ Mastered

A geometric series has second term 9 and sum to infinity 36.

(a) Show that $36r^2 - 36r + 9 = 0$. **(3)**

(b) Hence find the possible values of the common ratio, r, explaining why any value must be rejected. **(3)**

WORKING

Q3.1.17 **Modelling with series** ●●○ **[5 marks]** ☐ Todo ☐ Mastered

Sam's salary in year 1 is £26,000. It increases by 3% each year.

(a) Find the first year in which Sam's salary exceeds £40,000. **(3)**

(b) Find the total Sam earns over the first 10 years, to the nearest £. **(2)**

WORKING

Q3.1.18 **Recurrence relations, increasing / decreasing / periodic** ●●○ **[4 marks]** ☐ Todo ☐ Mastered

A sequence is defined by $u_1 = 3$, $u_{n+1} = \frac{2}{u_n}$.

(a) Show that the sequence is periodic and state its order. **(2)**

(b) Find $\sum_{r=1}^{41} u_r$. **(2)**

WORKING

Q3.1.19 Arithmetic series ●●○ [4 marks] ☐ Todo ☐ Mastered

An arithmetic series has first term 6 and common difference 4. The sum of the first n terms is 1680.
Show that $4n^2 + 8n - 3360 = 0$ and hence find n.

(4)

WORKING

Q3.1.20 Sum to infinity ●●○ [6 marks] ☐ Todo ☐ Mastered

A geometric series has second term -6 and sum to infinity 8.

(a) Show that $8r^2 - 8r - 6 = 0$. **(3)**

(b) Hence find the possible values of the common ratio, r, explaining why any value must be rejected. **(3)**

WORKING

Q3.1.21 Modelling with series ●●○ [5 marks] ☐ Todo ☐ Mastered

Sam's salary in year 1 is £26,000. It increases by 4% each year.

(a) Find the first year in which Sam's salary exceeds £45,000. **(3)**

(b) Find the total Sam earns over the first 10 years, to the nearest £. **(2)**

WORKING

Q3.1.22 Sigma notation ●●○ [3 marks] ☐ Todo ☐ Mastered

Find the exact value of $\sum_{k=1}^{11} 4\left(\frac{1}{2}\right)^k$.

(3)

WORKING

Q3.1.23 Arithmetic series ●●○ [4 marks] ☐ Todo ☐ Mastered

An arithmetic series has first term 10 and common difference 3. The sum of the first n terms is 413.
Show that $3n^2 + 17n - 826 = 0$ and hence find n.

(4)

WORKING

Q3.1.24 Sum to infinity ●●○ [6 marks] ☐ Todo ☐ Mastered

A geometric series has second term 4 and sum to infinity 18.

(a) Show that $18r^2 - 18r + 4 = 0$. **(3)**

(b) Hence find the possible values of the common ratio, r, explaining why any value must be rejected. **(3)**

WORKING

MY NOTES — 3.1 SEQUENCES AND SERIES: MISTAKES TO REMEMBER

3.2 Radians & Trigonometry

Radians, arcs and sectors, small angle approximations, sec/cosec/cot, Pythagorean identities and inverse trig functions.

Key facts

- π rad $= 180°$; arc $s = r\theta$, sector $A = \frac{1}{2}r^2\theta$, segment $= \frac{1}{2}r^2(\theta - \sin\theta)$
- Small θ (radians): $\sin\theta \approx \theta$, $\tan\theta \approx \theta$, $\cos\theta \approx 1 - \frac{\theta^2}{2}$
- $\sec x = \frac{1}{\cos x}$, $\operatorname{cosec} x = \frac{1}{\sin x}$, $\cot x = \frac{1}{\tan x}$
- $1 + \tan^2 x = \sec^2 x$, $1 + \cot^2 x = \operatorname{cosec}^2 x$
- arcsin: $[-1, 1] \to [-\frac{\pi}{2}, \frac{\pi}{2}]$; arccos: $[-1, 1] \to [0, \pi]$; arctan: $\mathbb{R} \to (-\frac{\pi}{2}, \frac{\pi}{2})$

Method

- Set calculators to radians whenever π or arc/sector formulas appear.
- Equations with $\sec^2$ and $\tan$: replace $\sec^2 x$ by $1 + \tan^2 x$ to get a quadratic in $\tan x$.
- Small angle: substitute, simplify, ignore powers of θ above the one asked for.

Exam traps

- Using degrees in $r\theta$.
- Missing solutions from the second quadrant (CAST).
- Small angle approximations only work in radians.

In the exam: Sector/segment questions are 5-7 marks; trig equations with identities appear on every paper.

Specification checklist

Skill	Learnt	Practised	Mastered
Radian measure	☐	☐	☐
Arc length	☐	☐	☐
Sector and segment areas	☐	☐	☐
Small angle approximations	☐	☐	☐
sec, cosec and cot	☐	☐	☐
Identities 1 + tan^2 = sec^2 and 1 + cot^2 = cosec^2	☐	☐	☐
Inverse trig functions arcsin, arccos, arctan	☐	☐	☐

Videos (click to open)

☐ **Recap** — Radians in less than 18 minutes • A-Level Maths, Pure Year 2, Chapter 5 (Bicen Maths)
☐ **Recap** — Trigonometry in 30 minutes • A-Level Maths, Pure Year 2, Chapters 6 & 7 (Bicen Maths)
☐ Radian measure — Trigonometry: Introducing Radians (TLMaths)
☐ Arc length — Trigonometry: Using the Formula for Arc Length in Radians (TLMaths)
☐ Sector and segment areas — Trigonometry: Using the Formula for Area of a Sector in Radians (TLMaths)
☐ Small angle approximations — Small-Angle Approximation: Geometrical Derivation (TLMaths)
☐ sec, cosec and cot — Trigonometry: Introducing cosec(x), sec(x) & cot(x) (TLMaths)
☐ Identities 1 + tan^2 = sec^2 and 1 + cot^2 = cosec^2 — Trigonometric Identities: Proving 1 + cot^2 θ = cosec^2 θ (TLMaths)
☐ Inverse trig functions arcsin, arccos, arctan — Trigonometry: Introducing arcsin(x) (TLMaths)

Q3.2.01 Radian measure ●○○ [2 marks] ☐ Todo ☐ Mastered

(a) Convert 150° to radians, giving your answer as an exact multiple of π. **(1)**

(b) Convert $\frac{5}{4}\pi$ radians to degrees. **(1)**

WORKING

Q3.2.02 sec, cosec and cot ●○○ [3 marks] ☐ Todo ☐ Mastered

Solve, for $0 \leq x < 360^\circ$, $\operatorname{cosec} x = -1.5$, giving your answers to 1 decimal place.

(3)

WORKING

Q3.2.03 Arc length ●○○ [4 marks] ☐ Todo ☐ Mastered

A sector OAB of a circle has radius 15 cm and angle 0.55 radians at the centre O.

(a) Find the length of the arc AB. **(1)**

(b) Find the perimeter of the sector. **(1)**

(c) Find the area of the sector. **(2)**

WORKING

Q3.2.04 Inverse trig functions arcsin, arccos, arctan ●○○ [3 marks] ☐ Todo ☐ Mastered

Find the exact value of $\arctan(-1)$ in radians, and state the domain and range of $y = \arctan x$.

(3)

WORKING

Q3.2.05 Radian measure ●○○ [2 marks] ☐ Todo ☐ Mastered

(a) Convert 315° to radians, giving your answer as an exact multiple of π. **(1)**

(b) Convert $\frac{3}{2}\pi$ radians to degrees. **(1)**

WORKING

Q3.2.06 sec, cosec and cot ●○○ [3 marks] ☐ Todo ☐ Mastered

Solve, for $0 \leq x < 360^\circ$, $\operatorname{cosec} x = 4$, giving your answers to 1 decimal place.

(3)

WORKING

Q3.2.07 Arc length ●○○ [4 marks] ☐ Todo ☐ Mastered

A sector OAB of a circle has radius 5 cm and angle 1.81 radians at the centre O.

(a) Find the length of the arc AB. **(1)**

(b) Find the perimeter of the sector. **(1)**

(c) Find the area of the sector. **(2)**

WORKING

Q3.2.08 Inverse trig functions arcsin, arccos, arctan ●○○ [3 marks] ☐ Todo ☐ Mastered

Find the exact value of $\arccos\left(\frac{\sqrt{3}}{2}\right)$ in radians, and state the domain and range of $y = \arccos x$.

(3)

WORKING

Q3.2.09 Radian measure ●○○ [2 marks] ☐ Todo ☐ Mastered

(a) Convert 270° to radians, giving your answer as an exact multiple of π. **(1)**

(b) Convert $\frac{2}{3}\pi$ radians to degrees. **(1)**

WORKING

Q3.2.10 sec, cosec and cot ●○○ [3 marks] ☐ Todo ☐ Mastered

Solve, for $0 \leq x < 360^\circ$, $\operatorname{cosec} x = 2$, giving your answers to 1 decimal place.

(3)

WORKING

Q3.2.11 Arc length ●○○ [4 marks] ☐ Todo ☐ Mastered

A sector OAB of a circle has radius 14 cm and angle 0.92 radians at the centre O.

(a) Find the length of the arc AB. **(1)**

(b) Find the perimeter of the sector. **(1)**

(c) Find the area of the sector. **(2)**

WORKING

Q3.2.12 Inverse trig functions arcsin, arccos, arctan ●○○ [3 marks] ☐ Todo ☐ Mastered

Find the exact value of $\arctan\left(\sqrt{3}\right)$ in radians, and state the domain and range of $y = \arctan x$.

(3)

WORKING

Q3.2.13 Identities 1 + tan^2 = sec^2 and 1 + cot^2 = cosec^2 ●●○ [5 marks] ☐ Todo ☐ Mastered

Solve, for $0 \leq x < 360^\circ$, $\sec^2 x - 2\tan x - 4 = 0$.

(5)

WORKING

Q3.2.14 Sector and segment areas ●●○ [4 marks] ☐ Todo ☐ Mastered

The chord AB subtends an angle of 2.28 radians at the centre O of a circle of radius 4 cm. Find the area of the minor segment cut off by AB, giving your answer to 3 significant figures.

(4)

WORKING

Q3.2.15 **Small angle approximations** ●●○ **[3 marks]** ☐ Todo ☐ Mastered

Given that θ is small, use the small angle approximations to find an approximation for $\frac{1-\cos 4\theta}{\theta \sin \theta}$ as a constant.

(3)

WORKING

Q3.2.16 **Identities 1 + tan^2 = sec^2 and 1 + cot^2 = cosec^2** ●●○ **[5 marks]** ☐ Todo ☐ Mastered

Solve, for $0 \leq x < 360^\circ$, $\sec^2 x - \frac{3}{2}\tan x - \frac{1}{2} = 0$.

(5)

WORKING

Q3.2.17 **Sector and segment areas** ●●○ **[4 marks]** ☐ Todo ☐ Mastered

The chord AB subtends an angle of 1.02 radians at the centre O of a circle of radius 5 cm. Find the area of the minor segment cut off by AB, giving your answer to 3 significant figures.

(4)

WORKING

Q3.2.18 Small angle approximations ●●○ [3 marks] ☐ Todo ☐ Mastered

Given that θ is small, use the small angle approximations to find an approximation for $\cos 5\theta + \theta \sin 2\theta$ in the form $p + q\theta^2$.

(3)

WORKING

Q3.2.19 Identities 1 + tan^2 = sec^2 and 1 + cot^2 = cosec^2 ●●○ [5 marks] ☐ Todo ☐ Mastered

Solve, for $0 \leq x < 360^\circ$, $\sec^2 x - 2 = 0$.

(5)

WORKING

Q3.2.20 Sector and segment areas ●●○ [4 marks] ☐ Todo ☐ Mastered

The chord AB subtends an angle of 1.19 radians at the centre O of a circle of radius 10 cm. Find the area of the minor segment cut off by AB, giving your answer to 3 significant figures.

(4)

WORKING

Q3.2.21 **Small angle approximations** ●●○ **[3 marks]** ☐ **Todo** ☐ **Mastered**

Given that θ is small, use the small angle approximations to find an approximation for $\cos 4\theta + \theta \sin 4\theta$ in the form $p + q\theta^2$.

(3)

WORKING

MY NOTES — 3.2 RADIANS & TRIGONOMETRY: MISTAKES TO REMEMBER

3.3 Advanced Trigonometry & Modelling

Addition and double angle formulae, the harmonic form $R\cos(x \pm \alpha)$, solving trig equations and proving identities, with modelling applications.

Key facts

- $\sin(A \pm B) = \sin A\cos B \pm \cos A\sin B$
- $\cos(A \pm B) = \cos A\cos B \mp \sin A\sin B$, $\tan(A \pm B) = \frac{\tan A \pm \tan B}{1 \mp \tan A\tan B}$
- $\sin 2A = 2\sin A\cos A$, $\cos 2A = \cos^2 A - \sin^2 A = 2\cos^2 A - 1 = 1 - 2\sin^2 A$
- $a\cos x + b\sin x = R\cos(x - \alpha)$ with $R = \sqrt{a^2 + b^2}$, $\tan\alpha = \frac{b}{a}$
- Maximum of $R\cos(x - \alpha)$ is R when $x = \alpha$; minimum $-R$

Method

- Solving $\sin(2x + 30^\circ) = k$: transform the interval first ($30 \le 2x + 30 < 750$), find all angles, then undo.
- Proofs: start from the more complicated side; convert everything to $\sin$ and $\cos$ if stuck.
- Harmonic form: expand $R\cos(x - \alpha)$, compare coefficients, divide to get $\tan\alpha$.

Exam traps

- Dividing by $\sin x$ or $\cos x$ and losing solutions (factorise instead).
- Giving α in degrees when the question works in radians (or vice versa).

In the exam: The R-form question is a Paper 2 regular (8-10 marks) often with a modelling part (tides, Ferris wheels, temperature).

Specification checklist

Skill	Learnt	Practised	Mastered
Addition formulae	☐	☐	☐
Double-angle formulae	☐	☐	☐
Solving trig equations	☐	☐	☐
Harmonic form R cos(x ± a), R sin(x ± a)	☐	☐	☐
Proving trig identities	☐	☐	☐
Modelling with trig functions	☐	☐	☐

Videos (click to open)

☐ **Recap** — Trigonometric Identities & Equations in less than 21 mins • A-Level Maths, Pure Year 1, Chapter 10 (Bicen Maths)
☐ **Recap** — Trigonometry in 30 minutes • A-Level Maths, Pure Year 2, Chapters 6 & 7 (Bicen Maths)
☐ Addition formulae — Compound Angles: Proving the Compound Angle Formulae (TLMaths)
☐ Double-angle formulae — Compound Angles: Introducing the Double Angle Formulae (TLMaths)
☐ Solving trig equations — Trig Equations: Solving Basic Trigonometric Equations in degrees (TLMaths)
☐ Harmonic form R cos(x ± a), R sin(x ± a) — Equivalent Forms: Writing acosθ + bsinθ in the form rcos(θ±α) or rsin(θ±α) (TLMaths)
☐ Proving trig identities — Proving Trigonometric Identities: The Methods (TLMaths)
☐ Modelling with trig functions — A level Edexcel June 2022 Paper 2 Q9 - Trigonometric functions - Modelling (ExamSolutions)

Q3.3.01 Addition formulae ●○○ [3 marks] ☐ Todo ☐ Mastered

Using an appropriate addition formula, find the exact value of $\cos 15^\circ$.

(3)

WORKING

Q3.3.02 Solving trig equations ●○○ [4 marks] ☐ Todo ☐ Mastered

Solve, for $0 \leq x < 180^\circ$, $\sin(2x) = 0.5$, giving your answers to 1 decimal place.

(4)

WORKING

Q3.3.03 Addition formulae ●○○ [3 marks] ☐ Todo ☐ Mastered

Using an appropriate addition formula, find the exact value of $\cos 105^\circ$.

(3)

WORKING

Q3.3.04 Proving trig identities ●●○ [4 marks] ☐ Todo ☐ Mastered

Prove that $\frac{\sin\theta}{1+\cos\theta} + \frac{1+\cos\theta}{\sin\theta} \equiv 2\operatorname{cosec}\theta$.

(4)

WORKING

Q3.3.05 Modelling with trig functions ●●○ [5 marks] ☐ Todo ☐ Mastered

The depth of water, D metres, in a harbour t hours after midnight is modelled by $D = 11 + 2.4\sin(30t)°$, $0 \leq t < 24$.

(a) State the maximum depth and find the first time it occurs. **(3)**

(b) A boat needs at least 11 m of water. For how long, in each 12-hour period, can it enter the harbour? **(2)**

WORKING

Q3.3.06 Double-angle formulae ●●○ [4 marks] ☐ Todo ☐ Mastered

Given that $\sin A = \frac{5}{13}$ and that A is acute, find the exact values of

(a) $\sin 2A$, (b) $\cos 2A$, (c) $\tan 2A$.

(a) **(2)**

(b) **(1)**

(c) **(1)**

WORKING

Q3.3.07 Solving trig equations ●●○ [6 marks] ☐ Todo ☐ Mastered

Solve, for $0 \leq x < 360°$, $3\sin^2 x - 2\cos x - 2 = 0$, giving your answers to 1 decimal place.

(6)

WORKING

Q3.3.08 Proving trig identities ●●○ [4 marks] ☐ Todo ☐ Mastered

Prove that $\cot\theta + \tan\theta \equiv 2\,\text{cosec}\,2\theta$.

(4)

WORKING

Q3.3.09 Modelling with trig functions ●●○ [5 marks] ☐ Todo ☐ Mastered

The depth of water, D metres, in a harbour t hours after midnight is modelled by $D = 8 + 4.1\sin(30t + 45)^\circ$, $0 \leq t < 24$.

(a) State the maximum depth and find the first time it occurs. **(3)**

(b) A boat needs at least 8 m of water. For how long, in each 12-hour period, can it enter the harbour? **(2)**

WORKING

Q3.3.10 Double-angle formulae ●●○ [4 marks] ☐ Todo ☐ Mastered

Given that $\sin A = \frac{7}{25}$ and that A is acute, find the exact values of
(a) $\sin 2A$, (b) $\cos 2A$, (c) $\tan 2A$.

(a) **(2)**

(b) **(1)**

(c) **(1)**

WORKING

Q3.3.11 Proving trig identities ●●○ [4 marks] ☐ Todo ☐ Mastered

Prove that $\frac{\cos 2\theta}{\cos\theta + \sin\theta} \equiv \cos\theta - \sin\theta$.

(4)

WORKING

Q3.3.12 Modelling with trig functions ●●○ [5 marks] ☐ Todo ☐ Mastered

The depth of water, D metres, in a harbour t hours after midnight is modelled by $D = 9 + 4.4\sin(30t)^\circ$, $0 \leq t < 24$.

(a) State the maximum depth and find the first time it occurs. **(3)**

(b) A boat needs at least 9 m of water. For how long, in each 12-hour period, can it enter the harbour? **(2)**

WORKING

Q3.3.13 Double-angle formulae ●●○ [4 marks] ☐ Todo ☐ Mastered

Given that $\sin A = \frac{20}{29}$ and that A is acute, find the exact values of
(a) $\sin 2A$, (b) $\cos 2A$, (c) $\tan 2A$.

(a) **(2)**

(b) **(1)**

(c) **(1)**

WORKING

Q3.3.14 Proving trig identities ●●○ [4 marks] ☐ Todo ☐ Mastered

Prove that $\sin 3\theta \equiv 3\sin\theta - 4\sin^3\theta$.

(4)

WORKING

Q3.3.15 Modelling with trig functions ●●○ [5 marks] ☐ Todo ☐ Mastered

The depth of water, D metres, in a harbour t hours after midnight is modelled by $D = 8 + 3.1\sin(30t + 30)^\circ$, $0 \leq t < 24$.

(a) State the maximum depth and find the first time it occurs. **(3)**

(b) A boat needs at least 8 m of water. For how long, in each 12-hour period, can it enter the harbour? **(2)**

WORKING

Q3.3.16 Harmonic form R cos(x ± a), R sin(x ± a) ●●● [8 marks] ☐ Todo ☐ Mastered

(a) Express $12\cos x - 5\sin x$ in the form $R\cos(x+\alpha)$, where $R > 0$ and $0 < \alpha < 90^\circ$. Give α to 2 decimal places. **(3)**

(b) Hence solve $12\cos x - 5\sin x = 6.4$ for $0 \leq x < 360^\circ$, giving answers to 1 decimal place. **(4)**

(c) State the maximum value of $12\cos x - 5\sin x$. **(1)**

WORKING

Q3.3.17 Harmonic form R cos(x ± a), R sin(x ± a) ●●● [8 marks] ☐ Todo ☐ Mastered

(a) Express $3\cos x + 4\sin x$ in the form $R\cos(x-\alpha)$, where $R > 0$ and $0 < \alpha < 90^\circ$. Give α to 2 decimal places. **(3)**

(b) Hence solve $3\cos x + 4\sin x = 2.0$ for $0 \leq x < 360^\circ$, giving answers to 1 decimal place. **(4)**

(c) State the maximum value of $3\cos x + 4\sin x$. **(1)**

WORKING

Q3.3.18 Harmonic form R cos(x ± a), R sin(x ± a) ●●● [8 marks] ☐ Todo ☐ Mastered

(a) Express $2\sin x - 3\cos x$ in the form $R\sin(x - \alpha)$, where $R > 0$ and $0 < \alpha < 90^\circ$. Give α to 2 decimal places. **(3)**

(b) Hence solve $2\sin x - 3\cos x = 2.7$ for $0 \leq x < 360^\circ$, giving answers to 1 decimal place. **(4)**

(c) State the maximum value of $2\sin x - 3\cos x$. **(1)**

WORKING

MY NOTES — 3.3 ADVANCED TRIGONOMETRY & MODELLING: MISTAKES TO REMEMBER

3.4 Parametric Equations

Parametric equations: converting to Cartesian form, sketching, finding intersections, and modelling motion with a parameter.

Key facts

- Eliminate t: make t the subject of one equation and substitute
- Trig parameters: use $\sin^2 t + \cos^2 t = 1$, $\sec^2 t = 1 + \tan^2 t$, $\cos 2t = 1 - 2\sin^2 t$
- $x = a\cos t + p$, $y = a\sin t + q$ is the circle $(x-p)^2 + (y-q)^2 = a^2$
- The domain of the Cartesian equation comes from the range of $x(t)$

Method

- For intersections with a line, substitute $x(t)$ and $y(t)$ into the line and solve for t.
- For axis crossings, set $x(t) = 0$ or $y(t) = 0$.
- In modelling, interpret t (often time) and check the realistic range of t.

Exam traps

- Forgetting to state the domain/range restrictions.
- Only giving t values when coordinates are asked for.

In the exam: Paper 2, 6-10 marks; often joined with parametric differentiation or integration.

Specification checklist

Skill	Learnt	Practised	Mastered
Parametric to Cartesian	☐	☐	☐
Trig parametric conversions	☐	☐	☐
Sketching parametric curves	☐	☐	☐
Intersections with parametric curves	☐	☐	☐
Modelling with parametric equations	☐	☐	☐

Videos (click to open)

☐ **Recap** — Parametric Equations in less than 21 minutes • A-Level Maths, Pure Year 2, Chapter 8 (Bicen Maths)
☐ Parametric to Cartesian — Parametric Equations: Parametric to Cartesian (TLMaths)
☐ Sketching parametric curves — Parametric Equations: Introducing Parametric Equations (TLMaths)
☐ Intersections with parametric curves — Finding points of Intersection between Parametric and Cartesian Equations (ExamSolutions)
☐ Modelling with parametric equations — Parametric Equation Modelling: A Projectiles Problem (TLMaths)

Q3.4.01 Parametric to Cartesian ●○○ [2 marks] ☐ Todo ☐ Mastered

A curve has parametric equations $x = 3t + 1$, $y = t^2 + 3$. Find a Cartesian equation of the curve in the form $y = f(x)$.

(2)

WORKING

Q3.4.02 Sketching parametric curves ●○○ [3 marks] ☐ Todo ☐ Mastered

A curve has parametric equations $x = 3t + 4$, $y = t^2 - 25$. Find the coordinates of the points where the curve meets the x-axis, and the point where $t = 0$.

(3)

WORKING

Q3.4.03 Parametric to Cartesian ●○○ [2 marks] ☐ Todo ☐ Mastered

A curve has parametric equations $x = t - 2$, $y = t^2 - 3$. Find a Cartesian equation of the curve in the form $y = f(x)$.

(2)

WORKING

Q3.4.04 Sketching parametric curves ●○○ [3 marks] ☐ Todo ☐ Mastered

A curve has parametric equations $x = 3 - 3t$, $y = t^2 - 25$. Find the coordinates of the points where the curve meets the x-axis, and the point where $t = 0$.

(3)

WORKING

Q3.4.05 Parametric to Cartesian ●○○ [2 marks] ☐ Todo ☐ Mastered

A curve has parametric equations $x = 4 - t$, $y = t^2 - 1$. Find a Cartesian equation of the curve in the form $y = f(x)$.

(2)

WORKING

Q3.4.06 Sketching parametric curves ●○○ [3 marks] ☐ Todo ☐ Mastered

A curve has parametric equations $x = 2t + 1$, $y = t^2 - 4$. Find the coordinates of the points where the curve meets the x-axis, and the point where $t = 0$.

(3)

WORKING

Q3.4.07 Intersections with parametric curves ●●○ [4 marks] ☐ Todo ☐ Mastered

The curve C has parametric equations $x = t^2$, $y = 2t$. The line $y = 1x - 8$ meets C at two points. Find the coordinates of these points.

(4)

WORKING

Q3.4.08 Modelling with parametric equations ●●○ [5 marks] ☐ Todo ☐ Mastered

A ball is kicked from ground level. Its position after t seconds is modelled by $x = 12.30t$, $y = 18.24t - 4.9t^2$ (metres).

(a) Find the time when the ball lands. **(2)**

(b) Find the horizontal distance travelled. **(1)**

(c) Find a Cartesian equation of its path. **(2)**

WORKING

Q3.4.09 Trig parametric conversions ●●○ [3 marks] ☐ Todo ☐ Mastered

A curve has parametric equations $x = \sin t$, $y = 4\cos 2t$, $-\frac{\pi}{2} \leq t \leq \frac{\pi}{2}$. Show that a Cartesian equation is $y = 4 - 8x^2$ and state the domain of x.

(3)

WORKING

Q3.4.10 Intersections with parametric curves ●●○ [4 marks] ☐ Todo ☐ Mastered

The curve C has parametric equations $x = t^2$, $y = 2t$. The line $y = -\frac{1}{3}x - \frac{8}{3}$ meets C at two points. Find the coordinates of these points.

(4)

WORKING

Q3.4.11 Modelling with parametric equations ●●○ [5 marks] ☐ Todo ☐ Mastered

A ball is kicked from ground level. Its position after t seconds is modelled by $x = 14.38t$, $y = 10.83t - 4.9t^2$ (metres).

(a) Find the time when the ball lands. **(2)**

(b) Find the horizontal distance travelled. **(1)**

(c) Find a Cartesian equation of its path. **(2)**

WORKING

Q3.4.12 Trig parametric conversions ●●○ [3 marks] ☐ Todo ☐ Mastered

A curve has parametric equations $x = 6\sec t$, $y = 5\tan t$. Find a Cartesian equation of the curve.

(3)

WORKING

Q3.4.13 Intersections with parametric curves ●●○ [4 marks] ☐ Todo ☐ Mastered

The curve C has parametric equations $x = t^2$, $y = 2t$. The line $y = -\frac{1}{2}x$ meets C at two points. Find the coordinates of these points.

(4)

WORKING

Q3.4.14 Modelling with parametric equations ●●○ [5 marks] ☐ Todo ☐ Mastered

A ball is kicked from ground level. Its position after t seconds is modelled by $x = 5.00t$, $y = 8.66t - 4.9t^2$ (metres).

(a) Find the time when the ball lands. **(2)**

(b) Find the horizontal distance travelled. **(1)**

(c) Find a Cartesian equation of its path. **(2)**

WORKING

Q3.4.15 Trig parametric conversions ●●○ [3 marks] ☐ Todo ☐ Mastered

A curve has parametric equations $x = 4\sec t$, $y = 3\tan t$. Find a Cartesian equation of the curve.

(3)

WORKING

MY NOTES — 3.4 PARAMETRIC EQUATIONS: MISTAKES TO REMEMBER

Answers and mark schemes

M = method mark, A = accuracy mark (depends on the preceding M), B = independent mark, ft = follow-through, * = given answer. Numerical answers are to 3 s.f. unless stated.

3.1 Sequences and Series

Q3.1.01 Geometric series · 2 marks

1720

M1 $\frac{29(1-1.2^{14})}{1-1.2}$ · A1

Q3.1.02 Arithmetic sequences · 5 marks

(a) $a=-3,\ d=8$

M1 $a+7d=53$ and $a+20d=157$ · A1 d · A1 a

(b) $S_{30}=3390$

M1 $\frac{30}{2}(2a+29d)$ · A1

Q3.1.03 Geometric sequences · 4 marks

$r=\frac{3}{4},\ a=4$

M1 $\frac{ar^4}{ar}=r^3$ · A1 $r=\frac{3}{4}$ · M1 substitutes back · A1 a

Q3.1.04 Geometric series · 2 marks

33.1

M1 $\frac{7(1-0.8^{13})}{1-0.8}$ · A1

Q3.1.05 Arithmetic sequences · 5 marks

(a) $a=-2,\ d=9$

M1 $a+4d=34$ and $a+12d=106$ · A1 d · A1 a

(b) $S_{47}=9635$

M1 $\frac{47}{2}(2a+46d)$ · A1

Q3.1.06 Sigma notation · 3 marks

4305

M1 recognises an arithmetic series with 35 terms · M1 first term 21, last term 225 · A1

Q3.1.07 Geometric sequences · 4 marks

$r=\frac{3}{4},\ a=29$

M1 $\frac{ar^4}{ar}=r^3$ · A1 $r=\frac{3}{4}$ · M1 substitutes back · A1 a

Q3.1.08 Recurrence relations, increasing / decreasing / periodic · 3 marks

$u_2=-7,\ u_3=20,\ u_4=-61$; neither

M1 uses the recurrence · A1 all three terms · B1 description

Q3.1.09 Geometric series · 2 marks

478

M1 $\frac{29(1-1.2^{8})}{1-1.2}$ · A1

Q3.1.10 Arithmetic sequences · 5 marks

(a) $a=8,\ d=7$

M1 $a+7d=57$ and $a+24d=176$ · A1 d · A1 a

(b) $S_{42}=6363$

M1 $\frac{42}{2}(2a+41d)$ · A1

Q3.1.11 Geometric sequences · 4 marks

$r=3,\ a=26$

M1 $\frac{ar^3}{ar}=r^2$ · A1 $r=3$ · M1 substitutes back · A1 a

Q3.1.12 Modelling with series · 5 marks

(a) year 30

M1 $20000(1.04)^{n-1}>60000$ · M1 logs · A1 year 30

(b) £240,122

M1 geometric sum · A1

Q3.1.13 Recurrence relations, increasing / decreasing / periodic · 4 marks

(a) $u_2=\frac{2}{5},\ u_3=5$; period 2

M1 finds $u_2,\ u_3$ · A1 order 2

(b) $\frac{349}{5}$

M1 counts complete cycles · A1

Q3.1.14 Sigma notation · 3 marks

78720

M1 geometric with $a=24$, $r=3$ · M1 sum formula · A1

Q3.1.15 Arithmetic series · 4 marks

$n=34$

M1 uses $S_n=\frac{n}{2}(2a+(n-1)d)$ · A1* rearranges to the given quadratic · M1 solves · A1 $n=34$ (reject negative)

Q3.1.16 Sum to infinity · 6 marks

(a) given

M1 $ar=9$ · M1 $\frac{a}{1-r}=36$ · A1* eliminates a

(b) $r=\frac{1}{2}$

M1 solves the quadratic · A1 roots · B1 requires $|r|<1$ for a sum to infinity

Q3.1.17 Modelling with series · 5 marks

(a) year 16

M1 $26000(1.03)^{n-1}>40000$ · M1 logs · A1 year 16

(b) £298,061

M1 geometric sum · A1

Q3.1.18 Recurrence relations, increasing / decreasing / periodic · 4 marks

(a) $u_2=\frac{2}{3}$, $u_3=3$; period 2

M1 finds u_2, u_3 · A1 order 2

(b) $\frac{229}{3}$

M1 counts complete cycles · A1

Q3.1.19 Arithmetic series · 4 marks

$n=28$

M1 uses $S_n=\frac{n}{2}(2a+(n-1)d)$ · A1* rearranges to the given quadratic · M1 solves · A1 $n=28$ (reject negative)

Q3.1.20 Sum to infinity · 6 marks

(a) given

M1 $ar=-6$ · M1 $\frac{a}{1-r}=8$ · A1* eliminates a

(b) $r=-\frac{1}{2}$ (reject $r=\frac{3}{2}$ since $|r|\geq 1$)

M1 solves the quadratic · A1 roots · B1 requires $|r|<1$ for a sum to infinity

Q3.1.21 Modelling with series · 5 marks

(a) year 15

M1 $26000(1.04)^{n-1}>45000$ · M1 logs · A1 year 15

(b) £312,159

M1 geometric sum · A1

Q3.1.22 Sigma notation · 3 marks

$\frac{2047}{512}$

M1 geometric with $a=2$, $r=\frac{1}{2}$ · M1 sum formula · A1

Q3.1.23 Arithmetic series · 4 marks

$n=14$

M1 uses $S_n=\frac{n}{2}(2a+(n-1)d)$ · A1* rearranges to the given quadratic · M1 solves · A1 $n=14$ (reject negative)

Q3.1.24 Sum to infinity · 6 marks

(a) given

M1 $ar=4$ · M1 $\frac{a}{1-r}=18$ · A1* eliminates a

(b) $r=\frac{1}{3}, \frac{2}{3}$

M1 solves the quadratic · A1 roots · B1 requires $|r|<1$ for a sum to infinity

3.2 Radians & Trigonometry

Q3.2.01 Radian measure · 2 marks

(a) $\frac{5}{6}\pi$

B1

(b) 225°

B1

Q3.2.02 sec, cosec and cot · 3 marks

$x=180.0^\circ, 221.8^\circ, 318.2^\circ$

M1 $\sin x=-\frac{2}{3}$ · A1 one value · A1 all values

Q3.2.03 **Arc length · 4 marks**

(a) 8.25 cm

B1 $r\theta$

(b) 38.2 cm

B1ft

(c) 61.9 cm^2

M1 $\frac{1}{2}r^2\theta$ · A1

Q3.2.04 **Inverse trig functions arcsin, arccos, arctan · 3 marks**

$-\frac{\pi}{4}$; domain $x \in \mathbb{R}$, range $-\frac{\pi}{2} < y < \frac{\pi}{2}$

B1 value · B1 domain · B1 range

Q3.2.05 **Radian measure · 2 marks**

(a) $\frac{7}{4}\pi$

B1

(b) 270°

B1

Q3.2.06 **sec, cosec and cot · 3 marks**

$x = 14.5^\circ, 165.5^\circ, 180.0^\circ$

M1 $\sin x = \frac{1}{4}$ · A1 one value · A1 all values

Q3.2.07 **Arc length · 4 marks**

(a) 9.05 cm

B1 $r\theta$

(b) 19.1 cm

B1ft

(c) 22.6 cm^2

M1 $\frac{1}{2}r^2\theta$ · A1

Q3.2.08 **Inverse trig functions arcsin, arccos, arctan · 3 marks**

$\frac{\pi}{6}$; domain $-1 \leq x \leq 1$, range $0 \leq y \leq \pi$

B1 value · B1 domain · B1 range

Q3.2.09 **Radian measure · 2 marks**

(a) $\frac{3}{2}\pi$

B1

(b) 120°

B1

Q3.2.10 **sec, cosec and cot · 3 marks**

$x = 30.0^\circ, 150.0^\circ, 180.0^\circ$

M1 $\sin x = \frac{1}{2}$ · A1 one value · A1 all values

Q3.2.11 **Arc length · 4 marks**

(a) 12.9 cm

B1 $r\theta$

(b) 40.9 cm

B1ft

(c) 90.2 cm^2

M1 $\frac{1}{2}r^2\theta$ · A1

Q3.2.12 **Inverse trig functions arcsin, arccos, arctan · 3 marks**

$\frac{\pi}{3}$; domain $x \in \mathbb{R}$, range $-\frac{\pi}{2} < y < \frac{\pi}{2}$

B1 value · B1 domain · B1 range

Q3.2.13 **Identities 1 + tan^2 = sec^2 and 1 + cot^2 = cosec^2 · 5 marks**

$x = 71.6^\circ, 135.0^\circ, 251.6^\circ, 315.0^\circ$

M1 uses $\sec^2 x = 1 + \tan^2 x$ · M1 $\tan^2 x - 2\tan x - 3 = 0$ · A1 $\tan x = -1, 3$ · M1 finds angles in range · A1 all

Q3.2.14 **Sector and segment areas · 4 marks**

12.2 cm^2

M1 sector area $\frac{1}{2}r^2\theta$ · M1 triangle area $\frac{1}{2}r^2\sin\theta$ · M1 subtracts · A1

Q3.2.15 **Small angle approximations · 3 marks**

8

M1 uses $\sin\theta \approx \theta$, $\cos\theta \approx 1 - \frac{\theta^2}{2}$, $\tan\theta \approx \theta$ · M1 simplifies · A1

Q3.2.16 **Identities 1 + tan^2 = sec^2 and 1 + cot^2 = cosec^2 · 5 marks**

$x = 26.6^\circ, 45.0^\circ, 206.6^\circ, 225.0^\circ$

M1 uses $\sec^2 x = 1 + \tan^2 x$ · M1 $\tan^2 x - \frac{3}{2}\tan x + \frac{1}{2} = 0$ · A1 $\tan x = \frac{1}{2}, 1$ · M1 finds angles in range · A1 all

Q3.2.17 **Sector and segment areas · 4 marks**

$2.10\ \text{cm}^2$

M1 sector area $\frac{1}{2}r^2\theta$ · M1 triangle area $\frac{1}{2}r^2\sin\theta$ · M1 subtracts · A1

Q3.2.18 **Small angle approximations · 3 marks**

$1-\frac{21\theta^2}{2}$

M1 uses $\sin\theta\approx\theta$, $\cos\theta\approx1-\frac{\theta^2}{2}$, $\tan\theta\approx\theta$ · M1 simplifies · A1

Q3.2.19 **Identities 1 + tan^2 = sec^2 and 1 + cot^2 = cosec^2 · 5 marks**

$x=45.0^\circ, 135.0^\circ, 225.0^\circ, 315.0^\circ$

M1 uses $\sec^2 x=1+\tan^2 x$ · M1 $\tan^2 x-1=0$ · A1 $\tan x=1,\ -1$ · M1 finds angles in range · A1 all

Q3.2.20 **Sector and segment areas · 4 marks**

$13.1\ \text{cm}^2$

M1 sector area $\frac{1}{2}r^2\theta$ · M1 triangle area $\frac{1}{2}r^2\sin\theta$ · M1 subtracts · A1

Q3.2.21 **Small angle approximations · 3 marks**

$1-4\theta^2$

M1 uses $\sin\theta\approx\theta$, $\cos\theta\approx1-\frac{\theta^2}{2}$, $\tan\theta\approx\theta$ · M1 simplifies · A1

3.3 Advanced Trigonometry & Modelling

Q3.3.01 **Addition formulae · 3 marks**

$\frac{\sqrt{6}+\sqrt{2}}{4}$

M1 writes $15=45-30$ · M1 correct expansion with exact values · A1

Q3.3.02 **Solving trig equations · 4 marks**

$x=15.0^\circ, 75.0^\circ$

M1 adjusts the interval for $2x$ · A1 one correct angle · M1 uses symmetry/period to find others · A1 all, no extras

Q3.3.03 **Addition formulae · 3 marks**

$\frac{\sqrt{2}-\sqrt{6}}{4}$

M1 writes $105=60+45$ · M1 correct expansion with exact values · A1

Q3.3.04 **Proving trig identities · 4 marks**

Proof using common denominator, $\sin^2+\cos^2=1$

M1 correct identity used · M1 manipulates towards RHS · M1 second identity/simplification · A1* fully correct proof

Q3.3.05 **Modelling with trig functions · 5 marks**

(a) 13.4 m at $t=3$ hours

B1 max · M1 sets the angle to 90° · A1 time

(b) 6 hours

M1 $\sin\geq0$ for half of each cycle · A1

Q3.3.06 **Double-angle formulae · 4 marks**

(a) $\frac{120}{169}$

M1 $\cos A=\frac{12}{13}$ · A1

(b) $\frac{119}{169}$

B1

(c) $\frac{120}{119}$

B1ft

Q3.3.07 **Solving trig equations · 6 marks**

$x=70.5^\circ, 180.0^\circ, 289.5^\circ$

M1 uses $\sin^2 x=1-\cos^2 x$ · A1 $3\cos^2 x+2\cos x-1=0$ · M1 solves quadratic · A1 $\cos x=\frac{1}{3},\ -1$ · M1 finds angles · A1 all, no extras

Q3.3.08 **Proving trig identities · 4 marks**

Proof using write in sin, cos; use $\sin 2\theta$

M1 correct identity used · M1 manipulates towards RHS · M1 second identity/simplification · A1* fully correct proof

Q3.3.09 **Modelling with trig functions · 5 marks**

(a) 12.1 m at $t=\frac{3}{2}$ hours

B1 max · M1 sets the angle to 90° · A1 time

(b) 6 hours

M1 $\sin\geq0$ for half of each cycle · A1

Q3.3.10 Double-angle formulae · 4 marks
(a) $\frac{336}{625}$
M1 $\cos A = \frac{24}{25}$ · A1
(b) $\frac{527}{625}$
B1
(c) $\frac{336}{527}$
B1ft

Q3.3.11 Proving trig identities · 4 marks
Proof using $\cos 2\theta = \cos^2 - \sin^2$, difference of two squares
M1 correct identity used · M1 manipulates towards RHS · M1 second identity/simplification · A1* fully correct proof

Q3.3.12 Modelling with trig functions · 5 marks
(a) 13.4 m at $t = 3$ hours
B1 max · M1 sets the angle to $90°$ · A1 time
(b) 6 hours
M1 $\sin \geq 0$ for half of each cycle · A1

Q3.3.13 Double-angle formulae · 4 marks
(a) $\frac{840}{841}$
M1 $\cos A = \frac{21}{29}$ · A1
(b) $\frac{41}{841}$
B1
(c) $\frac{840}{41}$
B1ft

Q3.3.14 Proving trig identities · 4 marks
Proof using $\sin(2\theta + \theta)$, then double-angle formulae
M1 correct identity used · M1 manipulates towards RHS · M1 second identity/simplification · A1* fully correct proof

Q3.3.15 Modelling with trig functions · 5 marks
(a) 11.1 m at $t = 2$ hours
B1 max · M1 sets the angle to $90°$ · A1 time
(b) 6 hours
M1 $\sin \geq 0$ for half of each cycle · A1

Q3.3.16 Harmonic form R cos(x ± a), R sin(x ± a) · 8 marks
(a) $R = 13$, $\alpha = 22.62°$
B1 R · M1 $\tan\alpha = \frac{5}{12}$ · A1 α
(b) $x = 37.9°, 276.9°$
M1 divides by R and uses inverse function · A1 first angle · M1 second angle in range · A1 both
(c) 13
B1ft

Q3.3.17 Harmonic form R cos(x ± a), R sin(x ± a) · 8 marks
(a) $R = 5$, $\alpha = 53.13°$
B1 R · M1 $\tan\alpha = \frac{4}{3}$ · A1 α
(b) $x = 119.6°, 346.7°$
M1 divides by R and uses inverse function · A1 first angle · M1 second angle in range · A1 both
(c) 5
B1ft

Q3.3.18 Harmonic form R cos(x ± a), R sin(x ± a) · 8 marks
(a) $R = \sqrt{13}$, $\alpha = 56.31°$
B1 R · M1 $\tan\alpha = \frac{3}{2}$ · A1 α
(b) $x = 104.8°, 187.8°$
M1 divides by R and uses inverse function · A1 first angle · M1 second angle in range · A1 both
(c) $\sqrt{13}$
B1ft

3.4 Parametric Equations

Q3.4.01 Parametric to Cartesian · 2 marks
$y = \frac{x^2}{9} - \frac{2x}{9} + \frac{28}{9}$
M1 $t = \frac{x-1}{3}$ · A1

Q3.4.02 Sketching parametric curves · 3 marks
$(-11, 0)$, $(19, 0)$; $t = 0$ gives $(4, -25)$
M1 $y = 0 \Rightarrow t = \pm 5$ · A1 both points · B1

Q3.4.03 **Parametric to Cartesian · 2 marks**

$y = x^2 + 4x + 1$

M1 $t = \frac{x+2}{1}$ · A1

Q3.4.04 **Sketching parametric curves · 3 marks**

$(-12, 0)$, $(18, 0)$; $t = 0$ gives $(3, -25)$

M1 $y = 0 \Rightarrow t = \pm 5$ · A1 both points · B1

Q3.4.05 **Parametric to Cartesian · 2 marks**

$y = x^2 - 8x + 15$

M1 $t = \frac{x-4}{-1}$ · A1

Q3.4.06 **Sketching parametric curves · 3 marks**

$(-3, 0)$, $(5, 0)$; $t = 0$ gives $(1, -4)$

M1 $y = 0 \Rightarrow t = \pm 2$ · A1 both points · B1

Q3.4.07 **Intersections with parametric curves · 4 marks**

$(4, -4)$ and $(16, 8)$

M1 substitutes parametric forms into the line · A1 quadratic in t · M1 solves · A1 both points

Q3.4.08 **Modelling with parametric equations · 5 marks**

(a) $t = 3.72$ s

M1 $y = 0$ · A1

(b) 45.8 m

B1ft

(c) $y = 1.48x - 0.0324x^2$

M1 eliminates t · A1

Q3.4.09 **Trig parametric conversions · 3 marks**

$y = 4(1 - 2\sin^2 t) = 4 - 8x^2, -1 \leq x \leq 1$

M1 $\cos 2t = 1 - 2\sin^2 t$ · A1* · B1 domain

Q3.4.10 **Intersections with parametric curves · 4 marks**

$(16, -8)$ and $(4, -4)$

M1 substitutes parametric forms into the line · A1 quadratic in t · M1 solves · A1 both points

Q3.4.11 **Modelling with parametric equations · 5 marks**

(a) $t = 2.21$ s

M1 $y = 0$ · A1

(b) 31.8 m

B1ft

(c) $y = 0.754x - 0.0237x^2$

M1 eliminates t · A1

Q3.4.12 **Trig parametric conversions · 3 marks**

$\frac{x^2}{36} - \frac{y^2}{25} = 1$

M1 $\sec^2 t - \tan^2 t = 1$ · M1 substitutes · A1

Q3.4.13 **Intersections with parametric curves · 4 marks**

$(16, -8)$ and $(0, 0)$

M1 substitutes parametric forms into the line · A1 quadratic in t · M1 solves · A1 both points

Q3.4.14 **Modelling with parametric equations · 5 marks**

(a) $t = 1.77$ s

M1 $y = 0$ · A1

(b) 8.84 m

B1ft

(c) $y = 1.73x - 0.196x^2$

M1 eliminates t · A1

Q3.4.15 **Trig parametric conversions · 3 marks**

$\frac{x^2}{16} - \frac{y^2}{9} = 1$

M1 $\sec^2 t - \tan^2 t = 1$ · M1 substitutes · A1

Pearson Edexcel A-Level Mathematics (9MA0)

Module 4

Calculus & Vectors

Paper 1 & 2 (Pure) · 108 practice questions · answers and mark schemes at the back

Subtopic	Questions	Done
4.1 Differentiation	36	____ / 36
4.2 Integration	36	____ / 36
4.3 Numerical Methods	18	____ / 18
4.4 Vectors (3D)	18	____ / 18

How to use this workbook

1. Read the key facts and watch one video for the subtopic. Tick the skills checklist as you go.
2. Attempt each question in the squared working space without notes. Mark it with the scheme at the back.
3. Tick **Todo** when you have attempted it and **Mastered** only when you can redo it cold a week later.
4. Copy any mistakes into the notes box at the end of the subtopic. Rebuild the workbook with a new seed for fresh questions.

Name _____ Started _____ Built 14 Oct 2026, seed 2026

4.1 Differentiation

Differentiation from first principles, standard derivatives, chain/product/quotient rules, implicit and parametric differentiation, stationary points, tangents, optimisation and rates of change.

Key facts

- $f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$
- $\frac{d}{dx}x^n = nx^{n-1}$, $\frac{d}{dx}e^{kx} = ke^{kx}$, $\frac{d}{dx}\ln x = \frac{1}{x}$, $\frac{d}{dx}a^x = a^x \ln a$
- $\frac{d}{dx}\sin kx = k\cos kx$, $\frac{d}{dx}\cos kx = -k\sin kx$, $\frac{d}{dx}\tan kx = k\sec^2 kx$
- Chain: $\frac{dy}{dx} = \frac{dy}{du}\frac{du}{dx}$; product: $u'v + uv'$; quotient: $\frac{vu' - uv'}{v^2}$
- Parametric: $\frac{dy}{dx} = \frac{dy/dt}{dx/dt}$; implicit: $\frac{d}{dx}y^2 = 2y\frac{dy}{dx}$
- Concave: $f''(x) < 0$; convex: $f''(x) > 0$; point of inflection: f'' changes sign

Method

- Stationary points: solve $f'(x) = 0$, then test with $f''(x)$ (or the gradient either side).
- Tangent: $y - y_1 = f'(x_1)(x - x_1)$; normal uses $-\frac{1}{f'(x_1)}$.
- Connected rates: chain the known rate with a derivative from a formula, e.g. $\frac{dr}{dt} = \frac{dV}{dt} \div \frac{dV}{dr}$.
- Optimisation: write the quantity in one variable using the constraint, differentiate, set to 0, justify max/min.

Exam traps

- Using degrees in trig derivatives (must be radians).
- Forgetting $\frac{dy}{dx}$ on y terms in implicit differentiation.
- Not justifying the nature of a stationary point when asked.

In the exam: The largest topic on Papers 1 and 2: expect 15-25 marks per paper across several questions.

Specification checklist

Skill	Learnt	Practised	Mastered
Differentiation from first principles	☐	☐	☐
Derivatives of sin x and cos x	☐	☐	☐
Exponentials and logarithms	☐	☐	☐
Chain rule	☐	☐	☐
Product rule	☐	☐	☐
Quotient rule	☐	☐	☐
Parametric differentiation	☐	☐	☐
Implicit differentiation	☐	☐	☐
Second derivatives, stationary points, convex / concave, inflection	☐	☐	☐
Tangents and normals	☐	☐	☐
Optimisation	☐	☐	☐
Connected rates of change	☐	☐	☐

Videos (click to open)

☐ **Recap** — Differentiation (Year 1) in less than 30 minutes • A-Level Maths, Pure Year 1, Chapter 12 (Bicen Maths)
☐ **Recap** — Differentiation (Year 2) in less than 42 minutes • A-Level Maths, Pure Year 2, Chapter 9 (Bicen Maths)
☐ Differentiation from first principles — Differentiation: Differentiation From First Principles (TLMaths)
☐ Derivatives of sin x and cos x — Differentiation: Differentiate sin(x) from First Principles (TLMaths)
☐ Exponentials and logarithms — Differentiation: Differentiating ln(kx) (TLMaths)
☐ Chain rule — Differentiation: Introducing the Chain Rule (TLMaths)
☐ Product rule — Differentiation: Introducing the Product Rule (TLMaths)
☐ Quotient rule — Differentiation: Introducing the Quotient Rule (TLMaths)
☐ Parametric differentiation — Differentiation: Introducing Parametric Differentiation (TLMaths)
☐ Implicit differentiation — Differentiation: Introducing Implicit Differentiation (TLMaths)
☐ Second derivatives, stationary points, convex / concave, inflection — Gradients: Finding where a Curve is Convex & Concave (TLMaths)

☐ Tangents and normals — Gradients: Introducing Tangents and Normals (TLMaths)
☐ Optimisation — Optimisation: Rectangular Fence Problem (TLMaths)
☐ Connected rates of change — Differentiation: Connected Rates of Change Example 1 (TLMaths)

Q4.1.01 Exponentials and logarithms ●○○ [3 marks] ☐ Todo ☐ Mastered

Given $y = 8^x + 4e^{3x} + \ln(4x)$, find $\frac{dy}{dx}$.

(3)

WORKING

Q4.1.02 Quotient rule ●○○ [3 marks] ☐ Todo ☐ Mastered

Use the quotient rule to find $\frac{dy}{dx}$ when $y = \frac{4x+2}{x+5}$, simplifying the numerator.

(3)

WORKING

Q4.1.03 Product rule ●○○ [3 marks] ☐ Todo ☐ Mastered

Find $\frac{dy}{dx}$ when $y = (x-4)^2 \cos 2x$.

(3)

WORKING

Q4.1.04 Tangents and normals ●○○ [4 marks] ☐ Todo ☐ Mastered

Find an equation of the tangent to the curve $y = 2x^2 + 3x - 1$ at the point where $x = 4$.

(4)

WORKING

Q4.1.05 Chain rule ●○○ [2 marks] ☐ Todo ☐ Mastered

Differentiate $y = \sqrt{x^3 - 3}$ with respect to x.

(2)

WORKING

Q4.1.06 Exponentials and logarithms ●○○ [3 marks] ☐ Todo ☐ Mastered

Given $y = 5^x + e^{2x} - 4\ln(4x)$, find $\frac{dy}{dx}$.

(3)

WORKING

Q4.1.07 Quotient rule ●○○ [3 marks] ☐ Todo ☐ Mastered

Use the quotient rule to find $\frac{dy}{dx}$ when $y = \frac{x^2 + 3}{e^x + 1}$, simplifying the numerator.

(3)

WORKING

Q4.1.08 Product rule ●○○ [3 marks] ☐ Todo ☐ Mastered

Find $\frac{dy}{dx}$ when $y = (x + 1)^2 \sin 2x$.

(3)

WORKING

Q4.1.09 Tangents and normals ●○○ [4 marks] ☐ Todo ☐ Mastered

Find an equation of the normal to the curve $y = 2\sqrt{x} + 2x^2 - x - 5$ at the point where $x = 4$.

(4)

WORKING

Q4.1.10 Chain rule ●○○ [2 marks] ☐ Todo ☐ Mastered

Differentiate $y = \ln(2x - 2)$ with respect to x.

(2)

WORKING

Q4.1.11 Exponentials and logarithms ●○○ [3 marks] ☐ Todo ☐ Mastered

Given $y = 3e^{4x} - 3\ln(4x)$, find $\frac{dy}{dx}$.

(3)

WORKING

Q4.1.12 Quotient rule ●○○ [3 marks] ☐ Todo ☐ Mastered

Use the quotient rule to find $\frac{dy}{dx}$ when $y = \frac{4x+2}{e^x+1}$, simplifying the numerator.

(3)

WORKING

Q4.1.13 **Product rule** ●○○ **[3 marks]** ☐ **Todo** ☐ **Mastered**

Find $\frac{dy}{dx}$ when $y = x^3 \cos 2x$.

(3)

WORKING

Q4.1.14 **Tangents and normals** ●○○ **[4 marks]** ☐ **Todo** ☐ **Mastered**

Find an equation of the tangent to the curve $y = 3x^2 - 5x + 1$ at the point where $x = 4$.

(4)

WORKING

Q4.1.15 **Chain rule** ●○○ **[2 marks]** ☐ **Todo** ☐ **Mastered**

Differentiate $y = (3x - 4)^6$ with respect to x.

(2)

WORKING

Q4.1.16 Second derivatives, stationary points, convex / concave, inflection ●●○ [8 marks] ☐ Todo ☐ Mastered

The curve C has equation $y = -x^3 + 6x^2 - 9x - 6$.

(a) Find the coordinates of the stationary points of C and determine their nature. **(6)**

(b) Find the range of values of x for which C is concave. **(2)**

WORKING

Q4.1.17 Connected rates of change ●●○ [4 marks] ☐ Todo ☐ Mastered

Air is pumped into a spherical balloon at $12\ \text{cm}^3\ \text{s}^{-1}$. Find the rate at which the radius is increasing when the radius is $10\ \text{cm}$.

(4)

WORKING

Q4.1.18 Parametric differentiation ●●○ [4 marks] ☐ Todo ☐ Mastered

A curve has parametric equations $x = t^3$, $y = t^3 - t$.
Find $\frac{dy}{dx}$ in terms of t, and its value when $t = -1$.

(4)

WORKING

Q4.1.19 **Implicit differentiation** ●●○ **[5 marks]** ☐ Todo ☐ Mastered

A curve has equation $x^2 - 3xy + 2y^2 - 3y = 5$.

(a) Find $\frac{dy}{dx}$ in terms of x and y. **(4)**

(b) Find the gradient of the curve at the point $(-3, -2)$. **(1)**

WORKING

Q4.1.20 **Differentiation from first principles** ●●○ **[4 marks]** ☐ Todo ☐ Mastered

Prove, from first principles, that the derivative of $x^2 - 6x - 1$ is $2x - 6$.

(4)

WORKING

Q4.1.21 **Second derivatives, stationary points, convex / concave, inflection** ●●○ **[8 marks]** ☐ Todo ☐ Mastered

The curve C has equation $y = -x^3 - 3x^2 + 24x - 6$.

(a) Find the coordinates of the stationary points of C and determine their nature. **(6)**

(b) Find the range of values of x for which C is concave. **(2)**

WORKING

Q4.1.22 Connected rates of change ●●○ [4 marks] ☐ Todo ☐ Mastered

Air is pumped into a spherical balloon at $2\text{ cm}^3\text{ s}^{-1}$. Find the rate at which the radius is increasing when the radius is 8 cm.

(4)

WORKING

Q4.1.23 Parametric differentiation ●●○ [4 marks] ☐ Todo ☐ Mastered

A curve has parametric equations $x = 2\sin t$, $y = 4t - 1$.
Find $\frac{dy}{dx}$ in terms of t, and its value when $t = \frac{\pi}{3}$.

(4)

WORKING

Q4.1.24 Implicit differentiation ●●○ [7 marks] ☐ Todo ☐ Mastered

A curve has equation $x^2 - 2xy + y^2 - 2y = 6$.

(a) Find $\frac{dy}{dx}$ in terms of x and y. **(4)**

(b) Find the gradient of the curve at the point $(-3, -1)$. **(1)**

(c) Find the equation of the normal at this point in the form $y = mx + c$. **(2)**

WORKING

Q4.1.25 Differentiation from first principles ●●○ [5 marks] ☐ Todo ☐ Mastered

Prove, from first principles, that the derivative of $-3x^3$ is $-9x^2$.

(5)

WORKING

Q4.1.26 Second derivatives, stationary points, convex / concave, inflection ●●○ [8 marks] ☐ Todo ☐ Mastered

The curve C has equation $y = x^3 - \frac{15x^2}{2} + 12x + 2$.

(a) Find the coordinates of the stationary points of C and determine their nature. **(6)**

(b) Find the range of values of x for which C is concave. **(2)**

WORKING

Q4.1.27 Connected rates of change ●●○ [3 marks] ☐ Todo ☐ Mastered

A circular ripple spreads so that its radius increases at 0.3 m s^{-1}. Find the rate of increase of its area when the radius is 4 m.

(3)

WORKING

Q4.1.28 Parametric differentiation ●●○ [4 marks] ☐ Todo ☐ Mastered

A curve has parametric equations $x = e^t + 2$, $y = 4t + 2$.
Find $\frac{dy}{dx}$ in terms of t, and its value when $t = 1$.

(4)

WORKING

Q4.1.29 Implicit differentiation ●●○ [7 marks] ☐ Todo ☐ Mastered

A curve has equation $x^2 + xy + 3y^2 - 2y = 37$.

(a) Find $\frac{dy}{dx}$ in terms of x and y. **(4)**

(b) Find the gradient of the curve at the point $(-1, -3)$. **(1)**

(c) Find the equation of the normal at this point in the form $y = mx + c$. **(2)**

WORKING

Q4.1.30 Differentiation from first principles ●●○ [4 marks] ☐ Todo ☐ Mastered

Prove, from first principles, that the derivative of $-2x^2 + 5x - 1$ is $5 - 4x$.

(4)

WORKING

Q4.1.31 Second derivatives, stationary points, convex / concave, inflection ●●○ [8 marks] ☐ Todo ☐ Mastered

The curve C has equation $y = x^3 + \frac{3x^2}{2} - 6x - 3$.

(a) Find the coordinates of the stationary points of C and determine their nature. **(6)**

(b) Find the range of values of x for which C is concave. **(2)**

WORKING

Q4.1.32 Connected rates of change ●●○ [4 marks] ☐ Todo ☐ Mastered

Air is pumped into a spherical balloon at $18\,\text{cm}^3\,\text{s}^{-1}$. Find the rate at which the radius is increasing when the radius is 7 cm.

(4)

WORKING

Q4.1.33 Derivatives of sin x and cos x ●●● [5 marks] ☐ Todo ☐ Mastered

Given that θ is in radians, prove, from first principles, that the derivative of $\cos 3x$ is $-3\sin 3x$.
You may assume that as $h \to 0$, $\frac{\sin h}{h} \to 1$ and $\frac{\cos h - 1}{h} \to 0$.

(5)

WORKING

Q4.1.34 Optimisation ●●● [9 marks] ☐ Todo ☐ Mastered

A closed cylindrical can has volume 2000 cm^3 and radius r cm.

(a) Show that the surface area is $A = 2\pi r^2 + \frac{4000}{r}$. **(3)**

(b) Find the value of r that minimises A and the minimum area, to 3 significant figures. **(4)**

(c) Justify that this is a minimum. **(2)**

WORKING

Q4.1.35 Optimisation ●●● [7 marks] ☐ Todo ☐ Mastered

An open box is made from a square sheet of card of side 12 cm by cutting a square of side x cm from each corner and folding up the sides.

(a) Show that the volume is $V = x(12 - 2x)^2$. **(2)**

(b) Find the value of x that maximises V and the maximum volume. **(5)**

WORKING

Q4.1.36 Derivatives of sin x and cos x ●●● [5 marks] ☐ Todo ☐ Mastered

Given that θ is in radians, prove, from first principles, that the derivative of $\sin x$ is $\cos x$.
You may assume that as $h \to 0$, $\frac{\sin h}{h} \to 1$ and $\frac{\cos h - 1}{h} \to 0$.

(5)

WORKING

MY NOTES — 4.1 DIFFERENTIATION: MISTAKES TO REMEMBER

4.2 Integration

Integration: standard integrals, $f(ax+b)$, reverse chain rule, trig identities, substitution, parts, partial fractions, areas, the trapezium rule and differential equations.

Key facts

- $\int x^n\,dx = \frac{x^{n+1}}{n+1} + c \ (n \neq -1)$, $\int \frac{1}{x}\,dx = \ln|x| + c$
- $\int e^{kx}\,dx = \frac{1}{k}e^{kx} + c$, $\int \cos kx\,dx = \frac{1}{k}\sin kx + c$, $\int \sin kx\,dx = -\frac{1}{k}\cos kx + c$
- $\int \frac{f'(x)}{f(x)}\,dx = \ln|f(x)| + c$
- By parts: $\int u\frac{dv}{dx}\,dx = uv - \int v\frac{du}{dx}\,dx$
- Trapezium rule: $\int_a^b y\,dx \approx \frac{h}{2}[y_0 + y_n + 2(y_1 + \ldots + y_{n-1})]$
- Separable: $\frac{dy}{dx} = f(x)g(y) \Rightarrow \int \frac{1}{g(y)}\,dy = \int f(x)\,dx$

Method

- Choose u in parts as the thing that gets simpler when differentiated (LATE: Logs, Algebra, Trig, Exponentials); with $\ln x$, always $u = \ln x$.
- Substitution: change dx and the limits; never mix x and u.
- Areas below the x-axis come out negative; split the integral at the roots.
- Area between curves: $\int(\text{top} - \text{bottom})\,dx$.

Exam traps

- Forgetting $+c$ (or the modulus in $\ln|x|$).
- Not changing limits in a substitution.
- Over/underestimate: trapezium rule overestimates for convex curves.

In the exam: Expect at least one 8-12 mark integration question per pure paper; DE modelling questions are common in Paper 2.

Specification checklist

Skill	Learnt	Practised	Mastered
Standard integrals	☐	☐	☐
Integrating f(ax + b)	☐	☐	☐
Reverse chain rule	☐	☐	☐
Using trig identities in integration	☐	☐	☐
Integration by substitution	☐	☐	☐
Integration by parts	☐	☐	☐
Integration with partial fractions	☐	☐	☐
Areas under curves and between curves	☐	☐	☐
Integration as the limit of a sum	☐	☐	☐
Trapezium rule	☐	☐	☐
Separable differential equations	☐	☐	☐
Modelling with differential equations	☐	☐	☐

Videos (click to open)

☐ **Recap** — Integration (Year 1) in less than 25 minutes • A-Level Maths, Pure Year 1, Chapter 13 (Bicen Maths)
☐ Standard integrals — Integration: Introducing the Fundamental Theorem of Calculus (TLMaths)
☐ Integrating f(ax + b) — Integration : $(ax+b)^n$ type functions (ExamSolutions)
☐ Reverse chain rule — Integration : $(ax+b)^n$ type functions (ExamSolutions)
☐ Using trig identities in integration — Integration using trigonometric identities (1) (ExamSolutions)
☐ Integration by substitution — How to Integrate by substitution (1) (ExamSolutions)
☐ Integration by parts — Integration by Parts - example 1 (ExamSolutions)
☐ Integration with partial fractions — Integration of Partial Fractions: Example 1 (TLMaths)
☐ Areas under curves and between curves — Integration: Finding the Area between Two Curves Example 1 (TLMaths)
☐ Integration as the limit of a sum — Integration: Integration as the Limit of a Sum (TLMaths)

☐ Trapezium rule — Numerical Integration: Introducing the Trapezium Rule (TLMaths)
☐ Separable differential equations — Differential Equations: The Method of Separation of Variables (TLMaths)
☐ Modelling with differential equations — Differential Equations: A Worded Example 1 (TLMaths)

Q4.2.01 Standard integrals ●○○ [3 marks] ☐ Todo ☐ Mastered

Find $\int\left(-6x^2 - 3 + \frac{3}{x^2}\right)dx$, simplifying each term.

(3)

WORKING

Q4.2.02 Trapezium rule ●○○ [4 marks] ☐ Todo ☐ Mastered

Use the trapezium rule with 4 strips of equal width to estimate $\int_0^2 10e^{-x^2}\,dx$, giving your answer to 3 significant figures. State whether your answer is an over- or under-estimate, given the shape of the curve.

(4)

WORKING

Q4.2.03 Integration as the limit of a sum ●○○ [3 marks] ☐ Todo ☐ Mastered

Write $\lim_{\delta x \to 0} \sum_{x=0}^{4} \sqrt{x}\,\delta x$ as an integral and hence find its exact value.

(3)

WORKING

Q4.2.04 Integrating f(ax + b) ●○○ [2 marks] ☐ Todo ☐ Mastered

Find $\int(4x-1)^5\,dx$.

(2)

WORKING

Q4.2.05 Standard integrals ●○○ [3 marks] ☐ Todo ☐ Mastered

Find $\int\left(2x^2 - 1 + \frac{4}{x^2}\right)dx$, simplifying each term.

(3)

WORKING

Q4.2.06 Trapezium rule ●○○ [4 marks] ☐ Todo ☐ Mastered

Use the trapezium rule with 5 strips of equal width to estimate $\int_0^{\frac{5}{2}} 10e^{-x^2}\,dx$, giving your answer to 3 significant figures. State whether your answer is an over- or under-estimate, given the shape of the curve.

(4)

WORKING

Q4.2.07 Integration as the limit of a sum ●○○ [3 marks] ☐ Todo ☐ Mastered

Write $\lim_{\delta x \to 0} \sum_{x=2}^{5} x^2\,\delta x$ as an integral and hence find its exact value.

(3)

WORKING

Q4.2.08 Integrating f(ax + b) ●○○ [2 marks] ☐ Todo ☐ Mastered

Find $\int (4x-7)^3\,dx$.

(2)

WORKING

Q4.2.09 Standard integrals ●○○ [4 marks] ☐ Todo ☐ Mastered

Find the exact value of $\int_2^4 \left(x^2+\frac{1}{x}-\frac{4}{x^2}\right)dx$.

(4)

WORKING

Q4.2.10 Trapezium rule ●○○ [4 marks] ☐ Todo ☐ Mastered

Use the trapezium rule with 6 strips of equal width to estimate $\int_0^3 x\sqrt{x+1}\,dx$, giving your answer to 3 significant figures. State whether your answer is an over- or under-estimate, given the shape of the curve.

(4)

WORKING

Q4.2.11 Integration as the limit of a sum ●○○ [3 marks] ☐ Todo ☐ Mastered

Write $\lim_{\delta x \to 0} \sum_{x=0}^{4} x^2\,\delta x$ as an integral and hence find its exact value.

(3)

WORKING

Q4.2.12 Integrating f(ax + b) ●○○ [2 marks] ☐ Todo ☐ Mastered

Find $\int \frac{1}{2x+5}\,dx$.

(2)

WORKING

Q4.2.13 Areas under curves and between curves ●●○ [6 marks] ☐ Todo ☐ Mastered

The line $y = 3x + 1$ and the curve $y = x^2 + 5x - 2$ intersect at two points.

(a) Find the x-coordinates of the points of intersection. **(2)**

(b) Find the exact area of the region enclosed between the line and the curve. **(4)**

WORKING

Q4.2.14 Reverse chain rule ●●○ [3 marks] ☐ Todo ☐ Mastered

Find $\int \frac{2x+1}{x^2+x+2}\,dx$.

(3)

WORKING

Q4.2.15 Modelling with differential equations ●●○ [2 marks] ☐ Todo ☐ Mastered

The rate at which the temperature θ of a liquid falls is proportional to the difference between θ and room temperature $20\,℃$. Write down a differential equation to model this situation, defining any constant you use.

(2)

WORKING

Q4.2.16 Integration with partial fractions ●●○ [5 marks] ☐ Todo ☐ Mastered

Find the exact value of $\int_0^3 \frac{4}{x^2+4x+3}\,dx$, giving your answer as a single logarithm.

(5)

WORKING

Q4.2.17 Using trig identities in integration ●●○ [3 marks] ☐ Todo ☐ Mastered

Find $\int \tan^2 x\,dx$.

(3)

WORKING

Q4.2.18 Separable differential equations ●●○ [4 marks] ☐ Todo ☐ Mastered

Find the particular solution of $\frac{dy}{dx} = \frac{1x^2}{y}$, $y > 0$, for which $y = 1$ when $x = 0$.

(4)

WORKING

Q4.2.19 Integration by parts ●●○ [4 marks] ☐ Todo ☐ Mastered

Use integration by parts to find $\int x\cos 2x\,dx$.

(4)

WORKING

Q4.2.20 Areas under curves and between curves ●●○ [6 marks] ☐ Todo ☐ Mastered

The line $y = -x - 2$ and the curve $y = 2x^2 - 13x + 8$ intersect at two points.

(a) Find the x-coordinates of the points of intersection. **(2)**

(b) Find the exact area of the region enclosed between the line and the curve. **(4)**

WORKING

Q4.2.21 Reverse chain rule ●●○ [3 marks] ☐ Todo ☐ Mastered

Find $\int x(x^2 + 4)^6\,dx$.

(3)

WORKING

Q4.2.22 **Modelling with differential equations** ●●○ **[2 marks]** ☐ **Todo** ☐ **Mastered**

The rate at which the volume V of a raindrop decreases is proportional to its surface area, which is proportional to $V^{\frac{2}{3}}$. Write down a differential equation to model this situation, defining any constant you use.

(2)

WORKING

Q4.2.23 **Integration with partial fractions** ●●○ **[5 marks]** ☐ **Todo** ☐ **Mastered**

Find the exact value of $\int_0^2 \frac{-5x-7}{x^2+3x+2}\,dx$, giving your answer as a single logarithm.

(5)

WORKING

Q4.2.24 **Using trig identities in integration** ●●○ **[3 marks]** ☐ **Todo** ☐ **Mastered**

Find $\int \tan^2 3x\,dx$.

(3)

WORKING

Q4.2.25 Separable differential equations ●●○ [5 marks] ☐ Todo ☐ Mastered

A cup of tea cools in a room at $21\,℃$. Its temperature θ after t minutes satisfies $\frac{d\theta}{dt} = -0.07(\theta - 21)$. Initially $\theta = 71$.

(a) Show that $\theta = 21 + 50e^{-0.07t}$. **(4)**

(b) Find the temperature after 12 minutes. **(1)**

WORKING

Q4.2.26 Integration by parts ●●○ [4 marks] ☐ Todo ☐ Mastered

Use integration by parts to find $\int xe^{3x}\,dx$.

(4)

WORKING

Q4.2.27 Areas under curves and between curves ●●○ [6 marks] ☐ Todo ☐ Mastered

The line $y = 2x - 1$ and the curve $y = x^2 - 5x + 9$ intersect at two points.

(a) Find the x-coordinates of the points of intersection. **(2)**

(b) Find the exact area of the region enclosed between the line and the curve. **(4)**

WORKING

Q4.2.28 Reverse chain rule ●●○ [3 marks] ☐ Todo ☐ Mastered

Find $\int x(x^2+3)^3\,dx$.

(3)

WORKING

Q4.2.29 Integration with partial fractions ●●○ [5 marks] ☐ Todo ☐ Mastered

Find the exact value of $\int_0^3 \frac{-5x-14}{x^2+6x+8}\,dx$, giving your answer as a single logarithm.

(5)

WORKING

Q4.2.30 Using trig identities in integration ●●○ [3 marks] ☐ Todo ☐ Mastered

Find $\int \tan^2 4x\,dx$.

(3)

WORKING

Q4.2.31 Separable differential equations ●●○ [5 marks] ☐ Todo ☐ Mastered

Solve $\frac{dy}{dx} = 2xy$, given $y = 3$ when $x = 0$. Give y in terms of x.

(5)

WORKING

Q4.2.32 Integration by parts ●●○ [4 marks] ☐ Todo ☐ Mastered

Use integration by parts to find $\int x \sin 3x \, dx$.

(4)

WORKING

Q4.2.33 Areas under curves and between curves ●●○ [6 marks] ☐ Todo ☐ Mastered

The line $y = 2x - 2$ and the curve $y = x^2 + 5x - 2$ intersect at two points.

(a) Find the x-coordinates of the points of intersection. **(2)**

(b) Find the exact area of the region enclosed between the line and the curve. **(4)**

WORKING

Q4.2.34 Integration by substitution ●●● [4 marks] ☐ Todo ☐ Mastered

Use the substitution $u = x^2 + 2$ to find $\int \frac{x}{(x^2+2)^2} \, dx$.

(4)

WORKING

Q4.2.35 Integration by substitution ●●● [6 marks] ☐ Todo ☐ Mastered

Use the substitution $u = x + 4$ to find the exact value of $\int_{-3}^{8} x\sqrt{x+4}\, dx$.

(6)

WORKING

Q4.2.36 Integration by substitution ●●● [6 marks] ☐ Todo ☐ Mastered

Use the substitution $u = x + 5$ to find the exact value of $\int_{-5}^{7} x\sqrt{x+5}\, dx$.

(6)

WORKING

MY NOTES — 4.2 INTEGRATION: MISTAKES TO REMEMBER

4.3 Numerical Methods

Numerical methods: locating roots by change of sign, fixed-point iteration (and its staircase/cobweb diagrams), and the Newton-Raphson method.

Key facts

- If f is continuous on $[a, b]$ and $f(a)f(b) < 0$, there is a root in (a, b)
- Iteration $x_{n+1} = g(x_n)$ converges near the root if $|g'(\alpha)| < 1$
- Staircase when $0 < g'(\alpha) < 1$; cobweb when $-1 < g'(\alpha) < 0$
- Newton-Raphson: $x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$

Method

- To show a root is 2.345 to 3 d.p., check the sign of $f(2.3445)$ and $f(2.3455)$.
- Always state 'sign change and f is continuous, so a root lies in the interval'.
- Newton-Raphson fails when $f'(x_n) = 0$ (horizontal tangent) or the start is near a turning point.

Exam traps

- Forgetting the continuity statement.
- Rounding iterates too early.
- Using degrees in trig iterations.

In the exam: Paper 2, 5-8 marks; often the end of a calculus or modelling question.

Specification checklist

Skill	Learnt	Practised	Mastered
Locating roots by change of sign	☐	☐	☐
When change of sign fails	☐	☐	☐
Fixed-point iteration x = g(x)	☐	☐	☐
Staircase and cobweb diagrams	☐	☐	☐
Newton-Raphson method	☐	☐	☐
Applications to modelling	☐	☐	☐

Videos (click to open)

☐ **Recap** — Numerical Methods in 21 minutes • A-Level Maths, Pure Year 2, Chapter 10 (Bicen Maths)
☐ Locating roots by change of sign — Locating Roots: Introducing the Change of Sign Method (TLMaths)
☐ When change of sign fails — Locating Roots: Failure of the Change of Sign Method (TLMaths)
☐ Fixed-point iteration x = g(x) — Locating Roots: Introducing the x=g(x) Method (TLMaths)
☐ Staircase and cobweb diagrams — Locating Roots: Cobweb and Staircase Diagrams (TLMaths)
☐ Newton-Raphson method — Locating Roots: Introducing the Newton-Raphson Method (TLMaths)
☐ Applications to modelling — Numerical Methods: Contextual Problem 2 (TLMaths)

Q4.3.01 Locating roots by change of sign ●○○ [2 marks] ☐ Todo ☐ Mastered

$f(x) = x^3 + 3x^2 + 5x + 1$. Show that the equation $f(x) = 0$ has a root α in the interval $[-1, 0]$.

(2)

WORKING

Q4.3.02 Locating roots by change of sign ●○○ [2 marks] ☐ Todo ☐ Mastered

$f(x) = x^3 + 4x^2 - 6x + 7$. Show that the equation $f(x) = 0$ has a root α in the interval $[-6, -5]$.

(2)

WORKING

Q4.3.03 Locating roots by change of sign ●○○ [2 marks] ☐ Todo ☐ Mastered

$f(x) = x^3 - 3x^2 + 6x + 2$. Show that the equation $f(x) = 0$ has a root α in the interval $[-1, 0]$.

(2)

WORKING

Q4.3.04 Locating roots by change of sign ●○○ [2 marks] ☐ Todo ☐ Mastered

$f(x) = x^3 + 2x^2 - x - 7$. Show that the equation $f(x) = 0$ has a root α in the interval $[1, 2]$.

(2)

WORKING

Q4.3.05 When change of sign fails ●●○ [2 marks] ☐ Todo ☐ Mastered

Explain why the change of sign method fails to locate the root of $f(x) = (x + 2)^2$ and why considering $f(x) = \frac{1}{x+2}$ on $[-3, -1]$ would be misleading.

(2)

WORKING

Q4.3.06 Staircase and cobweb diagrams ●●○ [2 marks] ☐ Todo ☐ Mastered

An iteration $x_{n+1} = g(x_n)$ has a fixed point α, and $g'(\alpha) = -2$. State whether the iteration converges to α for a starting value close to α, and whether the diagram is a staircase or a cobweb.

(2)

WORKING

Q4.3.07 Fixed-point iteration x = g(x) ●●○ [6 marks] ☐ Todo ☐ Mastered

(a) Show that $x^3 - 4x - 2 = 0$ can be written as $x = \sqrt[3]{4x+2}$. **(1)**

(b) Using $x_{n+1} = \sqrt[3]{4x_n + 2}$ with $x_0 = 2$, find x_1, x_2 and x_3 to 4 decimal places. **(3)**

(c) Show that the root is 2.214 to 3 decimal places. **(2)**

WORKING

Q4.3.08 Newton-Raphson method ●●○ [4 marks] ☐ Todo ☐ Mastered

$f(x) = x^3 + 3x^2 + 4x + 5$. Taking $x_0 = -2.5$ as a first approximation to a root of $f(x) = 0$, apply the Newton-Raphson method twice to find x_2 to 3 decimal places.

(4)

WORKING

Q4.3.09 **When change of sign fails** ●●○ **[2 marks]** ☐ Todo ☐ Mastered

Explain why the change of sign method fails to locate the root of $f(x) = (x-4)^2$ and why considering $f(x) = \frac{1}{x-4}$ on $[3, 5]$ would be misleading.

(2)

WORKING

Q4.3.10 **Staircase and cobweb diagrams** ●●○ **[2 marks]** ☐ Todo ☐ Mastered

An iteration $x_{n+1} = g(x_n)$ has a fixed point α, and $g'(\alpha) = -\frac{3}{4}$. State whether the iteration converges to α for a starting value close to α, and whether the diagram is a staircase or a cobweb.

(2)

WORKING

Q4.3.11 **Fixed-point iteration x = g(x)** ●●○ **[6 marks]** ☐ Todo ☐ Mastered

(a) Show that $x^3 - 4x - 3 = 0$ can be written as $x = \sqrt[3]{4x+3}$. **(1)**

(b) Using $x_{n+1} = \sqrt[3]{4x_n + 3}$ with $x_0 = 2$, find x_1, x_2 and x_3 to 4 decimal places. **(3)**

(c) Show that the root is 2.303 to 3 decimal places. **(2)**

WORKING

Q4.3.12 Newton-Raphson method ●●○ [4 marks] ☐ Todo ☐ Mastered

$f(x) = x^3 + 4x^2 - 4x - 9$. Taking $x_0 = -4.5$ as a first approximation to a root of $f(x) = 0$, apply the Newton-Raphson method twice to find x_2 to 3 decimal places.

(4)

WORKING

Q4.3.13 When change of sign fails ●●○ [2 marks] ☐ Todo ☐ Mastered

Explain why the change of sign method fails to locate the root of $f(x) = (x+4)^2$ and why considering $f(x) = \frac{1}{x+4}$ on $[-5, -3]$ would be misleading.

(2)

WORKING

Q4.3.14 Fixed-point iteration x = g(x) ●●○ [6 marks] ☐ Todo ☐ Mastered

(a) Show that $x^3 - 5x - 7 = 0$ can be written as $x = \sqrt[3]{5x+7}$. **(1)**

(b) Using $x_{n+1} = \sqrt[3]{5x_n + 7}$ with $x_0 = 3$, find x_1, x_2 and x_3 to 4 decimal places. **(3)**

(c) Show that the root is 2.747 to 3 decimal places. **(2)**

WORKING

Q4.3.15 Newton-Raphson method ●●○ [4 marks] ☐ Todo ☐ Mastered

$f(x) = x^3 - x^2 + 5x + 2$. Taking $x_0 = -0.5$ as a first approximation to a root of $f(x) = 0$, apply the Newton-Raphson method twice to find x_2 to 3 decimal places.

(4)

WORKING

Q4.3.16 Applications to modelling ●●● [4 marks] ☐ Todo ☐ Mastered

The height difference between two tanks is modelled by $f(t) = e^{0.5t} - 59 + 2t$, $t \geq 0$. The tanks are level when $f(t) = 0$.
Use the Newton-Raphson method with $t_0 = 6$ to find this time to 2 decimal places.

(4)

WORKING

Q4.3.17 Applications to modelling ●●● [4 marks] ☐ Todo ☐ Mastered

The height difference between two tanks is modelled by $f(t) = e^{0.5t} - 35 + 2t$, $t \geq 0$. The tanks are level when $f(t) = 0$.
Use the Newton-Raphson method with $t_0 = 6$ to find this time to 2 decimal places.

(4)

WORKING

Q4.3.18 Applications to modelling ●●● [4 marks] ☐ Todo ☐ Mastered

The height difference between two tanks is modelled by $f(t) = e^{0.5t} - 41 + 2t$, $t \geq 0$. The tanks are level when $f(t) = 0$.

Use the Newton-Raphson method with $t_0 = 6$ to find this time to 2 decimal places.

(4)

WORKING

MY NOTES — 4.3 NUMERICAL METHODS: MISTAKES TO REMEMBER

4.4 Vectors (3D)

Vectors in 2D and 3D: magnitude and direction, position vectors, distances, geometric problems (parallel, collinear, ratios) and vectors in mechanics.

Key facts

- $|a\mathbf{i} + b\mathbf{j} + c\mathbf{k}| = \sqrt{a^2 + b^2 + c^2}$; unit vector $\hat{\mathbf{a}} = \frac{\mathbf{a}}{|\mathbf{a}|}$
- $\overrightarrow{AB} = \mathbf{b} - \mathbf{a}$; midpoint $\frac{1}{2}(\mathbf{a} + \mathbf{b})$
- Angle with the x-axis: $\cos\theta_x = \frac{a}{|\mathbf{v}|}$ (similarly for y, z)
- Parallel vectors are scalar multiples; collinear points share a point and a parallel direction

Method

- For a parallelogram $ABCD$: $\overrightarrow{AD} = \overrightarrow{BC}$.
- For ratio problems, write a path in two ways and compare coefficients of non-parallel vectors.
- In mechanics, resultant force $= m\mathbf{a}$; speed is the magnitude of velocity.

Exam traps

- $\overrightarrow{AB} = \mathbf{b} - \mathbf{a}$, not $\mathbf{a} - \mathbf{b}$.
- Confusing position vector with displacement.

In the exam: Short 4-6 mark questions in pure; vector kinematics and forces in Paper 3 mechanics.

Specification checklist

Skill	Learnt	Practised	Mastered
3D coordinates and i, j, k notation	☐	☐	☐
Magnitude and direction	☐	☐	☐
Position vectors and distance	☐	☐	☐
Angles with axes	☐	☐	☐
Geometric problems with vectors	☐	☐	☐
Vectors in mechanics	☐	☐	☐

Videos (click to open)

☐ **Recap** — Vectors (Year 1) in less than 21 minutes • A-Level Maths, Pure Year 1, Chapter 11 (Bicen Maths)
☐ **Recap** — Vectors (Year 2) in less than 14 minutes • A-Level Maths, Pure Year 2, Chapter 12 (Bicen Maths)
☐ 3D coordinates and i, j, k notation — Vectors: Introducing 3D Vectors (TLMaths)
☐ Magnitude and direction — Vectors: Finding the Magnitude / Length of a 3D vector (TLMaths)
☐ Position vectors and distance — Vectors: Introducing Position Vectors (TLMaths)
☐ Angles with axes — Angle between a 3d vector and the positive axis (ExamSolutions)
☐ Geometric problems with vectors — Vectors: Vectors Problem 1 (TLMaths)
☐ Vectors in mechanics — Vectors: Kinematics Problem (TLMaths)

Q4.4.01 Position vectors and distance ●○○ [4 marks] ☐ Todo ☐ Mastered

Points A and B have position vectors $-\mathbf{j}-\mathbf{k}$ and $3\mathbf{i}+\mathbf{j}$.

(a) Find $\overrightarrow{AB}$. **(1)**

(b) Find the exact distance AB. **(2)**

(c) Find the position vector of the midpoint of AB. **(1)**

WORKING

Q4.4.02 3D coordinates and i, j, k notation ●○○ [3 marks] ☐ Todo ☐ Mastered

Show that the points $A(3, 1, 1)$, $B(6, 4, -1)$ and $C(12, 10, -5)$ are collinear, and find the ratio $AB : BC$.

(3)

WORKING

Q4.4.03 Magnitude and direction ●○○ [2 marks] ☐ Todo ☐ Mastered

$\mathbf{a} = -\mathbf{i} - 2\mathbf{j} + 2\mathbf{k}$. Find $|\mathbf{a}|$ and a unit vector in the direction of $\mathbf{a}$.

(2)

WORKING

Q4.4.04 Position vectors and distance ●○○ [4 marks] ☐ Todo ☐ Mastered

Points A and B have position vectors $3\mathbf{i}+6\mathbf{j}-6\mathbf{k}$ and $2\mathbf{i}+5\mathbf{j}+4\mathbf{k}$.

(a) Find $\overrightarrow{AB}$. **(1)**

(b) Find the exact distance AB. **(2)**

(c) Find the position vector of the midpoint of AB. **(1)**

WORKING

Q4.4.05 3D coordinates and i, j, k notation ●○○ [3 marks] ☐ Todo ☐ Mastered

Show that the points $A(4, 5, 5)$, $B(1, 7, 2)$ and $C(-8, 13, -7)$ are collinear, and find the ratio $AB : BC$.

(3)

WORKING

Q4.4.06 Magnitude and direction ●○○ [2 marks] ☐ Todo ☐ Mastered

$\mathbf{a} = -\mathbf{i} + 4\mathbf{j} - 8\mathbf{k}$. Find $|\mathbf{a}|$ and a unit vector in the direction of $\mathbf{a}$.

(2)

WORKING

Q4.4.07 Position vectors and distance ●○○ [4 marks] ☐ Todo ☐ Mastered

Points A and B have position vectors $-2\mathbf{i} - \mathbf{j} + 2\mathbf{k}$ and $-6\mathbf{i} - 4\mathbf{j} + 2\mathbf{k}$.

(a) Find $\overrightarrow{AB}$. **(1)**

(b) Find the exact distance AB. **(2)**

(c) Find the position vector of the midpoint of AB. **(1)**

WORKING

Q4.4.08 3D coordinates and i, j, k notation ●○○ [3 marks] ☐ Todo ☐ Mastered

Show that the points $A(4, -5, -2)$, $B(2, -6, -3)$ and $C(0, -7, -4)$ are collinear, and find the ratio $AB : BC$.

(3)

WORKING

Q4.4.09 Magnitude and direction ●○○ [2 marks] ☐ Todo ☐ Mastered

$\mathbf{a} = 3\mathbf{i} + 4\mathbf{j} + 12\mathbf{k}$. Find $|\mathbf{a}|$ and a unit vector in the direction of $\mathbf{a}$.

(2)

WORKING

Q4.4.10 Vectors in mechanics ●●○ [3 marks] ☐ Todo ☐ Mastered

Two forces $\mathbf{F}_1 = (-\mathbf{i} - \mathbf{j} - 8\mathbf{k})$ N and $\mathbf{F}_2 = (-3\mathbf{i} - 5\mathbf{j} + 3\mathbf{k})$ N act on a particle of mass 2 kg. Find the magnitude of the acceleration of the particle.

(3)

WORKING

Q4.4.11 Geometric problems with vectors ●●○ [3 marks] ☐ Todo ☐ Mastered

$ABCD$ is a parallelogram with $A(-3, 4, 4)$, $B(3, 0, 3)$ and $C(1, -3, 2)$. Find the coordinates of D.

(3)

WORKING

Q4.4.12 Angles with axes ●●○ [2 marks] ☐ Todo ☐ Mastered

Find the angle that the vector $4\mathbf{i} + \mathbf{j} - 2\mathbf{k}$ makes with the positive z-axis, to 1 decimal place.

(2)

WORKING

Q4.4.13 Vectors in mechanics ●●○ [3 marks] ☐ Todo ☐ Mastered

Two forces $\mathbf{F}_1 = (5\mathbf{i} - 4\mathbf{j} + 8\mathbf{k})$ N and $\mathbf{F}_2 = (3\mathbf{i} - 3\mathbf{j} - 7\mathbf{k})$ N act on a particle of mass 4 kg. Find the magnitude of the acceleration of the particle.

(3)

WORKING

Q4.4.14 Geometric problems with vectors ●●○ [3 marks] ☐ Todo ☐ Mastered

$ABCD$ is a parallelogram with $A(5, -2, -5)$, $B(2, -1, -3)$ and $C(-5, -5, 2)$. Find the coordinates of D.

(3)

WORKING

Q4.4.15 Angles with axes ●●○ [2 marks] ☐ Todo ☐ Mastered

Find the angle that the vector $2\mathbf{i} - 5\mathbf{j} - 3\mathbf{k}$ makes with the positive y-axis, to 1 decimal place.

(2)

WORKING

Q4.4.16 Vectors in mechanics ●●○ [3 marks] ☐ Todo ☐ Mastered

Two forces $\mathbf{F}_1 = (6\mathbf{i} - \mathbf{j} + 6\mathbf{k})$ N and $\mathbf{F}_2 = (3\mathbf{i} - \mathbf{j} - 8\mathbf{k})$ N act on a particle of mass 3 kg. Find the magnitude of the acceleration of the particle.

(3)

WORKING

Q4.4.17 Geometric problems with vectors ●●○ [3 marks] ☐ Todo ☐ Mastered

$ABCD$ is a parallelogram with $A(4, 4, -4)$, $B(-3, -5, 4)$ and $C(-1, 4, -4)$. Find the coordinates of D.

(3)

WORKING

Q4.4.18 Angles with axes ●●○ [2 marks] ☐ Todo ☐ Mastered

Find the angle that the vector $-4\mathbf{i} + 4\mathbf{j} - 6\mathbf{k}$ makes with the positive y-axis, to 1 decimal place.

(2)

WORKING

MY NOTES — 4.4 VECTORS (3D): MISTAKES TO REMEMBER

Answers and mark schemes

M = method mark, A = accuracy mark (depends on the preceding M), B = independent mark, ft = follow-through, * = given answer. Numerical answers are to 3 s.f. unless stated.

4.1 Differentiation

Q4.1.01 **Exponentials and logarithms · 3 marks**

$\frac{dy}{dx} = 8^x \ln 8 + 12e^{3x} + \frac{1}{x}$

B1 exponential term · B1 log term · B1 remaining term

Q4.1.02 **Quotient rule · 3 marks**

$\frac{18}{(x+5)^2}$

M1 $\frac{vu' - uv'}{v^2}$ · A1 correct unsimplified · A1 simplified

Q4.1.03 **Product rule · 3 marks**

$-2(x-4)^2 \sin 2x + (2x-8)\cos 2x$

M1 $u'v + uv'$ form · A1 one term · A1 fully correct

Q4.1.04 **Tangents and normals · 4 marks**

$y = 19x - 33$

M1 differentiates · A1 gradient 19 · M1 tangent and point · A1

Q4.1.05 **Chain rule · 2 marks**

$\frac{3x^2}{2\sqrt{x^3-3}}$

M1 chain rule form · A1

Q4.1.06 **Exponentials and logarithms · 3 marks**

$\frac{dy}{dx} = 5^x \ln 5 + 2e^{2x} - \frac{4}{x}$

B1 exponential term · B1 log term · B1 remaining term

Q4.1.07 **Quotient rule · 3 marks**

$\frac{-x^2e^x + 2xe^x + 2x - 3e^x}{(e^x+1)^2}$

M1 $\frac{vu' - uv'}{v^2}$ · A1 correct unsimplified · A1 simplified

Q4.1.08 **Product rule · 3 marks**

$2(x+1)^2 \cos 2x + (2x+2)\sin 2x$

M1 $u'v + uv'$ form · A1 one term · A1 fully correct

Q4.1.09 **Tangents and normals · 4 marks**

$y = \frac{845}{31} - \frac{2x}{31}$

M1 differentiates · A1 gradient $\frac{31}{2}$ · M1 uses $-1/m$ and point · A1

Q4.1.10 **Chain rule · 2 marks**

$\frac{2}{2x-2}$

M1 chain rule form · A1

Q4.1.11 **Exponentials and logarithms · 3 marks**

$\frac{dy}{dx} = 12e^{4x} - \frac{3}{x}$

B1 exponential term · B1 log term · B1 remaining term

Q4.1.12 **Quotient rule · 3 marks**

$\frac{-4xe^x + 2e^x + 4}{(e^x+1)^2}$

M1 $\frac{vu' - uv'}{v^2}$ · A1 correct unsimplified · A1 simplified

Q4.1.13 **Product rule · 3 marks**

$-2x^3 \sin 2x + 3x^2 \cos 2x$

M1 $u'v + uv'$ form · A1 one term · A1 fully correct

Q4.1.14 **Tangents and normals · 4 marks**

$y = 19x - 47$

M1 differentiates · A1 gradient 19 · M1 tangent and point · A1

Q4.1.15 **Chain rule · 2 marks**

$18(3x-4)^5$

M1 chain rule form · A1

Q4.1.16 **Second derivatives, stationary points, convex / concave, inflection · 8 marks**

(a) $(1,\ -10)$ minimum; $(3,\ -6)$ maximum

M1 differentiates · M1 sets $\frac{dy}{dx}=0$ and solves · A1 x values · A1 y values · M1 second derivative test · A1 natures

(b) $x>2$

M1 $\frac{d^2y}{dx^2}<0$ · A1

Q4.1.17 **Connected rates of change · 4 marks**

0.00955 cm s^{-1}

B1 $\frac{dV}{dr}=4\pi r^2$ · M1 chain rule $\frac{dr}{dt}=\frac{dV}{dt}\div\frac{dV}{dr}$ · M1 substitutes · A1

Q4.1.18 **Parametric differentiation · 4 marks**

$\frac{dy}{dx}=1-\frac{1}{3t^2}$; at $t=-1$: $\frac{2}{3}$

B1 $\frac{dx}{dt}$ · B1 $\frac{dy}{dt}$ · M1 divides correctly · A1 value

Q4.1.19 **Implicit differentiation · 5 marks**

(a) $\frac{dy}{dx}=\frac{2x-3y}{3x-4y+3}$

M1 differentiates y terms with $\frac{dy}{dx}$ · B1 product rule on xy · M1 collects $\frac{dy}{dx}$ · A1

(b) 0

B1ft

Q4.1.20 **Differentiation from first principles · 4 marks**

$\lim_{h\to0}\frac{f(x+h)-f(x)}{h}=\lim_{h\to0}(2x-6+h)=2x-6$

B1 $f(x+h)$ expanded correctly · M1 $\frac{f(x+h)-f(x)}{h}$ simplified · M1 limit as $h\to0$ taken with correct notation · A1* complete proof

Q4.1.21 **Second derivatives, stationary points, convex / concave, inflection · 8 marks**

(a) $(-4,\ -86)$ minimum; $(2,\ 22)$ maximum

M1 differentiates · M1 sets $\frac{dy}{dx}=0$ and solves · A1 x values · A1 y values · M1 second derivative test · A1 natures

(b) $x>-1$

M1 $\frac{d^2y}{dx^2}<0$ · A1

Q4.1.22 **Connected rates of change · 4 marks**

0.00249 cm s^{-1}

B1 $\frac{dV}{dr}=4\pi r^2$ · M1 chain rule $\frac{dr}{dt}=\frac{dV}{dt}\div\frac{dV}{dr}$ · M1 substitutes · A1

Q4.1.23 **Parametric differentiation · 4 marks**

$\frac{dy}{dx}=\frac{2}{\cos t}$; at $t=\frac{\pi}{3}$: 4

B1 $\frac{dx}{dt}$ · B1 $\frac{dy}{dt}$ · M1 divides correctly · A1 value

Q4.1.24 **Implicit differentiation · 7 marks**

(a) $\frac{dy}{dx}=\frac{x-y}{x-y+1}$

M1 differentiates y terms with $\frac{dy}{dx}$ · B1 product rule on xy · M1 collects $\frac{dy}{dx}$ · A1

(b) 2

B1ft

(c) $y=-\frac{x}{2}-\frac{5}{2}$

M1 normal gradient and point · A1

Q4.1.25 **Differentiation from first principles · 5 marks**

$\frac{-3(3x^2h+3xh^2+h^3)}{h}\to-9x^2$ as $h\to0$

B1 $(x+h)^3$ expanded · M1 difference quotient · M1 divides by h · M1 limit · A1*

Q4.1.26 **Second derivatives, stationary points, convex / concave, inflection · 8 marks**

(a) $(1,\ \frac{15}{2})$ maximum; $(4,\ -6)$ minimum

M1 differentiates · M1 sets $\frac{dy}{dx}=0$ and solves · A1 x values · A1 y values · M1 second derivative test · A1 natures

(b) $x<\frac{5}{2}$

M1 $\frac{d^2y}{dx^2}<0$ · A1

Q4.1.27 **Connected rates of change · 3 marks**

$7.54\text{ m}^2\text{ s}^{-1}$

B1 $\frac{dA}{dr}=2\pi r$ · M1 · A1

Q4.1.28 **Parametric differentiation · 4 marks**

$\frac{dy}{dx}=4e^{-t}$; at $t=1$: $\frac{4}{e}$

B1 $\frac{dx}{dt}$ · B1 $\frac{dy}{dt}$ · M1 divides correctly · A1 value

Q4.1.29 **Implicit differentiation · 7 marks**

(a) $\frac{dy}{dx}=\frac{-2x-y}{x+6y-2}$

M1 differentiates y terms with $\frac{dy}{dx}$ · B1 product rule on xy · M1 collects $\frac{dy}{dx}$ · A1

(b) $-\frac{5}{21}$

B1ft

(c) $y=\frac{21x}{5}+\frac{6}{5}$

M1 normal gradient and point · A1

Q4.1.30 Differentiation from first principles · 4 marks

$\lim_{h\to 0}\frac{f(x+h)-f(x)}{h} = \lim_{h\to 0}(5-4x-2h) = 5-4x$

B1 $f(x+h)$ expanded correctly · M1 $\frac{f(x+h)-f(x)}{h}$ simplified · M1 limit as $h\to 0$ taken with correct notation · A1* complete proof

Q4.1.31 Second derivatives, stationary points, convex / concave, inflection · 8 marks

(a) $(-2, 7)$ maximum; $(1, -\frac{13}{2})$ minimum

M1 differentiates · M1 sets $\frac{dy}{dx}=0$ and solves · A1 x values · A1 y values · M1 second derivative test · A1 natures

(b) $x < -\frac{1}{2}$

M1 $\frac{d^2y}{dx^2} < 0$ · A1

Q4.1.32 Connected rates of change · 4 marks

0.0292 cm s^{-1}

B1 $\frac{dV}{dr} = 4\pi r^2$ · M1 chain rule $\frac{dr}{dt} = \frac{dV}{dt} \div \frac{dV}{dr}$ · M1 substitutes · A1

Q4.1.33 Derivatives of sin x and cos x · 5 marks

uses the addition formula, splits into the two standard limits

B1 uses $\cos(A+B)$ expansion · M1 difference quotient · M1 groups terms with $\sin 3h$ and $\cos 3h - 1$ · M1 applies the limits · A1* complete

Q4.1.34 Optimisation · 9 marks

(a) $h = \frac{2000}{\pi r^2}$ substituted into $2\pi r^2 + 2\pi rh$

B1 volume equation · M1 substitutes h · A1*

(b) $r = 6.83$, $A = 879\text{ cm}^2$

M1 differentiates · M1 sets $=0$ · A1 r · A1 A

(c) $\frac{d^2A}{dr^2} = 4\pi + \frac{8000}{r^3} > 0$

M1 · A1

Q4.1.35 Optimisation · 7 marks

(a) base side $12-2x$, height x

M1 · A1*

(b) $x = 2$, $V = 128\text{ cm}^3$

M1 expands and differentiates · A1 · M1 solves $V' = 0$ (rejects $x=6$) · A1 x · A1 V

Q4.1.36 Derivatives of sin x and cos x · 5 marks

uses the addition formula, splits into the two standard limits

B1 uses $\sin(A+B)$ expansion · M1 difference quotient · M1 groups terms with $\sin h$ and $\cos h - 1$ · M1 applies the limits · A1* complete

4.2 Integration

Q4.2.01 Standard integrals · 3 marks

$-2x^3 - 3x - \frac{3}{x} + c$

M1 raises a power by 1 · A1 two terms · A1 all terms and $+c$

Q4.2.02 Trapezium rule · 4 marks

$x=0$: $y=10.0000$, $x=\frac{1}{2}$: $y=7.7880$, $x=1$: $y=3.6788$, $x=\frac{3}{2}$: $y=1.0540$, $x=2$: $y=0.1832$; estimate 8.81 (underestimate; true value 8.821)

B1 $h=\frac{1}{2}$ · B1 y values · M1 $\frac{h}{2}[y_0+y_n+2(\ldots)]$ · A1

Q4.2.03 Integration as the limit of a sum · 3 marks

$\int_0^4 \sqrt{x}\,dx = \frac{16}{3}$

B1 integral form · M1 integrates · A1

Q4.2.04 Integrating f(ax + b) · 2 marks

$\frac{(4x-1)^6}{24} + c$

M1 correct form, divides by 4 · A1

Q4.2.05 Standard integrals · 3 marks

$\frac{2x^3}{3} - x - \frac{4}{x} + c$

M1 raises a power by 1 · A1 two terms · A1 all terms and $+c$

Q4.2.06 Trapezium rule · 4 marks

$x=0$: $y=10.0000$, $x=\frac{1}{2}$: $y=7.7880$, $x=1$: $y=3.6788$, $x=\frac{3}{2}$: $y=1.0540$, $x=2$: $y=0.1832$, $x=\frac{5}{2}$: $y=0.0193$; estimate 8.86 (underestimate; true value 8.859)

B1 $h=\frac{1}{2}$ · B1 y values · M1 $\frac{h}{2}[y_0+y_n+2(\ldots)]$ · A1

Q4.2.07 Integration as the limit of a sum · 3 marks

$\int_2^5 x^2\,dx = 39$

B1 integral form · M1 integrates · A1

Q4.2.08 **Integrating f(ax + b) · 2 marks**

$\frac{(4x-7)^4}{16}+c$

M1 correct form, divides by 4 · A1

Q4.2.09 **Standard integrals · 4 marks**

$\ln 2+\frac{53}{3}$

M1 integrates · A1 · M1 substitutes limits · A1 exact

Q4.2.10 **Trapezium rule · 4 marks**

$x=0$: $y=0.0000$, $x=\frac{1}{2}$: $y=0.6124$, $x=1$: $y=1.4142$, $x=\frac{3}{2}$: $y=2.3717$, $x=2$: $y=3.4641$, $x=\frac{5}{2}$: $y=4.6771$, $x=3$: $y=6.0000$; estimate 7.77 (overestimate; true value 7.733)

B1 $h=\frac{1}{2}$ · B1 y values · M1 $\frac{h}{2}[y_0+y_n+2(\ldots)]$ · A1

Q4.2.11 **Integration as the limit of a sum · 3 marks**

$\int_0^4 x^2\,dx=\frac{64}{3}$

B1 integral form · M1 integrates · A1

Q4.2.12 **Integrating f(ax + b) · 2 marks**

$\frac{\ln|2x+5|}{2}+c$

M1 correct form, divides by 2 · A1

Q4.2.13 **Areas under curves and between curves · 6 marks**

(a) $x=-3$, $x=1$

M1 equates · A1

(b) $\frac{32}{3}$

M1 $\int(\text{line}-\text{curve})\,dx$ · M1 integrates · M1 limits · A1

Q4.2.14 **Reverse chain rule · 3 marks**

$\ln(x^2+x+2)+c$

M1 recognises $f'(x)g(f(x))$ form · M1 adjusts constant · A1

Q4.2.15 **Modelling with differential equations · 2 marks**

$\frac{d\theta}{dt}=-k(\theta-20)$, $k>0$ a constant

M1 correct derivative and proportionality · A1 sign and constant defined

Q4.2.16 **Integration with partial fractions · 5 marks**

$\ln 4$

M1 partial fractions · A1 · M1 integrates to $\ln$ terms · M1 limits · A1

Q4.2.17 **Using trig identities in integration · 3 marks**

$-x+\tan x+c$

M1 uses $\tan^2 A=\sec^2 A-1$ · M1 integrates · A1

Q4.2.18 **Separable differential equations · 4 marks**

$y=\sqrt{\frac{2x^3}{3}+1}$

M1 $\int y\,dy=\int 1x^2\,dx$ · A1 $\frac{y^2}{2}=\frac{1x^3}{3}+c$ · M1 condition · A1

Q4.2.19 **Integration by parts · 4 marks**

$\frac{x\sin 2x}{2}+\frac{\cos 2x}{4}+c$

M1 $u=x$ · A1 correct first stage · M1 integrates second term · A1

Q4.2.20 **Areas under curves and between curves · 6 marks**

(a) $x=1$, $x=5$

M1 equates · A1

(b) $\frac{64}{3}$

M1 $\int(\text{line}-\text{curve})\,dx$ · M1 integrates · M1 limits · A1

Q4.2.21 **Reverse chain rule · 3 marks**

$\frac{(x^2+4)^7}{14}+c$

M1 recognises $f'(x)g(f(x))$ form · M1 adjusts constant · A1

Q4.2.22 **Modelling with differential equations · 2 marks**

$\frac{dV}{dt}=-kV^{\frac{2}{3}}$, $k>0$ a constant

M1 correct derivative and proportionality · A1 sign and constant defined

Q4.2.23 **Integration with partial fractions · 5 marks**

$-\ln 72$

M1 partial fractions · A1 · M1 integrates to $\ln$ terms · M1 limits · A1

Q4.2.24 **Using trig identities in integration · 3 marks**

$-x+\frac{\tan 3x}{3}+c$

M1 uses $\tan^2 A=\sec^2 A-1$ · M1 integrates · A1

Q4.2.25 **Separable differential equations · 5 marks**

(a) separate, integrate to $\ln(\theta - 21) = -0.07t + c$, use $t = 0$

M1 separates · A1 · M1 condition · A1*

(b) 42.6 ℃

B1

Q4.2.26 **Integration by parts · 4 marks**

$\frac{(3x-1)e^{3x}}{9} + c$

M1 $u = x$ · A1 correct first stage · M1 integrates second term · A1

Q4.2.27 **Areas under curves and between curves · 6 marks**

(a) $x = 2$, $x = 5$

M1 equates · A1

(b) $\frac{9}{2}$

M1 $\int(\text{line} - \text{curve})\,dx$ · M1 integrates · M1 limits · A1

Q4.2.28 **Reverse chain rule · 3 marks**

$\frac{(x^2+3)^4}{8} + c$

M1 recognises $f'(x)g(f(x))$ form · M1 adjusts constant · A1

Q4.2.29 **Integration with partial fractions · 5 marks**

$\ln\left(\frac{256}{8575}\right)$

M1 partial fractions · A1 · M1 integrates to ln terms · M1 limits · A1

Q4.2.30 **Using trig identities in integration · 3 marks**

$-x + \frac{\tan 4x}{4} + c$

M1 uses $\tan^2 A = \sec^2 A - 1$ · M1 integrates · A1

Q4.2.31 **Separable differential equations · 5 marks**

$y = 3e^{x^2}$

M1 separates $\int \frac{1}{y}\,dy = \int 2x\,dx$ · A1 $\ln y = x^2 + c$ · M1 uses condition · M1 exponentiates · A1

Q4.2.32 **Integration by parts · 4 marks**

$-\frac{x\cos 3x}{3} + \frac{\sin 3x}{9} + c$

M1 $u = x$ · A1 correct first stage · M1 integrates second term · A1

Q4.2.33 **Areas under curves and between curves · 6 marks**

(a) $x = -3$, $x = 0$

M1 equates · A1

(b) $\frac{9}{2}$

M1 $\int(\text{line} - \text{curve})\,dx$ · M1 integrates · M1 limits · A1

Q4.2.34 **Integration by substitution · 4 marks**

$-\frac{1}{2x^2+4} + c$

B1 $du = 2x\,dx$ · M1 $\frac{1}{2}\int u^{-2}\,du$ · M1 integrates · A1 in terms of x

Q4.2.35 **Integration by substitution · 6 marks**

$\frac{34}{15} + \frac{256\sqrt{3}}{5}$

B1 $dx = du$, new limits · M1 $\int(u-4)u^{\frac{1}{2}}\,du$ · M1 integrates · A1 · M1 limits · A1 exact

Q4.2.36 **Integration by substitution · 6 marks**

$\frac{176\sqrt{3}}{5}$

B1 $dx = du$, new limits · M1 $\int(u-5)u^{\frac{1}{2}}\,du$ · M1 integrates · A1 · M1 limits · A1 exact

4.3 Numerical Methods

Q4.3.01 **Locating roots by change of sign · 2 marks**

$f(-1) = -2$, $f(0) = 1$; sign change and f is continuous, so a root lies in the interval

M1 evaluates both · A1 sign change, continuity, conclusion

Q4.3.02 **Locating roots by change of sign · 2 marks**

$f(-6) = -29$, $f(-5) = 12$; sign change and f is continuous, so a root lies in the interval

M1 evaluates both · A1 sign change, continuity, conclusion

Q4.3.03 **Locating roots by change of sign · 2 marks**

$f(-1) = -8$, $f(0) = 2$; sign change and f is continuous, so a root lies in the interval

M1 evaluates both · A1 sign change, continuity, conclusion

Q4.3.04 **Locating roots by change of sign · 2 marks**

$f(1) = -5$, $f(2) = 7$; sign change and f is continuous, so a root lies in the interval

M1 evaluates both · A1 sign change, continuity, conclusion

Q4.3.05 **When change of sign fails · 2 marks**
repeated root at $x = -2$: no sign change; $\frac{1}{x - -2}$ changes sign across an asymptote, not a root
B1 · B1

Q4.3.06 **Staircase and cobweb diagrams · 2 marks**
diverges ($|g'(\alpha)| > 1$); cobweb diagram
B1 convergence with reason · B1 shape

Q4.3.07 **Fixed-point iteration x = g(x) · 6 marks**
(a) rearranges
B1*
(b) $x_1 = 2.1544$, $x_2 = 2.1979$, $x_3 = 2.2098$
M1 · A1 x_1 · A1 x_2, x_3
(c) $f(2.2138)$ and $f(2.2148)$ have opposite signs
M1 interval bounds · A1 conclusion

Q4.3.08 **Newton-Raphson method · 4 marks**
$x_1 = -2.2581$, $x_2 = -2.215$
B1 $f'(x) = 3x^2 + 6x + 4$ · M1 $x_0 - \frac{f(x_0)}{f'(x_0)}$ · A1 x_1 · A1 x_2

Q4.3.09 **When change of sign fails · 2 marks**
repeated root at $x = 4$: no sign change; $\frac{1}{x - 4}$ changes sign across an asymptote, not a root
B1 · B1

Q4.3.10 **Staircase and cobweb diagrams · 2 marks**
converges ($|g'(\alpha)| < 1$); cobweb diagram
B1 convergence with reason · B1 shape

Q4.3.11 **Fixed-point iteration x = g(x) · 6 marks**
(a) rearranges
B1*
(b) $x_1 = 2.2240$, $x_2 = 2.2828$, $x_3 = 2.2977$
M1 · A1 x_1 · A1 x_2, x_3
(c) $f(2.3023)$ and $f(2.3033)$ have opposite signs
M1 interval bounds · A1 conclusion

Q4.3.12 **Newton-Raphson method · 4 marks**
$x_1 = -4.4458$, $x_2 = -4.444$
B1 $f'(x) = 3x^2 + 8x - 4$ · M1 $x_0 - \frac{f(x_0)}{f'(x_0)}$ · A1 x_1 · A1 x_2

Q4.3.13 **When change of sign fails · 2 marks**
repeated root at $x = -4$: no sign change; $\frac{1}{x - -4}$ changes sign across an asymptote, not a root
B1 · B1

Q4.3.14 **Fixed-point iteration x = g(x) · 6 marks**
(a) rearranges
B1*
(b) $x_1 = 2.8020$, $x_2 = 2.7594$, $x_3 = 2.7500$
M1 · A1 x_1 · A1 x_2, x_3
(c) $f(2.7468)$ and $f(2.7478)$ have opposite signs
M1 interval bounds · A1 conclusion

Q4.3.15 **Newton-Raphson method · 4 marks**
$x_1 = -0.3704$, $x_2 = -0.364$
B1 $f'(x) = 3x^2 - 2x + 5$ · M1 $x_0 - \frac{f(x_0)}{f'(x_0)}$ · A1 x_1 · A1 x_2

Q4.3.16 **Applications to modelling · 4 marks**
$t = 7.56$
B1 $f'(t) = 0.5e^{0.5t} + 2$ · M1 applies the formula · A1 first iterate · A1 converged value

Q4.3.17 **Applications to modelling · 4 marks**
$t = 6.23$
B1 $f'(t) = 0.5e^{0.5t} + 2$ · M1 applies the formula · A1 first iterate · A1 converged value

Q4.3.18 **Applications to modelling · 4 marks**
$t = 6.64$
B1 $f'(t) = 0.5e^{0.5t} + 2$ · M1 applies the formula · A1 first iterate · A1 converged value

4.4 Vectors (3D)

Q4.4.01 **Position vectors and distance · 4 marks**

(a) $3\mathbf{i}+2\mathbf{j}+\mathbf{k}$

B1

(b) $\sqrt{14}$

M1 · A1

(c) $\frac{1}{2}(3\mathbf{i}-\mathbf{k})$

B1

Q4.4.02 **3D coordinates and i, j, k notation · 3 marks**

$\overrightarrow{AC}=3\overrightarrow{AB}$ with common point A; $AB:BC=1:2$

M1 finds two vectors · A1 scalar multiple + common point · B1 ratio

Q4.4.03 **Magnitude and direction · 2 marks**

$|\mathbf{a}|=3$; $\frac{1}{3}(-\mathbf{i}-2\mathbf{j}+2\mathbf{k})$

B1 · B1ft

Q4.4.04 **Position vectors and distance · 4 marks**

(a) $-\mathbf{i}-\mathbf{j}+10\mathbf{k}$

B1

(b) $\sqrt{102}$

M1 · A1

(c) $\frac{1}{2}(5\mathbf{i}+11\mathbf{j}-2\mathbf{k})$

B1

Q4.4.05 **3D coordinates and i, j, k notation · 3 marks**

$\overrightarrow{AC}=4\overrightarrow{AB}$ with common point A; $AB:BC=1:3$

M1 finds two vectors · A1 scalar multiple + common point · B1 ratio

Q4.4.06 **Magnitude and direction · 2 marks**

$|\mathbf{a}|=9$; $\frac{1}{9}(-\mathbf{i}+4\mathbf{j}-8\mathbf{k})$

B1 · B1ft

Q4.4.07 **Position vectors and distance · 4 marks**

(a) $-4\mathbf{i}-3\mathbf{j}$

B1

(b) 5

M1 · A1

(c) $\frac{1}{2}(-8\mathbf{i}-5\mathbf{j}+4\mathbf{k})$

B1

Q4.4.08 **3D coordinates and i, j, k notation · 3 marks**

$\overrightarrow{AC}=2\overrightarrow{AB}$ with common point A; $AB:BC=1:1$

M1 finds two vectors · A1 scalar multiple + common point · B1 ratio

Q4.4.09 **Magnitude and direction · 2 marks**

$|\mathbf{a}|=13$; $\frac{1}{13}(3\mathbf{i}+4\mathbf{j}+12\mathbf{k})$

B1 · B1ft

Q4.4.10 **Vectors in mechanics · 3 marks**

4.39 m s^{-2}

M1 resultant force · M1 $\mathbf{F}=m\mathbf{a}$ · A1

Q4.4.11 **Geometric problems with vectors · 3 marks**

$D(-5,\,1,\,3)$

M1 $\overrightarrow{AD}=\overrightarrow{BC}$ · M1 $\overrightarrow{OD}=\overrightarrow{OA}+\overrightarrow{BC}$ · A1

Q4.4.12 **Angles with axes · 2 marks**

115.9°

M1 $\cos\theta=\frac{-2}{\sqrt{21}}$ · A1

Q4.4.13 **Vectors in mechanics · 3 marks**

2.67 m s^{-2}

M1 resultant force · M1 $\mathbf{F}=m\mathbf{a}$ · A1

Q4.4.14 **Geometric problems with vectors · 3 marks**

$D(-2,\,-6,\,0)$

M1 $\overrightarrow{AD}=\overrightarrow{BC}$ · M1 $\overrightarrow{OD}=\overrightarrow{OA}+\overrightarrow{BC}$ · A1

Q4.4.15 **Angles with axes · 2 marks**

144.2°

M1 $\cos\theta=\frac{-5}{\sqrt{38}}$ · A1

Q4.4.16 **Vectors in mechanics · 3 marks**

3.14 m s^{-2}

M1 resultant force · M1 $\mathbf{F} = m\mathbf{a}$ · A1

Q4.4.17 **Geometric problems with vectors · 3 marks**

$D(6, 13, -12)$

M1 $\overrightarrow{AD} = \overrightarrow{BC}$ · M1 $\overrightarrow{OD} = \overrightarrow{OA} + \overrightarrow{BC}$ · A1

Q4.4.18 **Angles with axes · 2 marks**

61.0°

M1 $\cos\theta = \frac{4}{\sqrt{68}}$ · A1

Pearson Edexcel A-Level Mathematics (9MA0)

Module 5

Applied: Statistics & Mechanics

Paper 3 · 69 practice questions · answers and mark schemes at the back

Subtopic	Questions	Done
5.1 Statistics	36	____ / 36
5.2 Mechanics	33	____ / 33

How to use this workbook

1. Read the key facts and watch one video for the subtopic. Tick the skills checklist as you go.
2. Attempt each question in the squared working space without notes. Mark it with the scheme at the back.
3. Tick **Todo** when you have attempted it and **Mastered** only when you can redo it cold a week later.
4. Copy any mistakes into the notes box at the end of the subtopic. Rebuild the workbook with a new seed for fresh questions.

Name _____ Started _____ Built 14 Oct 2026, seed 2026

5.1 Statistics

Statistics (Paper 3 Section A): sampling, data presentation, the large data set, probability, binomial and normal distributions, regression/correlation and hypothesis testing.

Key facts

- $X \sim \mathrm{B}(n, p)$: $\mathrm{P}(X = r) = \binom{n}{r} p^r (1-p)^{n-r}$, mean np, variance $np(1-p)$
- $X \sim \mathrm{N}(\mu, \sigma^2)$: $Z = \frac{X-\mu}{\sigma} \sim \mathrm{N}(0, 1)$
- Normal approximation to $\mathrm{B}(n, p)$: $\mathrm{N}(np, np(1-p))$ with continuity correction
- $\bar{X} \sim \mathrm{N}\left(\mu, \frac{\sigma^2}{n}\right)$ for a sample mean
- $\mathrm{P}(A \mid B) = \frac{\mathrm{P}(A \cap B)}{\mathrm{P}(B)}$; independent iff $\mathrm{P}(A \cap B) = \mathrm{P}(A)\mathrm{P}(B)$
- $\sigma^2 = \frac{\sum x^2}{n} - \bar{x}^2$; outlier rule: beyond $Q_1 - 1.5\,\mathrm{IQR}$ or $Q_3 + 1.5\,\mathrm{IQR}$

Method

- Hypothesis test template: hypotheses (parameter!), distribution under H_0, probability or critical region, compare, conclusion in context.
- Critical region: find the largest/smallest value whose tail probability is below the significance level.
- Histograms: frequency density = frequency ÷ class width; area represents frequency.

Exam traps

- Writing hypotheses with $\bar{x}$ or X instead of p, μ or ρ.
- Conclusions with no context or that 'prove' something.
- Forgetting the continuity correction.
- Not knowing the large data set (stations, variables, 'tr', 'n/a').

In the exam: Paper 3 Section A is 50 marks: around 5 questions mixing the topics above.

Specification checklist

Skill	Learnt	Practised	Mastered
Populations and sampling methods	☐	☐	☐
Measures of location and spread, coding	☐	☐	☐
Histograms, box plots, cumulative frequency, outliers	☐	☐	☐
Correlation and regression	☐	☐	☐
The large data set	☐	☐	☐
Probability: Venn diagrams and set notation	☐	☐	☐
Conditional probability and tree diagrams	☐	☐	☐
Binomial distribution	☐	☐	☐
Hypothesis testing: critical regions and p-values	☐	☐	☐
Normal distribution	☐	☐	☐
Normal approximation to binomial	☐	☐	☐
Hypothesis testing: normal mean and correlation	☐	☐	☐

Videos (click to open)

☐ **Recap** — Probability (Year 1) in less than 20 minutes • A-Level Maths, Statistics Year 1, Chapter 5 (Bicen Maths)
☐ **Recap** — Statistical Distributions (+ binomial) in 29 minutes • A-Level Maths, Statistics Year 1, Chapter 6 (Bicen Maths)
☐ **Recap** — Hypothesis Testing (Binomial) in 24 minutes • A-Level Maths, Statistics Year 1, Chapter 7 (Bicen Maths)
☐ Populations and sampling methods — Sampling: Population vs Sample (TLMaths)
☐ Measures of location and spread, coding — Data: Introducing the Variance and Standard Deviation (TLMaths)
☐ Histograms, box plots, cumulative frequency, outliers — Data: Histograms (TLMaths)
☐ Correlation and regression — Scatter Graphs: The Product Moment Correlation Coefficient (TLMaths)
☐ The large data set — Edexcel Large Data Set - Weather (TLMaths)
☐ Probability: Venn diagrams and set notation — Probability: Venn Diagrams - Union and Intersection (TLMaths)
☐ Conditional probability and tree diagrams — Probability: Introducing Conditional Probability (TLMaths)
☐ Binomial distribution — Binomial Distribution: Introducing the Binomial Distribution (TLMaths)
☐ Hypothesis testing: critical regions and p-values — Binomial Hypothesis Testing: Finding the Critical Region (TLMaths)

☐ Normal distribution — Normal Distribution: Introducing the Bell Curve (TLMaths)
☐ Normal approximation to binomial — Normal Distribution: Approximating a Binomial Distribution (TLMaths)
☐ Hypothesis testing: normal mean and correlation — Introducing Sample Means Hypothesis Testing (TLMaths)

Q5.1.01 Histograms, box plots, cumulative frequency, outliers ●○○ [4 marks] ☐ Todo ☐ Mastered

The times, t minutes, taken by some students are grouped as: $0 \le t < 10$: 5, $10 \le t < 20$: 13, $20 \le t < 30$: 16, $30 \le t < 50$: 22, $50 \le t < 80$: 13.

(a) Find the frequency density of each class for a histogram. **(2)**

(b) For a different data set, $Q_1 = 29$ and $Q_3 = 54$. Outliers are values more than $1.5 \times$ IQR beyond the quartiles. Find the outlier boundaries. **(2)**

WORKING

Q5.1.02 Measures of location and spread, coding ●○○ [5 marks] ☐ Todo ☐ Mastered

For 17 values of x: $\sum x = 635$ and $\sum x^2 = 27329$.

(a) Find the mean and standard deviation of x. **(3)**

(b) The data are coded using $y = \frac{x-5}{10}$. Find the mean and standard deviation of y. **(2)**

WORKING

Q5.1.03 The large data set ●○○ [2 marks] ☐ Todo ☐ Mastered

Explain why data from the large data set for Perth (Australia) in May might not be suitable for estimating UK summer rainfall.

(2)

WORKING

Q5.1.04 Binomial distribution ●○○ [7 marks] ☐ Todo ☐ Mastered

$X \sim \mathrm{B}(10, 0.35)$. Find

(a) $\mathrm{P}(X = 3)$, (b) $\mathrm{P}(X \leq 6)$, (c) $\mathrm{P}(X > 3)$, (d) $\mathrm{P}(3 < X \leq 6)$.

(a) **(2)**

(b) **(1)**

(c) **(2)**

(d) **(2)**

WORKING

Q5.1.05 Probability: Venn diagrams and set notation ●○○ [9 marks] ☐ Todo ☐ Mastered

In a group of 50 students, 19 study Art, 13 study Biology and 26 study neither.

(a) Draw a Venn diagram. **(3)**

(b) Find $\mathrm{P}(A \cap B)$, $\mathrm{P}(A \cup B')$ and $\mathrm{P}(A \mid B)$. **(4)**

(c) Determine whether A and B are independent. **(2)**

WORKING

Q5.1.06 Populations and sampling methods ●○○ [2 marks] ☐ Todo ☐ Mastered

Give one advantage and one disadvantage of quota sampling compared with stratified sampling.

(2)

WORKING

Q5.1.07 Correlation and regression ●○○ [3 marks] ☐ Todo ☐ Mastered

The regression line of y, ice creams sold, on x, daily mean temperature (°C), is $y = 15.4 + 0.31x$.

(a) Interpret the gradient in context. **(1)**

(b) Estimate y when $x = 26$ and comment on reliability if the data ranged from 2 to 20. **(2)**

WORKING

Q5.1.08 Histograms, box plots, cumulative frequency, outliers ●○○ [4 marks] ☐ Todo ☐ Mastered

The times, t minutes, taken by some students are grouped as: $0 \le t < 10$: 14, $10 \le t < 20$: 11, $20 \le t < 30$: 29, $30 \le t < 50$: 11, $50 \le t < 80$: 26.

(a) Find the frequency density of each class for a histogram. **(2)**

(b) For a different data set, $Q_1 = 23$ and $Q_3 = 41$. Outliers are values more than $1.5 \times$ IQR beyond the quartiles. Find the outlier boundaries. **(2)**

WORKING

Q5.1.09 Measures of location and spread, coding ●○○ [5 marks] ☐ Todo ☐ Mastered

For 23 values of x: $\sum x = 828$ and $\sum x^2 = 35852$.

(a) Find the mean and standard deviation of x. **(3)**

(b) The data are coded using $y = \frac{x - 8}{10}$. Find the mean and standard deviation of y. **(2)**

WORKING

Q5.1.10 The large data set ●○○ [2 marks] ☐ Todo ☐ Mastered

The large data set records 'tr' for daily total rainfall. State what this means and the value used in calculations.

(2)

WORKING

Q5.1.11 Binomial distribution ●○○ [7 marks] ☐ Todo ☐ Mastered

$X \sim \mathrm{B}(14, 0.25)$. Find

(a) $\mathrm{P}(X = 3)$, (b) $\mathrm{P}(X \leq 3)$, (c) $\mathrm{P}(X > 3)$, (d) $\mathrm{P}(3 < X \leq 3)$.

(a) **(2)**

(b) **(1)**

(c) **(2)**

(d) **(2)**

WORKING

Q5.1.12 Probability: Venn diagrams and set notation ●○○ [9 marks] ☐ Todo ☐ Mastered

In a group of 60 students, 20 study Art, 19 study Biology and 28 study neither.

(a) Draw a Venn diagram. **(3)**

(b) Find $P(A \cap B)$, $P(A \cup B')$ and $P(A \mid B)$. **(4)**

(c) Determine whether A and B are independent. **(2)**

WORKING

Q5.1.13 Populations and sampling methods ●○○ [3 marks] ☐ Todo ☐ Mastered

Describe how to take a simple random sample of 40 students from a school of 900.

(3)

WORKING

Q5.1.14 Correlation and regression ●○○ [3 marks] ☐ Todo ☐ Mastered

The regression line of y, ice creams sold, on x, daily mean temperature (°C), is $y = 30.1 + 3.92x$.

(a) Interpret the gradient in context. **(1)**

(b) Estimate y when $x = 17$ and comment on reliability if the data ranged from 2 to 20. **(2)**

WORKING

Q5.1.15 Histograms, box plots, cumulative frequency, outliers ●○○ [4 marks] ☐ Todo ☐ Mastered

The times, t minutes, taken by some students are grouped as: $0 \le t < 10$: 12, $10 \le t < 20$: 8, $20 \le t < 30$: 25, $30 \le t < 50$: 13, $50 \le t < 80$: 28.

(a) Find the frequency density of each class for a histogram. **(2)**

(b) For a different data set, $Q_1 = 26$ and $Q_3 = 45$. Outliers are values more than $1.5 \times$ IQR beyond the quartiles. Find the outlier boundaries. **(2)**

WORKING

Q5.1.16 Measures of location and spread, coding ●○○ [5 marks] ☐ Todo ☐ Mastered

For 29 values of x: $\sum x = 1063$ and $\sum x^2 = 45661$.

(a) Find the mean and standard deviation of x. **(3)**

(b) The data are coded using $y = \frac{x - 10}{5}$. Find the mean and standard deviation of y. **(2)**

WORKING

Q5.1.17 The large data set ●○○ [2 marks] ☐ Todo ☐ Mastered

Using your knowledge of the large data set, explain why daily mean windspeed for Leeming may contain the entry 'n/a' and how you would clean such data.

(2)

WORKING

Q5.1.18 Binomial distribution ●○○ [7 marks] ☐ Todo ☐ Mastered

$X \sim \mathrm{B}(24, 0.5)$. Find

(a) $\mathrm{P}(X = 5)$, (b) $\mathrm{P}(X \leq 17)$, (c) $\mathrm{P}(X > 5)$, (d) $\mathrm{P}(5 < X \leq 17)$.

(a) **(2)**

(b) **(1)**

(c) **(2)**

(d) **(2)**

WORKING

Q5.1.19 Probability: Venn diagrams and set notation ●○○ [9 marks] ☐ Todo ☐ Mastered

In a group of 60 students, 23 study Art, 22 study Biology and 24 study neither.

(a) Draw a Venn diagram. **(3)**

(b) Find $\mathrm{P}(A \cap B)$, $\mathrm{P}(A \cup B')$ and $\mathrm{P}(A \mid B)$. **(4)**

(c) Determine whether A and B are independent. **(2)**

WORKING

Q5.1.20 Populations and sampling methods ●○○ [2 marks] ☐ Todo ☐ Mastered

Explain the difference between a census and a sample, giving one advantage of each.

(2)

WORKING

Q5.1.21 Correlation and regression ●○○ [3 marks] ☐ Todo ☐ Mastered

The regression line of y, test score, on x, hours of revision, is $y = 13.1 + 3.36x$.

(a) Interpret the gradient in context. **(1)**

(b) Estimate y when $x = 23$ and comment on reliability if the data ranged from 2 to 20. **(2)**

WORKING

Q5.1.22 Normal approximation to binomial ●●○ [5 marks] ☐ Todo ☐ Mastered

$X \sim \mathrm{B}(177, 0.4)$.

(a) Explain why a normal approximation is suitable. **(1)**

(b) Use it to estimate $\mathrm{P}(X \leq 72)$. **(4)**

WORKING

Q5.1.23 Hypothesis testing: critical regions and p-values ●●○ [5 marks] ☐ Todo ☐ Mastered

Historically, a seed company claims that 40% of its seeds produce red flowers. After a change, a random sample of 40 is taken and 13 are found to have the property. Test, at the 10% level, whether there is evidence that the proportion has decreased. State your hypotheses clearly.

(5)

WORKING

Q5.1.24 Conditional probability and tree diagrams ●●○ [6 marks] ☐ Todo ☐ Mastered

The probability that a person has a condition is $\frac{1}{3}$. If so, the probability that a person tests positive is $\frac{4}{5}$; otherwise it is $\frac{1}{5}$.

(a) Find the probability that a person tests positive. **(3)**

(b) Given that a person tests positive, find the probability that the person has the condition. **(3)**

WORKING

Q5.1.25 Normal distribution ●●○ [3 marks] ☐ Todo ☐ Mastered

$X \sim \mathrm{N}(88, 4^2)$. Find the value of a such that $\mathrm{P}(X < a) = 0.8$.

(3)

WORKING

Q5.1.26 Normal approximation to binomial ●●○ [5 marks] ☐ Todo ☐ Mastered

$X \sim \mathrm{B}(62, 0.5)$.

(a) Explain why a normal approximation is suitable. **(1)**

(b) Use it to estimate $\mathrm{P}(X \leq 36)$. **(4)**

WORKING

Q5.1.27 **Hypothesis testing: critical regions and p-values** ●●○ **[5 marks]** ☐ **Todo** ☐ **Mastered**

Historically, 50% of patients normally recover within a week. After a change, a random sample of 20 is taken and 5 are found to have the property. Test, at the 5% level, whether there is evidence that the proportion has decreased. State your hypotheses clearly.

(5)

WORKING

Q5.1.28 **Conditional probability and tree diagrams** ●●○ **[6 marks]** ☐ **Todo** ☐ **Mastered**

The probability that a person has a condition is $\frac{2}{5}$. If so, the probability that a person tests positive is $\frac{2}{3}$; otherwise it is $\frac{1}{4}$.

(a) Find the probability that a person tests positive. **(3)**

(b) Given that a person tests positive, find the probability that the person has the condition. **(3)**

WORKING

Q5.1.29 Normal approximation to binomial ●●○ [5 marks] ☐ Todo ☐ Mastered

$X \sim \mathrm{B}(65, 0.55)$.

(a) Explain why a normal approximation is suitable. **(1)**

(b) Use it to estimate $\mathrm{P}(X \leq 31)$. **(4)**

WORKING

Q5.1.30 Hypothesis testing: critical regions and p-values ●●○ [5 marks] ☐ Todo ☐ Mastered

Historically, a website claims that 20% of visitors make a purchase. After a change, a random sample of 20 is taken and 6 are found to have the property. Test, at the 10% level, whether there is evidence that the proportion has increased. State your hypotheses clearly.

(5)

WORKING

Q5.1.31 Conditional probability and tree diagrams ●●○ [6 marks] ☐ Todo ☐ Mastered

The probability that it rains on a given day is $\frac{3}{10}$. If so, the probability that Ben is late for school is $\frac{4}{5}$; otherwise it is $\frac{1}{10}$.

(a) Find the probability that Ben is late for school. **(3)**

(b) Given that Ben is late for school, find the probability that it rained. **(3)**

WORKING

Q5.1.32 Normal distribution ●●● [6 marks] ☐ Todo ☐ Mastered

$X \sim \mathrm{N}(\mu, \sigma^2)$. Given $\mathrm{P}(X < 114.4) = 0.2$ and $\mathrm{P}(X > 133.5) = 0.1$, find μ and σ.

(6)

WORKING

Q5.1.33 Hypothesis testing: normal mean and correlation ●●● [6 marks] ☐ Todo ☐ Mastered

The masses of items are normally distributed with standard deviation 17 g. The mean is claimed to be 90 g. A random sample of 36 items has mean 86.9 g. Test, at the 1% level, whether the mean is less than 90 g.

(6)

WORKING

Q5.1.34 Hypothesis testing: normal mean and correlation ●●● [6 marks] ☐ Todo ☐ Mastered

The masses of items are normally distributed with standard deviation 15 g. The mean is claimed to be 54 g. A random sample of 20 items has mean 46.0 g. Test, at the 5% level, whether the mean is less than 54 g.

(6)

WORKING

Q5.1.35 Normal distribution ●●● [6 marks] ☐ Todo ☐ Mastered

$X \sim \mathrm{N}(\mu, \sigma^2)$. Given $\mathrm{P}(X < 120.5) = 0.1$ and $\mathrm{P}(X > 139.6) = 0.2$, find μ and σ.

(6)

WORKING

Q5.1.36 Hypothesis testing: normal mean and correlation ●●● [6 marks] ☐ Todo ☐ Mastered

The masses of items are normally distributed with standard deviation 9 g. The mean is claimed to be 125 g. A random sample of 20 items has mean 128.3 g. Test, at the 5% level, whether the mean is greater than 125 g.

(6)

WORKING

MY NOTES — 5.1 STATISTICS: MISTAKES TO REMEMBER

5.2 Mechanics

Mechanics (Paper 3 Section B): modelling, kinematics (suvat, graphs, calculus, vectors), forces and Newton's laws, friction, connected particles, moments and projectiles.

Key facts

- $v = u + at,\ \ s = ut + \frac{1}{2}at^2,\ \ s = vt - \frac{1}{2}at^2,\ \ v^2 = u^2 + 2as,\ \ s = \frac{1}{2}(u+v)t$
- $v = \frac{dr}{dt},\ \ a = \frac{dv}{dt};\ \ r = \int v\,dt,\ \ v = \int a\,dt$
- $\mathbf{F} = m\mathbf{a}$; weight $= mg$ with $g = 9.8$ m s^{-2}
- Friction $F \le \mu R$; limiting equilibrium/moving: $F = \mu R$
- Moment = force $\times$ perpendicular distance; equilibrium: resultant force 0 and total moment 0
- Projectiles: horizontal $x = U\cos\alpha\, t$; vertical $y = U\sin\alpha\, t - \frac{1}{2}gt^2$

Method

- Draw a force diagram for every body; resolve parallel and perpendicular to the motion.
- Connected particles: write $F = ma$ for each particle separately (same a, same T for a light inextensible string).
- Moments: take moments about the point with the most unknown forces.

Exam traps

- Using $g = 9.81$ or 10 instead of 9.8.
- Giving too many significant figures after using $g = 9.8$ (2 or 3 s.f.).
- Forgetting that friction opposes the direction of (potential) motion.

In the exam: Paper 3 Section B is 50 marks; projectiles, connected particles and moments are each worth 8-12 marks.

Specification checklist

Skill	Learnt	Practised	Mastered
Quantities and units, modelling assumptions	☐	☐	☐
Constant acceleration: suvat	☐	☐	☐
Motion graphs	☐	☐	☐
Variable acceleration with calculus	☐	☐	☐
2D motion with vectors	☐	☐	☐
Newton's laws and resolving forces	☐	☐	☐
Friction	☐	☐	☐
Connected particles and pulleys	☐	☐	☐
Inclined planes	☐	☐	☐
Moments, equilibrium and centre of mass	☐	☐	☐
Projectiles	☐	☐	☐

Videos (click to open)

☐ **Recap** — Constant Acceleration (SUVAT) in 31 minutes • A-Level Maths, Mechanics Year 1, Chapter 9 (Bicen Maths)
☐ **Recap** — Variable Acceleration in less than 12 minutes • A-Level Maths, Mechanics Year 1, Chapter 11 (Bicen Maths)
☐ **Recap** — Forces & Motion (Year 1) in less than 37 minutes • A-Level Maths, Mechanics Year 1, Chapter 10 (Bicen Maths)
☐ Quantities and units, modelling assumptions — Introduction to Modelling (TLMaths)
☐ Constant acceleration: suvat — SUVAT: Deriving the Constant Acceleration Formulae (TLMaths)
☐ Motion graphs — Kinematics: Velocity / Time Graphs (TLMaths)
☐ Variable acceleration with calculus — variable acceleration - calculus methods Example (1) (ExamSolutions)
☐ 2D motion with vectors — Vectors: Kinematics Problem (TLMaths)
☐ Newton's laws and resolving forces — Forces: Resolving Forces Example 1 (TLMaths)
☐ Friction — Forces: Introducing the Coefficient of Friction (TLMaths)
☐ Connected particles and pulleys — Forces: Pulley Example 1 (TLMaths)
☐ Inclined planes — Forces: Introducing Smooth Inclined Planes (TLMaths)
☐ Moments, equilibrium and centre of mass — Moments: What is a Moment? (TLMaths)
☐ Projectiles — Projectiles: Introducing Projectiles (TLMaths)

Q5.2.01 Friction ●○○ [4 marks] ☐ Todo ☐ Mastered

A box of mass 8 kg on rough horizontal ground ($\mu = 0.3$) is pulled by a horizontal force of 41 N. Find its acceleration.

(4)

WORKING

Q5.2.02 Quantities and units, modelling assumptions ●○○ [2 marks] ☐ Todo ☐ Mastered

In mechanics, explain the meaning of the modelling term 'particle' and its effect on calculations.

(2)

WORKING

Q5.2.03 Motion graphs ●○○ [5 marks] ☐ Todo ☐ Mastered

A train starts from rest and accelerates uniformly for 8 s to 12 m s^{-1}, travels at this speed for 17 s, then decelerates uniformly to rest in 5 s.

(a) Sketch a velocity-time graph. **(2)**

(b) Find the total distance travelled. **(2)**

(c) Find the deceleration. **(1)**

WORKING

Q5.2.04 Friction ●○○ [4 marks] ☐ Todo ☐ Mastered

A box of mass 3 kg on rough horizontal ground ($\mu = 0.4$) is pulled by a horizontal force of 71 N. Find its acceleration.

(4)

WORKING

Q5.2.05 Quantities and units, modelling assumptions ●○○ [2 marks] ☐ Todo ☐ Mastered

In mechanics, explain the meaning of the modelling term 'rigid' and its effect on calculations.

(2)

WORKING

Q5.2.06 Motion graphs ●○○ [5 marks] ☐ Todo ☐ Mastered

A train starts from rest and accelerates uniformly for 8 s to 9 m s^{-1}, travels at this speed for 25 s, then decelerates uniformly to rest in 2 s.

(a) Sketch a velocity-time graph. **(2)**

(b) Find the total distance travelled. **(2)**

(c) Find the deceleration. **(1)**

WORKING

Q5.2.07 Friction ●○○ [4 marks] ☐ Todo ☐ Mastered

A box of mass 5 kg on rough horizontal ground ($\mu = 0.1$) is pulled by a horizontal force of 40 N. Find its acceleration.

(4)

WORKING

Q5.2.08 Quantities and units, modelling assumptions ●○○ [2 marks] ☐ Todo ☐ Mastered

In mechanics, explain the meaning of the modelling term 'uniform rod' and its effect on calculations.

(2)

WORKING

Q5.2.09 Motion graphs ●○○ [5 marks] ☐ Todo ☐ Mastered

A train starts from rest and accelerates uniformly for 5 s to 18 m s^{-1}, travels at this speed for 21 s, then decelerates uniformly to rest in 8 s.

(a) Sketch a velocity-time graph. **(2)**

(b) Find the total distance travelled. **(2)**

(c) Find the deceleration. **(1)**

WORKING

Q5.2.10 Connected particles and pulleys ●●○ [8 marks] ☐ Todo ☐ Mastered

Particles A (2 kg) and B (6 kg) are connected by a light inextensible string over a smooth fixed pulley. The system is released from rest with B 4 m above the ground.

(a) Find the acceleration of the system and the tension in the string. **(5)**

(b) Find the speed with which B hits the ground. **(2)**

(c) State how you used the fact that the string is inextensible. **(1)**

WORKING

Q5.2.11 Moments, equilibrium and centre of mass ●●○ [5 marks] ☐ Todo ☐ Mastered

A uniform beam AB of length 4 m and weight 80 N rests horizontally on supports at A and B. A load of 72 N is placed 2 m from A. Find the reactions at the supports, and state one modelling assumption.

(5)

WORKING

Q5.2.12 Newton's laws and resolving forces ●●○ [5 marks] ☐ Todo ☐ Mastered

A particle of weight 19 N hangs in equilibrium from two light strings inclined at 30° and 60° to the horizontal. Find the tension in each string.

(5)

WORKING

Q5.2.13 Inclined planes ●●○ [6 marks] ☐ Todo ☐ Mastered

A particle of mass 5 kg slides down a rough plane inclined at α to the horizontal, where $\tan\alpha = \frac{7}{24}$. The coefficient of friction is 0.1.

Find the normal reaction and the acceleration of the particle.

(6)

WORKING

Q5.2.14 Variable acceleration with calculus ●●○ [5 marks] ☐ Todo ☐ Mastered

A particle moves in a straight line with velocity $v = -t^2 - 3t + 4$ m s^{-1} at time t s.

(a) Find the acceleration when $t = 5$. **(2)**

(b) Find the displacement from the starting point when $t = 5$. **(3)**

WORKING

Q5.2.15 Projectiles ●●○ [7 marks] ☐ Todo ☐ Mastered

A particle is projected from a point O on horizontal ground with speed 23 m s^{-1} at angle α above the horizontal, where $\tan\alpha = \frac{5}{12}$.

Find (a) the time of flight, (b) the range, (c) the greatest height.

(a) **(3)**

(b) **(2)**

(c) **(2)**

WORKING

Q5.2.16 2D motion with vectors ●●○ [4 marks] ☐ Todo ☐ Mastered

A particle has initial velocity $(5\mathbf{i} - \mathbf{j})$ m s^{-1} and constant acceleration $(-2\mathbf{i} - \mathbf{j})$ m s^{-2}. Find its speed after 4 s and the angle its direction of motion makes with $\mathbf{i}$.

(4)

WORKING

Q5.2.17 Constant acceleration: suvat ●●○ [7 marks] ☐ Todo ☐ Mastered

A ball is projected vertically upwards at 15 m s^{-1} from a point 30 m above the ground.

(a) Find the greatest height above the ground. **(3)**

(b) Find the time taken to reach the ground. **(3)**

(c) State one modelling assumption. **(1)**

WORKING

Q5.2.18 Connected particles and pulleys ●●○ [8 marks] ☐ Todo ☐ Mastered

Particles A (3 kg) and B (5 kg) are connected by a light inextensible string over a smooth fixed pulley. The system is released from rest with B 3 m above the ground.

(a) Find the acceleration of the system and the tension in the string. **(5)**

(b) Find the speed with which B hits the ground. **(2)**

(c) State how you used the fact that the string is inextensible. **(1)**

WORKING

Q5.2.19 Moments, equilibrium and centre of mass ●●○ [5 marks] ☐ Todo ☐ Mastered

A uniform beam AB of length 5 m and weight 169 N rests horizontally on supports at A and B. A load of 52 N is placed 0.5 m from A. Find the reactions at the supports, and state one modelling assumption.

(5)

WORKING

Q5.2.20 Newton's laws and resolving forces ●●○ [5 marks] ☐ Todo ☐ Mastered

A particle of weight 42 N hangs in equilibrium from two light strings inclined at 40° and 50° to the horizontal. Find the tension in each string.

(5)

WORKING

Q5.2.21 Inclined planes ●●○ [6 marks] ☐ Todo ☐ Mastered

A particle of mass 5 kg slides down a rough plane inclined at α to the horizontal, where $\tan\alpha = \frac{5}{12}$. The coefficient of friction is 0.2.
Find the normal reaction and the acceleration of the particle.

(6)

WORKING

Q5.2.22 Variable acceleration with calculus ●●○ [5 marks] ☐ Todo ☐ Mastered

A particle moves in a straight line with velocity $v = -3t^2 - 10t + 5$ m s^{-1} at time t s.

(a) Find the acceleration when $t = 3$. **(2)**

(b) Find the displacement from the starting point when $t = 3$. **(3)**

WORKING

Q5.2.23 Projectiles ●●○ [7 marks] ☐ Todo ☐ Mastered

A particle is projected from a point O on horizontal ground with speed 16 m s^{-1} at angle α above the horizontal, where $\tan\alpha = \frac{4}{3}$.

Find (a) the time of flight, (b) the range, (c) the greatest height.

(a) **(3)**

(b) **(2)**

(c) **(2)**

WORKING

Q5.2.24 2D motion with vectors ●●○ [4 marks] ☐ Todo ☐ Mastered

A particle has initial velocity $(2\mathbf{i})$ m s^{-1} and constant acceleration $(-3\mathbf{i} - 3\mathbf{j})$ m s^{-2}. Find its speed after 4 s and the angle its direction of motion makes with $\mathbf{i}$.

(4)

WORKING

Q5.2.25 **Constant acceleration: suvat** ●●○ **[7 marks]** ☐ Todo ☐ Mastered

A ball is projected vertically upwards at 11 m s^{-1} from a point 30 m above the ground.

(a) Find the greatest height above the ground. **(3)**

(b) Find the time taken to reach the ground. **(3)**

(c) State one modelling assumption. **(1)**

WORKING

Q5.2.26 **Connected particles and pulleys** ●●○ **[8 marks]** ☐ Todo ☐ Mastered

Particles A (1 kg) and B (7 kg) are connected by a light inextensible string over a smooth fixed pulley. The system is released from rest with B 2 m above the ground.

(a) Find the acceleration of the system and the tension in the string. **(5)**

(b) Find the speed with which B hits the ground. **(2)**

(c) State how you used the fact that the string is inextensible. **(1)**

WORKING

Q5.2.27 **Moments, equilibrium and centre of mass** ●●○ **[5 marks]** ☐ **Todo** ☐ **Mastered**

A uniform beam AB of length 4 m and weight 187 N rests horizontally on supports at A and B. A load of 85 N is placed 3.5 m from A. Find the reactions at the supports, and state one modelling assumption.

(5)

WORKING

Q5.2.28 **Newton's laws and resolving forces** ●●○ **[5 marks]** ☐ **Todo** ☐ **Mastered**

A particle of weight 18 N hangs in equilibrium from two light strings inclined at 45° and 45° to the horizontal. Find the tension in each string.

(5)

WORKING

Q5.2.29 Inclined planes ●●○ [6 marks] ☐ Todo ☐ Mastered

A particle of mass 9 kg slides down a rough plane inclined at α to the horizontal, where $\tan\alpha = \frac{5}{12}$. The coefficient of friction is 0.1.

Find the normal reaction and the acceleration of the particle.

(6)

WORKING

Q5.2.30 Variable acceleration with calculus ●●○ [5 marks] ☐ Todo ☐ Mastered

A particle moves in a straight line with velocity $v = 3t^2 - 8t + 7$ m s^{-1} at time t s.

(a) Find the acceleration when $t = 2$. **(2)**

(b) Find the displacement from the starting point when $t = 2$. **(3)**

WORKING

Q5.2.31 Projectiles ●●○ [7 marks] ☐ Todo ☐ Mastered

A particle is projected from a point O on horizontal ground with speed 10 m s^{-1} at angle α above the horizontal, where $\tan\alpha = \frac{3}{4}$.

Find (a) the time of flight, (b) the range, (c) the greatest height.

(a) **(3)**

(b) **(2)**

(c) **(2)**

WORKING

Q5.2.32 2D motion with vectors ●●○ [4 marks] ☐ Todo ☐ Mastered

A particle has initial velocity $(-3\mathbf{i} + 2\mathbf{j})$ m s^{-1} and constant acceleration $(1\mathbf{i} - 2\mathbf{j})$ m s^{-2}. Find its speed after 6 s and the angle its direction of motion makes with $\mathbf{i}$.

(4)

WORKING

Q5.2.33 Constant acceleration: suvat ●●○ [7 marks] ☐ Todo ☐ Mastered

A ball is projected vertically upwards at 18 m s^{-1} from a point 11 m above the ground.

(a) Find the greatest height above the ground. **(3)**

(b) Find the time taken to reach the ground. **(3)**

(c) State one modelling assumption. **(1)**

WORKING

MY NOTES — 5.2 MECHANICS: MISTAKES TO REMEMBER

Answers and mark schemes

M = method mark, A = accuracy mark (depends on the preceding M), B = independent mark, ft = follow-through, * = given answer. Numerical answers are to 3 s.f. unless stated.

5.1 Statistics

Q5.1.01 Histograms, box plots, cumulative frequency, outliers · 4 marks

(a) 0.500, 1.30, 1.60, 1.10, 0.433

M1 frequency ÷ class width · A1

(b) below -8.5 or above 91.5

M1 IQR $= 25$ · A1

Q5.1.02 Measures of location and spread, coding · 5 marks

(a) mean 37.35, s.d. 14.57

B1 mean · M1 $\sqrt{\frac{\Sigma x^2}{n} - \bar{x}^2}$ · A1

(b) mean 3.235, s.d. 1.457

B1ft · B1ft

Q5.1.03 The large data set · 2 marks

different hemisphere/season and climate; not representative of UK summer

B1 per valid point · B1 per valid point

Q5.1.04 Binomial distribution · 7 marks

(a) 0.2522

M1 ${}^{10}C_3(0.35)^3(0.65)^7$ · A1

(b) 0.9740

B1

(c) 0.4862

M1 $1 - \mathrm{P}(X \leq 3)$ · A1

(d) 0.4601

M1 $\mathrm{P}(X \leq 6) - \mathrm{P}(X \leq 3)$ · A1

Q5.1.05 Probability: Venn diagrams and set notation · 9 marks

(a) Art only 11, both 8, Biology only 5, neither 26

B1 both · B1 only regions · B1 neither

(b) $\frac{4}{25}$; $\frac{9}{10}$; $\frac{8}{13}$

B1 · M1A1 · B1

(c) $\mathrm{P}(A)\mathrm{P}(B) = \frac{247}{2500} \neq \frac{4}{25}$ so not independent

M1 · A1

Q5.1.06 Populations and sampling methods · 2 marks

advantage: no sampling frame needed, quick; disadvantage: not random, interviewer bias

B1 per valid point · B1 per valid point

Q5.1.07 Correlation and regression · 3 marks

(a) each extra unit of x changes ice creams sold by 0.31

B1

(b) 23.5; extrapolation, so unreliable

B1 · B1

Q5.1.08 Histograms, box plots, cumulative frequency, outliers · 4 marks

(a) 1.40, 1.10, 2.90, 0.550, 0.867

M1 frequency ÷ class width · A1

(b) below -4 or above 68

M1 IQR $= 18$ · A1

Q5.1.09 Measures of location and spread, coding · 5 marks

(a) mean 36.00, s.d. 16.21

B1 mean · M1 $\sqrt{\frac{\Sigma x^2}{n} - \bar{x}^2}$ · A1

(b) mean 2.800, s.d. 1.621

B1ft · B1ft

Q5.1.10 The large data set · 2 marks

trace: less than 0.05 mm; treated as 0

B1 per valid point · B1 per valid point

Q5.1.11 Binomial distribution · 7 marks

(a) 0.2402

M1 ${}^{14}C_3(0.25)^3(0.75)^{11}$ · A1

(b) 0.5213

B1

(c) 0.4787

M1 $1 - \mathrm{P}(X \leq 3)$ · A1

(d) 0.0000

M1 $\mathrm{P}(X \leq 3) - \mathrm{P}(X \leq 3)$ · A1

Q5.1.12 Probability: Venn diagrams and set notation · 9 marks

(a) Art only 13, both 7, Biology only 12, neither 28

B1 both · B1 only regions · B1 neither

(b) $\frac{7}{60}$; $\frac{4}{5}$; $\frac{7}{19}$

B1 · M1A1 · B1

(c) $\mathrm{P}(A)\mathrm{P}(B) = \frac{19}{180} \neq \frac{7}{60}$ so not independent

M1 · A1

Q5.1.13 Populations and sampling methods · 3 marks

number students 1 to 900 (sampling frame); use a random number generator to pick 40 distinct numbers; ignore repeats

B1 per valid point · B1 per valid point · B1 per valid point

Q5.1.14 Correlation and regression · 3 marks

(a) each extra unit of x changes ice creams sold by 3.92

B1

(b) 96.7; interpolation, so reliable

B1 · B1

Q5.1.15 Histograms, box plots, cumulative frequency, outliers · 4 marks

(a) 1.20, 0.800, 2.50, 0.650, 0.933

M1 frequency ÷ class width · A1

(b) below -2.5 or above 73.5

M1 IQR $= 19$ · A1

Q5.1.16 Measures of location and spread, coding · 5 marks

(a) mean 36.66, s.d. 15.20

B1 mean · M1 $\sqrt{\frac{\Sigma x^2}{n} - \bar{x}^2}$ · A1

(b) mean 5.331, s.d. 3.039

B1ft · B1ft

Q5.1.17 The large data set · 2 marks

data not recorded/available; remove those days before calculating statistics

B1 per valid point · B1 per valid point

Q5.1.18 Binomial distribution · 7 marks

(a) 0.0025

M1 ${}^{24}C_5(0.5)^5(0.5)^{19}$ · A1

(b) 0.9887

B1

(c) 0.9967

M1 $1 - \mathrm{P}(X \leq 5)$ · A1

(d) 0.9854

M1 $\mathrm{P}(X \leq 17) - \mathrm{P}(X \leq 5)$ · A1

Q5.1.19 Probability: Venn diagrams and set notation · 9 marks

(a) Art only 14, both 9, Biology only 13, neither 24

B1 both · B1 only regions · B1 neither

(b) $\frac{3}{20}$; $\frac{47}{60}$; $\frac{9}{22}$

B1 · M1A1 · B1

(c) $\mathrm{P}(A)\mathrm{P}(B) = \frac{253}{1800} \neq \frac{3}{20}$ so not independent

M1 · A1

Q5.1.20 Populations and sampling methods · 2 marks

census: every member, fully accurate; sample: subset, quicker/cheaper/less data

B1 per valid point · B1 per valid point

Q5.1.21 Correlation and regression · 3 marks

(a) each extra unit of x changes test score by 3.36

B1

(b) 90.4; extrapolation, so unreliable

B1 · B1

Q5.1.22 Normal approximation to binomial · 5 marks

(a) n is large and $np = 70.8$, $n(1-p) = 106$ are both greater than 5 (p not close to 0 or 1)

B1

(b) $Y \sim \mathrm{N}(70.80, 42.48)$, $\mathrm{P}(Y < 72.5) = 0.6029$ (exact 0.6047)

B1 mean and variance · M1 continuity correction · M1 standardises · A1

Q5.1.23 Hypothesis testing: critical regions and p-values · 5 marks

$\mathrm{H}_0 : p = 0.4$, $\mathrm{H}_1 : p < 0.4$; $\mathrm{P}(X \leq 13) = 0.2112 > 0.1$; do not reject H_0: insufficient evidence that the proportion has decreased

B1 both hypotheses in terms of p · M1 $\mathrm{B}(40, 0.4)$ used · A1 p-value · M1 compares with significance level · A1 conclusion in context

Q5.1.24 Conditional probability and tree diagrams · 6 marks

(a) $\frac{2}{5}$

M1 tree diagram / sum of two products · A1 one product · A1

(b) $\frac{2}{3}$

M1 $\frac{\mathrm{P}(A \cap B)}{\mathrm{P}(B)}$ · A1ft numerator · A1

Q5.1.25 Normal distribution · 3 marks

$a = 91.37$

M1 $z = 0.8416$ · M1 $\frac{a-88}{4} = z$ · A1

Q5.1.26 Normal approximation to binomial · 5 marks

(a) n is large and $np = 31.0$, $n(1-p) = 31.0$ are both greater than 5 (p not close to 0 or 1)

B1

(b) $Y \sim \mathrm{N}(31.00, 15.50)$, $\mathrm{P}(Y < 36.5) = 0.9188$ (exact 0.9191)

B1 mean and variance · M1 continuity correction · M1 standardises · A1

Q5.1.27 Hypothesis testing: critical regions and p-values · 5 marks

$\mathrm{H}_0 : p = 0.5$, $\mathrm{H}_1 : p < 0.5$; $\mathrm{P}(X \leq 5) = 0.0207 < 0.05$; reject H_0: evidence that the proportion has decreased

B1 both hypotheses in terms of p · M1 $\mathrm{B}(20, 0.5)$ used · A1 p-value · M1 compares with significance level · A1 conclusion in context

Q5.1.28 Conditional probability and tree diagrams · 6 marks

(a) $\frac{5}{12}$

M1 tree diagram / sum of two products · A1 one product · A1

(b) $\frac{16}{25}$

M1 $\frac{\mathrm{P}(A \cap B)}{\mathrm{P}(B)}$ · A1ft numerator · A1

Q5.1.29 Normal approximation to binomial · 5 marks

(a) n is large and $np = 35.8$, $n(1-p) = 29.2$ are both greater than 5 (p not close to 0 or 1)

B1

(b) $Y \sim \mathrm{N}(35.75, 16.09)$, $\mathrm{P}(Y < 31.5) = 0.1447$ (exact 0.1447)

B1 mean and variance · M1 continuity correction · M1 standardises · A1

Q5.1.30 Hypothesis testing: critical regions and p-values · 5 marks

$\mathrm{H}_0 : p = 0.2$, $\mathrm{H}_1 : p > 0.2$; $\mathrm{P}(X \geq 6) = 0.1958 > 0.1$; do not reject H_0: insufficient evidence that the proportion has increased

B1 both hypotheses in terms of p · M1 $\mathrm{B}(20, 0.2)$ used · A1 p-value · M1 compares with significance level · A1 conclusion in context

Q5.1.31 Conditional probability and tree diagrams · 6 marks

(a) $\frac{31}{100}$

M1 tree diagram / sum of two products · A1 one product · A1

(b) $\frac{24}{31}$

M1 $\frac{\mathrm{P}(A \cap B)}{\mathrm{P}(B)}$ · A1ft numerator · A1

Q5.1.32 Normal distribution · 6 marks

$\mu = 122.0$, $\sigma = 9.00$

B1 $z = -0.8416$ · B1 $z = 1.2816$ · M1 $\frac{114.4-\mu}{\sigma} = -0.8416$ · M1 second equation · M1 solves simultaneously · A1 both

Q5.1.33 Hypothesis testing: normal mean and correlation · 6 marks

$\bar{X} \sim \mathrm{N}(90, \frac{17^2}{36})$; $z = -1.094$, p-value 0.1370; do not reject H_0

B1 $\mathrm{H}_0 : \mu = 90$, $\mathrm{H}_1 : \mu < 90$ · M1 distribution of $\bar{X}$ · M1 standardises · A1 p-value or z vs critical value · M1 comparison · A1 conclusion in context

Q5.1.34 Hypothesis testing: normal mean and correlation · 6 marks

$\bar{X} \sim \mathrm{N}(54, \frac{15^2}{20})$; $z = -2.385$, p-value 0.0085; reject H_0

B1 $\mathrm{H}_0 : \mu = 54$, $\mathrm{H}_1 : \mu < 54$ · M1 distribution of $\bar{X}$ · M1 standardises · A1 p-value or z vs critical value · M1 comparison · A1 conclusion in context

Q5.1.35 Normal distribution · 6 marks

$\mu = 132.0$, $\sigma = 9.00$

B1 $z = -1.2816$ · B1 $z = 0.8416$ · M1 $\frac{120.5-\mu}{\sigma} = -1.2816$ · M1 second equation · M1 solves simultaneously · A1 both

Q5.1.36 **Hypothesis testing: normal mean and correlation · 6 marks**

$\bar{X} \sim \text{N}(125, \frac{9^2}{20})$; $z = 1.640$, p-value 0.0505; do not reject H_0

B1 $\text{H}_0 : \mu = 125$, $\text{H}_1 : \mu > 125$ · M1 distribution of $\bar{X}$ · M1 standardises · A1 p-value or z vs critical value · M1 comparison · A1 conclusion in context

5.2 Mechanics

Q5.2.01 **Friction · 4 marks**

$2.19\ \text{m s}^{-2}$

M1 $R = 8g$ · M1 $F = \mu R$ · M1 $P - F = ma$ · A1

Q5.2.02 **Quantities and units, modelling assumptions · 2 marks**

dimensions negligible; no rotation, mass at a point

B1 meaning · B1 effect

Q5.2.03 **Motion graphs · 5 marks**

(a) trapezium shape

B1 shape · B1 values

(b) 282 m

M1 area of trapezium · A1

(c) $2.40\ \text{m s}^{-2}$

B1

Q5.2.04 **Friction · 4 marks**

$19.7\ \text{m s}^{-2}$

M1 $R = 3g$ · M1 $F = \mu R$ · M1 $P - F = ma$ · A1

Q5.2.05 **Quantities and units, modelling assumptions · 2 marks**

does not bend

B1 meaning · B1 effect

Q5.2.06 **Motion graphs · 5 marks**

(a) trapezium shape

B1 shape · B1 values

(b) 270 m

M1 area of trapezium · A1

(c) $4.50\ \text{m s}^{-2}$

B1

Q5.2.07 **Friction · 4 marks**

$7.02\ \text{m s}^{-2}$

M1 $R = 5g$ · M1 $F = \mu R$ · M1 $P - F = ma$ · A1

Q5.2.08 **Quantities and units, modelling assumptions · 2 marks**

mass evenly distributed so weight acts at the midpoint

B1 meaning · B1 effect

Q5.2.09 **Motion graphs · 5 marks**

(a) trapezium shape

B1 shape · B1 values

(b) 495 m

M1 area of trapezium · A1

(c) $2.25\ \text{m s}^{-2}$

B1

Q5.2.10 **Connected particles and pulleys · 8 marks**

(a) $a = 4.90\ \text{m s}^{-2}$, $T = 29.4\ \text{N}$

M1 equation for B: $6g - T = 6a$ · M1 equation for A: $T - 2g = 2a$ · A1 both correct · A1 a · A1 T

(b) $6.26\ \text{m s}^{-1}$

M1 $v^2 = 2as$ · A1

(c) both particles have the same acceleration (magnitude)

B1

Q5.2.11 **Moments, equilibrium and centre of mass · 5 marks**

$R_A = 76.0\ \text{N}$, $R_B = 76.0\ \text{N}$

M1 moments about A · A1 · M1 resolves vertically · A1 · B1 uniform rod / load as particle

Q5.2.12 **Newton's laws and resolving forces · 5 marks**

$T_1 = 9.50\ \text{N}$ (at 30°), $T_2 = 16.5\ \text{N}$ (at 60°)

M1 resolves horizontally · M1 resolves vertically · A1 equations · M1 solves · A1 both

Q5.2.13 **Inclined planes · 6 marks**

$R = 47.0$ N, $a = 1.80$ m s^{-2}

B1 $\sin\alpha = \frac{7}{25}$, $\cos\alpha = \frac{24}{25}$ · M1 resolves perpendicular · A1 R · M1 $F = \mu R$ · M1 Newton II along the plane · A1 a

Q5.2.14 **Variable acceleration with calculus · 5 marks**

(a) -13 m s^{-2}

M1 differentiates · A1

(b) $-\frac{355}{6}$ m

M1 integrates · M1 limits / $c = 0$ · A1

Q5.2.15 **Projectiles · 7 marks**

(a) 1.81 s

B1 components 21.2308 and 8.84615 · M1 $0 = U\sin\alpha\, t - 4.9t^2$ · A1

(b) 38.3 m

M1 horizontal distance · A1

(c) 3.99 m

M1 $v^2 = u^2 - 2gs$ · A1

Q5.2.16 **2D motion with vectors · 4 marks**

$\mathbf{v} = -3\mathbf{i} - 5\mathbf{j}$; speed 5.83 m s^{-1}; angle 121.0° from $\mathbf{i}$

M1 $\mathbf{v} = \mathbf{u} + \mathbf{a}t$ · A1 · M1 magnitude · A1 angle

Q5.2.17 **Constant acceleration: suvat · 7 marks**

(a) 41.5 m

M1 $v^2 = u^2 + 2as$ with $v = 0$ · A1 height above start · A1

(b) 4.44 s

M1 $-30 = 15t - 4.9t^2$ · M1 solves · A1

(c) particle; no air resistance; g constant

B1

Q5.2.18 **Connected particles and pulleys · 8 marks**

(a) $a = 2.45$ m s^{-2}, $T = 36.8$ N

M1 equation for B: $5g - T = 5a$ · M1 equation for A: $T - 3g = 3a$ · A1 both correct · A1 a · A1 T

(b) 3.83 m s^{-1}

M1 $v^2 = 2as$ · A1

(c) both particles have the same acceleration (magnitude)

B1

Q5.2.19 **Moments, equilibrium and centre of mass · 5 marks**

$R_A = 131$ N, $R_B = 89.7$ N

M1 moments about A · A1 · M1 resolves vertically · A1 · B1 uniform rod / load as particle

Q5.2.20 **Newton's laws and resolving forces · 5 marks**

$T_1 = 27.0$ N (at 40°), $T_2 = 32.2$ N (at 50°)

M1 resolves horizontally · M1 resolves vertically · A1 equations · M1 solves · A1 both

Q5.2.21 **Inclined planes · 6 marks**

$R = 45.2$ N, $a = 1.96$ m s^{-2}

B1 $\sin\alpha = \frac{5}{13}$, $\cos\alpha = \frac{12}{13}$ · M1 resolves perpendicular · A1 R · M1 $F = \mu R$ · M1 Newton II along the plane · A1 a

Q5.2.22 **Variable acceleration with calculus · 5 marks**

(a) -28 m s^{-2}

M1 differentiates · A1

(b) -57 m

M1 integrates · M1 limits / $c = 0$ · A1

Q5.2.23 **Projectiles · 7 marks**

(a) 2.61 s

B1 components 9.6 and 12.8 · M1 $0 = U\sin\alpha\, t - 4.9t^2$ · A1

(b) 25.1 m

M1 horizontal distance · A1

(c) 8.36 m

M1 $v^2 = u^2 - 2gs$ · A1

Q5.2.24 **2D motion with vectors · 4 marks**

$\mathbf{v} = -10\mathbf{i} - 12\mathbf{j}$; speed 15.6 m s^{-1}; angle 129.8° from $\mathbf{i}$

M1 $\mathbf{v} = \mathbf{u} + \mathbf{a}t$ · A1 · M1 magnitude · A1 angle

Q5.2.25 Constant acceleration: suvat · 7 marks

(a) 36.2 m

M1 $v^2 = u^2 + 2as$ with $v = 0$ · A1 height above start · A1

(b) 3.84 s

M1 $-30 = 11t - 4.9t^2$ · M1 solves · A1

(c) particle; no air resistance; g constant

B1

Q5.2.26 Connected particles and pulleys · 8 marks

(a) $a = 7.35$ m s^{-2}, $T = 17.2$ N

M1 equation for B: $7g - T = 7a$ · M1 equation for A: $T - 1g = 1a$ · A1 both correct · A1 a · A1 T

(b) 5.42 m s^{-1}

M1 $v^2 = 2as$ · A1

(c) both particles have the same acceleration (magnitude)

B1

Q5.2.27 Moments, equilibrium and centre of mass · 5 marks

$R_A = 104$ N, $R_B = 168$ N

M1 moments about A · A1 · M1 resolves vertically · A1 · B1 uniform rod / load as particle

Q5.2.28 Newton's laws and resolving forces · 5 marks

$T_1 = 12.7$ N (at 45°), $T_2 = 12.7$ N (at 45°)

M1 resolves horizontally · M1 resolves vertically · A1 equations · M1 solves · A1 both

Q5.2.29 Inclined planes · 6 marks

$R = 81.4$ N, $a = 2.86$ m s^{-2}

B1 $\sin\alpha = \frac{5}{13}$, $\cos\alpha = \frac{12}{13}$ · M1 resolves perpendicular · A1 R · M1 $F = \mu R$ · M1 Newton II along the plane · A1 a

Q5.2.30 Variable acceleration with calculus · 5 marks

(a) 4 m s^{-2}

M1 differentiates · A1

(b) 6 m

M1 integrates · M1 limits / $c = 0$ · A1

Q5.2.31 Projectiles · 7 marks

(a) 1.22 s

B1 components 8 and 6 · M1 $0 = U\sin\alpha\, t - 4.9t^2$ · A1

(b) 9.80 m

M1 horizontal distance · A1

(c) 1.84 m

M1 $v^2 = u^2 - 2gs$ · A1

Q5.2.32 2D motion with vectors · 4 marks

$\mathbf{v} = 3\mathbf{i} - 10\mathbf{j}$; speed 10.4 m s^{-1}; angle 73.3° from $\mathbf{i}$

M1 $\mathbf{v} = \mathbf{u} + \mathbf{a}t$ · A1 · M1 magnitude · A1 angle

Q5.2.33 Constant acceleration: suvat · 7 marks

(a) 27.5 m

M1 $v^2 = u^2 + 2as$ with $v = 0$ · A1 height above start · A1

(b) 4.21 s

M1 $-11 = 18t - 4.9t^2$ · M1 solves · A1

(c) particle; no air resistance; g constant

B1